CyberSeat Team 3 Consolidated Report

Information System for Managing Gaming Station Reservations in Esports Centres

Astana IT University | Software Engineering | SE-2404

Aibekuly Alibi - shared database and reservation module
Akhmetov Nurali - administrative functionality and reporting
Mansur Urynbassar - visitor and administrator interfaces

Supervisor: Myrzakerimova Alua

October 24, 2026 | Analysis, design, and planning submission

Revised review copy. The supervisor response is now documented verbatim with an English translation. Alibi and Mansur reviewed the preceding full PDF; the changed sections in this rebuilt file require revision-specific confirmation. Separate release/access controls and university receipt are not claimed.

The source chapters reproduce the current repository materials. Claims about interviews, earlier reviews, and tests retain their source provenance and limitations. The supervisor response is supplied evidence, and the submitted approval of PR 13 is linked. This compilation creates no interview, review, implementation, or application-test result.

Repository: https://github.com/NeryBR/cyberseat-booking-system

Pinned source baseline: 0ac6fb0f1b662896910127040ba6a3e0876ddb45

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Reading guide

Chapters 1-11 cover the committee components. Architecture includes all three readable diagrams. UX includes current v2 documentation and clearly dated v1.1 screen plates. The audit, source-review records, and manifest follow the main chapters. The feedback chapter records the actual supervisor response, the original wording, English translation, date, channel, and provenance; the private screenshot is retained separately.

Source approval does not constitute approval of this compiled report. Historical open checkboxes are reproduced as outstanding source actions and must be read together with the later closure record. The task audit is a point-in-time observation; live statuses may change.

1. Project proposal and scope

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/report/project-proposal.md

Version 1.1 | October 23, 2026 | Owner: Aibekuly Alibi | ECR-2

Status: documentation completed; team agreement confirmed by Alibi on October 23, 2026. This is a design-stage proposal, not evidence of an implemented or evaluated product.

1. Project identity

Title: Information System for Managing Gaming Station Reservations in Esports Centres.

Programme and group: Software Engineering, Astana IT University, SE-2404. **Supervisor:** Myrzakerimova Alua. The names and responsibilities below follow the current repository README.

Aibekuly Alibi

Diploma responsibility: Shared database and reservation module

Initial-evaluation contribution: Proposal, literature review, technology comparison, booking and non-functional requirements, system context and component diagrams

Mansur Urynbassar

Diploma responsibility: Visitor and administrator interfaces

Initial-evaluation contribution: User evidence, existing-system comparison, scenarios, UX mockups and use-case diagram

Akhmetov Nurali

Diploma responsibility: Administrative server functionality

Initial-evaluation contribution: Administrative requirements, access matrix, evaluation plan, risks and supervisor-feedback record

2. Problem and evidence boundary

The working problem hypothesis is that separate telephone, messaging and spreadsheet records make it difficult to give visitors consistent station availability and administrators a consolidated schedule. Two users may both see a station as available before either reservation is saved. Independent administrative changes can also invalidate a reservation if schedule and maintenance rules are checked inconsistently.

The ECR-3 evidence record now documents three anonymised visitor discussions supplied by its owner. They support the exploratory visitor-need findings; they do not establish a representative market estimate, administrator validation, client endorsement or measured reduction in manual work. The ECR-5 system comparison provides the product comparison. Literature supports the concurrency problem and design methods; local visitor evidence comes from ECR-3. The records retain their provenance and validation limits.

Target users: guests comparing clubs; registered visitors reserving PC or console stations; administrators operating assigned clubs; a system administrator maintaining clubs and administrator assignments. A multi-club manager can be represented through assignments to several clubs, without an unrestricted extra role.

3. Aim, research questions and objectives

Aim: design, implement and evaluate a web information system that combines gaming-station discovery, interval reservations and club administration while preserving reservation consistency and role-based access boundaries.

RQ1: Can the proposed transaction protocol preserve non-overlapping active reservations and maintenance intervals under concurrent operations and retries? **RQ2:** Can target users complete discovery, reservation and cancellation tasks through the proposed interface? **RQ3:** Does a shared schedule provide administrators with correct, reproducible utilisation information?

O1 Identify users, needs and scope

Deliverable: Evidence record and bounded proposal

Planned evidence of achievement: Trace each claimed need to a real interview/client record or label it an assumption

O2 Examine research and alternatives

Deliverable: Critical review, product comparison, technology decisions

Planned evidence of achievement: Cited sources, limitations and explicit requirement implications

O3 Specify verifiable behaviour

Deliverable: Requirements, access matrix and scenarios

Planned evidence of achievement: IDs, priorities, acceptance criteria and cross-document traceability

O4 Design interactions and architecture

Deliverable: UX mockups, context, components and use cases

Planned evidence of achievement: Consistency check against scenarios, data ownership and failure flows

O5 Build the core workflow after initial evaluation

Deliverable: Web client, API, relational schema and migrations

Planned evidence of achievement: Demonstration of discovery, creation, viewing and cancellation

O6 Evaluate correctness and usability

Deliverable: Reproducible tests and participant task observations

Planned evidence of achievement: Invariant checks, access tests, load conditions and task results; no invented outcomes

4. Included and excluded scope

Included	Boundary
Club and station catalogue	PC and console specifications, club address, working hours, filters and station layout
Availability search	One station and a finite visit interval; club-local display; server decides at commit
Personal reservations	Authenticated creation, own-record viewing, cancellation before start and retained history
Club administration	Assigned-club stations, schedules, maintenance intervals, reservations, events and reports
Role-based access	Guest, visitor, club administrator and system administrator; enforced by the server
Event calendar	Informational published events; browsing does not create a reservation or tournament entry
Reporting	Defined utilisation measures from controlled booking and schedule data; no financial settlement

Excluded: online payments, loyalty programmes, hardware/device control, native mobile applications, automatic tournament registration and paid ticketing. External tournament feeds, email/SMS and social login are not required

dependencies. Administrators can enter event information manually with its source and publication status. Group booking across multiple stations is deferred; the core transaction reserves one station.

5. Proposed solution and critical constraints

A React/TypeScript web client communicates with a FastAPI REST API. The API is a modular monolith with access, catalogue, reservation, administration, events and reporting boundaries backed by PostgreSQL. These are proposed choices, justified in the technology comparison, not installed or benchmarked services.

The central invariant is that confirmed reservations for the same station never overlap. A reservation must also respect working hours and maintenance. Checking the browser display alone cannot preserve this invariant. Creation, cancellation, maintenance and schedule changes must share a database transaction protocol. Repeated submission of one logical creation operation must not create another booking. An unknown network outcome must remain unknown until reconciled.

The design uses half-open intervals `[start, end)`, explicit club time zones and UTC instants in storage. A booking ending at 15:00 may be followed by one starting at 15:00. A detailed schema is deferred to ECR-12; these semantics are already required for the October design.

6. Delivery stages and evaluation

Stage	Exit condition
Initial submission, October 24, 2026 before 18:00 according to the university's stated time	Analysis, cited research, requirements, diagrams, UX and evaluation plan assembled; outstanding evidence/review clearly identified
Foundation	Schema, migrations, authentication and catalogue can be demonstrated on synthetic data
Core MVP	Reservation creation, own history, cancellation, maintenance coordination and assigned-club access pass integration tests
Diploma extension	Event calendar, station layout, defined reports and audit trail are integrated
Final evaluation	Reproducible correctness and load results, usability observations, limitations and demonstration are reported

No final experiment results are required or claimed at the initial submission. Planned correctness checks include simultaneous conflicting requests, adjacent intervals, retries, cancellation races, maintenance races and cross-club access attempts. Performance targets and workload conditions are specified in ECR-7 requirements; Nurali's ECR-13 plan should adopt or explicitly revise them. Usability assessment should combine task success, time, observed errors and a validated questionnaire, rather than treating a questionnaire score as proof of all quality attributes.

7. Assumptions, dependencies and review

The project depends on access to target users for ECR-3, team agreement on administrative rules in ECR-8, a hosting environment for the eventual prototype and supervisor review recorded in ECR-16. The main technical risks are interval races, inconsistent authorisation and scope growth; mitigations are shared transactional writes, negative access tests and the exclusions above. Real personal data is unnecessary for the October package.

This proposal aligns with the current README and approved ECR-1 scenarios. Reviewer requested: Mansur for user flow/event scope and Nurali for administrative scope and evaluation alignment. Alibi subsequently confirmed his teammates' agreement in the working chat on October 23, 2026. Supervisor approval is not implied.

Related deliverables

- Critical literature review and source matrix

- Technology alternatives and decisions
- Booking and non-functional specification
- Architecture, diagrams and transaction protocol
- User and administrator scenarios

2. User needs and relevance evidence

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/research/user-needs-evidence.md

Version 1.0 · October 23, 2026 · ECR-3

Document owner: Mansur Urynbassar Status: Three anonymised user discussions completed and synthesised

Purpose

This document records evidence used to assess the relevance of CyberSeat and to inform its requirements. It distinguishes findings supported by discussions with representatives of the intended user group from assumptions that still require validation. It does not claim that the proposed system has already been implemented or evaluated.

Method

Three semi-structured asynchronous text discussions were conducted on October 23, 2026, with people who have direct experience visiting gaming clubs. All participants gave permission for their responses to be used anonymously in the diploma project. No names, contact details, account identifiers, or other unnecessary personal data were retained.

The same eleven core questions were used in each discussion. They covered current availability checking and reservation practices, recurring problems, information needed before confirmation, conflict and cancellation handling, the value of an event calendar, mobile access, and factors affecting trust.

The sample is exploratory and is not statistically representative. It provides initial evidence from visitors, while administrative workflows and reporting needs still require validation with a club administrator or manager.

Participant Overview

R1

User role: Regular gaming-club visitor

Relevant experience: Regular CS2 player who visits clubs with friends and uses higher-performance club equipment

Date: October 23, 2026

Method: Asynchronous text discussion

Consent: Anonymous use approved

R2

User role: Occasional gaming-club visitor

Relevant experience: Visits clubs with friends to play CS2, Dota 2, and Valorant

Date: October 23, 2026

Method: Asynchronous text discussion

Consent: Anonymous use approved

R3

User role: Regular gaming-club visitor

Relevant experience: Competitive CS2 player who prioritises powerful PCs and high-refresh-rate monitors

Date: October 23, 2026

Method: Asynchronous text discussion

Consent: Anonymous use approved

Discussion Questions

1. What is your role or experience with gaming clubs?
2. How do you currently check whether a suitable station is available?
3. How do you currently make or receive reservations?
4. What problems occur most often?
5. What information should be visible before confirming a reservation?
6. What should happen if the selected station becomes unavailable?
7. How should reservation cancellation work?
8. Would an event or tournament calendar be useful? Why?
9. What information must be convenient to view on a phone?
10. What would make you trust or distrust such a system?
11. May the answers be used as anonymous research notes for the diploma project?

Discussion Records

R1 Regular Gaming-Club Visitor

Role and experience. R1 regularly plays CS2 and sometimes visits gaming clubs with friends to use better equipment.

Current process. Availability is normally checked through WhatsApp, Instagram, or an in-person request to the administrator. Reservations are made by message or telephone call, specifying the number of PCs and the desired time.

Problems observed. Higher-performance PCs may already be occupied, group members may be unable to obtain adjacent stations, and the administrator may take time to respond.

Information required before confirmation. R1 expects to see PC specifications, monitor refresh rate, hourly price, available intervals, station location, and whether stations are adjacent.

Conflict and cancellation expectations. If a station becomes unavailable, the system should notify the visitor immediately and offer a station with similar specifications or another time. Cancellation should be available in the application, with rules and any applicable refund conditions explained before booking.

Events and mobile access. R1 considers a calendar useful for discovering CS2 tournaments, checking dates, and potentially registering for events. On mobile devices, R1 expects availability, price, specifications, reservation time and status, and cancellation options.

Trust factors. Real-time availability, transparent prices, and reliable confirmation would increase trust. Incorrect availability and unexpected cancellation would reduce trust.

Consent. R1 approved anonymous use of the responses.

R2 Occasional Group Visitor

Role and experience. R2 occasionally visits gaming clubs with friends to play CS2, Dota 2, and Valorant.

Current process. Availability is checked through friends or WhatsApp. One person normally calls the club and books stations for the entire group.

Problems observed. The club may not have enough free PCs for the group, or the available stations may be far apart.

Information required before confirmation. R2 expects to see the number of available PCs, hourly prices, PC specifications, and whether stations are adjacent.

Conflict and cancellation expectations. If a selected station becomes unavailable, the system should offer another station or allow the reservation time to be changed. Visitors should be able to cancel through the application without calling the administrator.

Events and mobile access. R2 considers a local tournament calendar useful for group participation. On mobile devices, R2 expects availability, prices, booking confirmation, and reservation history.

Trust factors. Accurate reservations and ease of use would increase trust. Incorrect information would reduce trust.

Consent. R2 approved anonymous use of the responses.

R3 Competitive Gaming-Club Visitor

Role and experience. R3 regularly plays competitive games, particularly CS2, and prefers clubs with powerful PCs and high-refresh-rate monitors.

Current process. Availability is checked by contacting the administrator or reviewing the club's social-media pages. A specific station is reserved by message for a defined number of hours.

Problems observed. A preferred PC may already be occupied, and current availability cannot be checked without contacting the club.

Information required before confirmation. R3 expects to see GPU, CPU, monitor refresh rate, peripherals, hourly price, and exact availability.

Conflict and cancellation expectations. If a station becomes unavailable, the system should notify the visitor and offer another PC with similar performance. Cancellation should use a simple action with clearly stated rules and deadlines.

Events and mobile access. R3 considers a tournament calendar useful for following upcoming CS2 events and planning practice sessions. On mobile devices, R3 expects specifications, available intervals, prices, and upcoming reservations.

Trust factors. Accurate specifications, reliable reservations, transparent pricing, and prompt notifications would increase trust.

Consent. R3 approved anonymous use of the responses and explicitly requested that no name be disclosed.

Evidence Synthesis

F1

Finding: Visitors currently depend on messaging, calls, social media, or in-person requests to check availability and reserve stations.

Evidence: R1, R2, R3

Project implication: Provide self-service availability search and reservation.

Status: Validated across all three discussions

F2

Finding: Visitors cannot reliably inspect current station availability without contacting club staff.

Evidence: R1, R3; indirectly supported by R2

Project implication: Display current availability and revalidate it on the server before confirmation.

Status: Validated

F3

Finding: Groups need enough available stations and prefer adjacent seating.

Evidence: R1, R2

Project implication: Represent station location or adjacency and support group-oriented selection in later design work.

Status: Validated by two discussions

F4

Finding: Equipment specifications influence station choice.

Evidence: R1, R2, R3

Project implication: Display CPU, GPU, monitor refresh rate, peripherals, and other relevant station attributes.

Status: Validated across all three discussions

F5

Finding: Price and exact available intervals are required before confirmation.

Evidence: R1, R2, R3

Project implication: Show transparent hourly prices, time zone, and available intervals before confirmation.

Status: Validated across all three discussions

F6

Finding: When availability changes, visitors expect prompt notification and an alternative station or time.

Evidence: R1, R2, R3

Project implication: Provide a clear conflict state and offer comparable stations or another interval.

Status: Validated across all three discussions

F7

Finding: Visitors expect self-service cancellation with clear rules.

Evidence: R1, R2, R3

Project implication: Provide cancellation in the personal account and display applicable rules and deadlines.

Status: Validated across all three discussions

F8

Finding: A local tournament and event calendar is considered useful.

Evidence: R1, R2, R3

Project implication: Retain the event calendar in the planned scope. Treat event registration as a separate scope decision.

Status: Validated across all three discussions

F9

Finding: Key reservation functions must be convenient on mobile devices.

Evidence: R1, R2, R3

Project implication: Use responsive layouts for availability, specifications, prices, confirmation, status, history, and cancellation.

Status: Validated across all three discussions

F10

Finding: Trust depends on accurate availability, specifications and prices, reliable confirmation, and timely notifications.

Evidence: R1, R2, R3

Project implication: Prioritise data consistency, explicit confirmation, transparent information, and visible error handling.

Status: Validated across all three discussions

Requirements Informed by Evidence

The system shall allow visitors to view station availability without contacting club staff.
Evidence reference: F1, F2

Resulting decision: Retain as a mandatory functional requirement.

The server shall revalidate availability before confirming a reservation.
Evidence reference: F2, F6, F10

Resulting decision: Retain as a mandatory integrity requirement.

Station details shall include relevant PC, monitor, and peripheral specifications.
Evidence reference: F4

Resulting decision: Expand the station-information requirement.

The interface shall display hourly price and exact available intervals before confirmation.
Evidence reference: F5, F10

Resulting decision: Add as mandatory information requirements.

The station-selection design shall communicate location or adjacency.
Evidence reference: F3

Resulting decision: Include in UX prototype; evaluate a floor layout against a structured list.

An availability conflict shall offer another comparable station or time.
Evidence reference: F6

Resulting decision: Retain as an acceptance criterion for reservation confirmation.

A visitor shall be able to cancel an eligible personal reservation without contacting the administrator.
Evidence reference: F7

Resulting decision: Retain; define deadlines and club-specific rules before implementation.

The system shall provide a calendar of club events and tournaments.
Evidence reference: F8

Resulting decision: Retain in scope. Event registration remains outside the confirmed mandatory scope.

Principal visitor workflows shall support mobile screens.
Evidence reference: F9

Resulting decision: Retain as a non-functional usability requirement.

The interface shall distinguish confirmed, failed, and uncertain reservation outcomes.

Evidence reference: F6, F10

Resulting decision: Retain as a reliability and UX requirement.

Verified Findings and Remaining Assumptions

Verified by the Discussions

- Existing availability checking and reservation processes rely heavily on direct communication with club staff.
- Visitors need accurate station availability, specifications, price, and time information.
- Group visitors value adjacent stations.
- Visitors expect self-service cancellation and explicit conflict handling.
- Participants considered an event calendar useful.
- Mobile access and data accuracy are important trust factors.

Assumptions Requiring Further Validation

Club administrators need the proposed schedule, station, event, and reporting workflows.

Why it remains unverified: No administrator or manager participated in this sample.

Planned validation: Interview at least one club administrator or manager and evaluate the administrative prototype.

Utilisation reports should use the currently proposed indicators and calculation rules.

Why it remains unverified: Visitors were not asked to define administrative reporting metrics.

Planned validation: Administrator discussion and comparison with a controlled data set.

Event registration should be integrated into CyberSeat.

Why it remains unverified: Participants expressed interest, but the current scope confirms only event discovery.

Planned validation: Treat registration as optional and evaluate it during scope review.

Refund processing should be part of cancellation.

Why it remains unverified: R1 mentioned refund conditions, but online payment is outside the mandatory scope.

Planned validation: Define cancellation information without implementing payments; revisit only if payment enters scope.

A visual floor layout is more usable than a structured station list.

Why it remains unverified: Two participants need adjacency information, but no interface alternative was evaluated.

Planned validation: Compare both representations during prototype evaluation.

Minimum reservation duration and cancellation deadlines are acceptable.

Why it remains unverified: Participants requested clear rules but did not agree on concrete values.

Planned validation: Validate club-specific rules with an administrator or manager.

Conflicting and Divergent Views

No direct contradiction was identified in this small sample. Participants differed mainly in visit frequency and the amount of technical detail they expected. R3 emphasised specific hardware characteristics, while R2 focused more on group availability and ease of use. The proposed interface should therefore provide essential information at a glance and allow users to inspect detailed specifications when needed.

The absence of disagreement in three discussions does not imply agreement across the wider user population.

Effect on the Project

The discussions support the relevance of CyberSeat's central problem: visitors currently rely on manual communication and cannot independently verify accurate availability. The evidence strengthens the requirements for server-validated availability, transparent station information, conflict alternatives, self-service cancellation, responsive mobile access, and an event calendar.

The evidence also changes the level of detail required by the interface. Station information should include hardware and monitor characteristics, price, exact time availability, and location or adjacency. Group seating should be represented in the UX prototype rather than treated only as a future idea.

No evidence from club administrators was collected in this round. Administrative workflows, reporting metrics, maintenance rules, and concrete cancellation deadlines therefore remain assumptions and must not be presented as user-validated findings.

Limitations

- The sample contains three participants and is exploratory rather than statistically representative.

- All participants represent visitors; no club administrator, operator, manager, or client participated.

- All three participants play competitive games, especially CS2, which may overemphasise performance specifications and tournament-related needs.

- The discussions were conducted through text rather than direct observation of the reservation process.

- Self-reported practices may differ from observed behavior.

- Findings should be revisited during UX prototype evaluation and later system testing.

Conclusion

Three anonymised discussions provide initial evidence that CyberSeat addresses a relevant user problem. All participants currently rely on staff communication, social media, or calls to determine availability and make reservations. They consistently requested accurate availability, detailed station information, transparent pricing, clear confirmation, alternatives during conflicts, self-service cancellation, mobile access, and an event calendar.

These findings support the planned visitor-facing functionality and refine its acceptance criteria. They do not validate administrative and reporting requirements, which require a separate discussion with a club administrator or manager. The project shall preserve this distinction between evidence and assumptions in subsequent requirements, diagrams, and UX materials.

Evidence provenance and verification boundary

The discussion summaries and consent statements above were supplied by the document owner. The documentation consistency review did not contact participants or independently verify the original exchanges. It verifies coverage and traceability, not interview authenticity. The owner should retain the original responses and consent evidence securely and provide anonymised extracts to the supervisor if requested; do not publish names, contact details or private conversations in this repository.

Confirmed findings in this document mean findings reported within this small visitor sample, not statistically established claims about all clubs. Administrator research and prototype evaluation remain outstanding.

Team Review

Recorded in source: Mansur Urynbassar recorded the anonymised discussions and prepared the initial synthesis.

Recorded in source: Alibi approved the correction revision in PR #7, including this evidence document; see the October 24 review and agreement record (new package record; see chapter 11).

Recorded in source: Team agreement with the documented administrative/reporting evidence limitations was reported by Mansur on behalf of all three participants on October 24; see the October 24 review and agreement record (new package record; see chapter 11). This is reported agreement, not a separately observed review by Nurali.

Recorded in source: Validated findings are explicitly separated from assumptions.

Publication and review status

The initial version was published through PR #4. The correction revision is covered by Alibi-stack's approval of PR #7 (commit `d7bae8ea9931bbdb8e4caec01c243f33c8cea493`). Additional team agreement was reported by Mansur on October 24; see the October 24 review and agreement record (new package record; see chapter 11). This does not independently verify the original interviews.

3. Critical literature review

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/research/literature-review.md

Version 1.1 | October 23, 2026 | Owner: Aibekuly Alibi | ECR-4

Status: documentation completed; team agreement confirmed by Alibi on October 23, 2026. This focused engineering review supports the booking module and its quality requirements. It is not a systematic review, a market study or evidence of user interviews.

1. Questions and selection method

The review asks: how can overlapping reservations be prevented under concurrency; how should correctness be tested; and how can access control and usability be evaluated without overstating the evidence?

Searches on October 23, 2026 used publisher/conference pages, author-hosted papers and official technical specifications. Query families included database isolation verification, concurrent application integrity, snapshot isolation, interval constraints, object-level authorisation and SUS analysis. Papers were included when they provided a method, empirical finding or explicit limitation relevant to a requirement. Vendor marketing, unsourced summaries and publications whose identity could not be verified were excluded from the source matrix. No exhaustive search count or screening flow is claimed.

The matrix contains **13 substantively used sources: 9 scientific publications and 4 technical/standards sources**. Six of nine scientific publications are from 2022-2025, within the recent five-year window. Three older papers are retained for foundational isolation and application-integrity arguments. Current specifications are used for concrete implementation behaviour; papers about older releases are not treated as descriptions of current software versions. Where a source was available only as a publisher abstract, the matrix says so and avoids unsupported detailed performance claims.

2. Critical source matrix

S01 Berenson et al. (1995), *A Critique of ANSI SQL Isolation Levels*, SIGMOD, pp. 1-10. Author-deposited paper, DOI 10.1145/223784.223785

Approach and reported result: Analyses anomaly-based isolation definitions and defines snapshot isolation; shows why isolation labels alone are insufficient to characterise behaviour

Limitation / transfer boundary: Foundational semantics, not a booking-system evaluation; 2007 arXiv deposit is not the original publication year

CyberSeat implication: Specify forbidden committed states explicitly rather than saying the database is merely ACID; BR-04, NFR-01

S02 Ports and Grittner (2012), *Serializable Snapshot Isolation in PostgreSQL*, PVLDB 5(12), pp. 1850-1861. Paper

Approach and reported result: Describes PostgreSQL SSI implementation and benchmark evaluation; conflicting executions may abort to preserve serialisability

Limitation / transfer boundary: Concerns the introduction of SSI in PostgreSQL 9.1; benchmark outcomes do not predict CyberSeat latency

CyberSeat implication: Full-transaction retry is necessary if SSI is selected; explicit guard locking is another candidate, not equivalent to assuming no aborts; NFR-05

S03 Bailis et al. (2015), *Feral Concurrency Control: An Empirical Investigation of Modern Application Integrity*, SIGMOD, pp. 1327-1342. Publisher abstract

Approach and reported result: Empirically studies ORM-backed integrity mechanisms and identifies concurrency-related integrity failures in some application checks

Limitation / transfer boundary: Ruby on Rails sample; the publisher abstract supports the general finding, not a claim about this Python project

CyberSeat implication: A check followed by an independent insert is insufficient protection; add database enforcement and simultaneous-operation tests; BR-04/09

S04 Huang et al. (2023), *Efficient Black-box Checking of Snapshot Isolation in Databases*, PVLDB 16(6), pp. 1264-1276. Paper, DOI 10.14778/3583140.3583145

Approach and reported result: PolySI characterises snapshot isolation using generalised polygraphs, checks histories with SMT and reports counterexamples; evaluation reproduces known anomalies and finds new violations

Limitation / transfer boundary: Checks an isolation property under its history model; does not establish business invariants or model the complete reservation API

CyberSeat implication: Preserve operation histories and make counterexamples understandable; checking HTTP status counts alone cannot establish consistency; NFR-01/03

S05 Liu et al. (2024), *Leopard: A General Test Suite for Isolation Level Verification*, CIDR. Paper

Approach and reported result: Collects client-side interval traces and mirrors concurrency-control mechanisms for black-box checking; reports bugs in commercial DBMSs

Limitation / transfer boundary: Verification depends on the model and collected observations; building a general DBMS checker exceeds diploma scope

CyberSeat implication: Record start/end times, results and final DB state for controlled races; use a small domain oracle rather than claiming formal DBMS verification; NFR-01

S06 Li et al. (2024), *DBStorm: Generating Various Effective Workloads for Testing Isolation Levels*, ISSTA. Conference abstract, DOI 10.1145/3650212.3680318

Approach and reported result: Generates diverse conflict patterns and anomaly-sensitive workloads; reports finding DBMS implementation bugs

Limitation / transfer boundary: Conference abstract was used; no full-method replication or numerical speedup is claimed. DB engine anomalies differ from application rule violations

CyberSeat implication: Vary conflict shapes: identical, partial, containing, adjacent and cross-operation intervals; repeating one happy path is inadequate; BR-04/05/09

S07 Liu, Gu, Wei and Basin (2024), *Plume: Efficient and Complete Black-Box Checking of Weak Isolation Levels*, PACMPL 8(OOPSLA2), Article 302. Author paper, DOI 10.1145/3689742

Approach and reported result: Uses fine-grained anomaly patterns and clock techniques to check representative weak-isolation guarantees, with informative counterexamples

Limitation / transfer boundary: Read committed, read atomicity and causal-consistency checks are not proofs of serialisability or of non-overlapping bookings

CyberSeat implication: Name the property being checked. Domain invariant tests must remain even if the database satisfies its configured isolation contract; NFR-01

S08 Møldrup and Pavlogiannis (2025), *AWDIT: An Optimal Weak Database Isolation Tester*, PACMPL 9(PLDI), Article 209. Author paper, DOI 10.1145/3742465

Approach and reported result: Develops algorithms and complexity bounds for history checking of RC, RA and CC, with comparative evaluation

Limitation / transfer boundary: Optimality is conditional on the paper's problem model and complexity assumptions; weak-isolation validity is weaker than application correctness

CyberSeat implication: Keep repeatable, finite histories and explicit oracles; do not infer no-overlap merely from selecting a named isolation level; NFR-01

S09 Blattgerste, Behrends and Pfeiffer (2022), *A Web-Based Analysis Toolkit for the System Usability Scale*, PETRA, pp. 237-246. Author paper, DOI 10.1145/3529190.3529216

Approach and reported result: Presents an open-source toolkit for calculating, plotting and contextualising SUS scores

Limitation / transfer boundary: A toolkit is not an esports-club study. SUS reflects perceived usability; its 0-100 scale is not a percentage of successful tasks

CyberSeat implication: Combine SUS with observed task completion, time and errors; disclose participant selection and small sample limitations; NFR-07

S10 PostgreSQL Global Development Group, *Range Types*, PostgreSQL 18 documentation. Specification

Approach and reported result: Documents range bounds, overlap operators and exclusion constraints, with scheduling examples

Limitation / transfer boundary: Enforces configured constraints, not all cross-table rules; does not establish that the application uses them correctly

CyberSeat implication: Use half-open timestamp ranges and a same-station overlap constraint; separately coordinate maintenance and working-hours changes; BR-04/05/09/10

S11 PostgreSQL Global Development Group, *Explicit Locking*, PostgreSQL 18 documentation. Documentation

Approach and reported result: Describes row locks, transaction-scoped lock lifetimes and deadlock considerations

Limitation / transfer boundary: A lock protocol works only if every relevant writer follows it; more contention can increase latency

CyberSeat implication: Define club/station guard rows, universal lock order and bounded retries; a process-local lock is insufficient across API workers; BR-09/10, NFR-05

S12 OWASP (2023), *API1: Broken Object Level Authorization*. Guidance

Approach and reported result: Explains resource-ID manipulation and the need for object-level authorisation on requested objects

Limitation / transfer boundary: Risk guidance, not an exhaustive security certification or measured probability for this system

CyberSeat implication: Check ownership and club assignment in every server operation; test foreign IDs and cross-club requests; BR-07/08, NFR-02

S13 W3C (2024), *Web Content Accessibility Guidelines 2.2*, Recommendation of December 12, 2024. Standard

Approach and reported result: Provides testable accessibility criteria, including keyboard interaction, text alternatives and visible focus

Limitation / transfer boundary: Meeting selected criteria is not proof of full AA conformance or general usability; complete assessment remains future work

CyberSeat implication: Use text in addition to colour, labelled errors and keyboard checks for core screens; BR-13, NFR-06

All links were consulted on October 23, 2026. The matrix is a critical engineering synthesis; its CyberSeat implications are proposed design conclusions, not claims made by the original authors about CyberSeat.

3. Synthesis: integrity is a business property

S01 explains the semantic gap between isolation labels and the actual anomalies an application must prevent. S03 provides an empirical reason to distrust uncoordinated application checks. For a station reservation, two individually valid reads of an empty schedule can precede conflicting inserts. The resulting design conclusion is to state an invariant over the committed records, enforce it in the database and test it through independent connections.

S02 and S11 offer different coordination mechanisms: serialisable transactions with retry, or explicit transaction-scoped locks. CyberSeat provisionally chooses a shared lock protocol plus a database overlap constraint. This reduces the amount

of implicit predicate-level reasoning for the first implementation, at the cost of serialising some operations. It is an engineering choice to benchmark later, not a demonstrated superiority claim.

S10 directly supports interval representation. However, an exclusion constraint on reservations cannot, by itself, prevent a reservation from overlapping a maintenance record stored in another table. The architecture therefore requires reservation and maintenance writers to lock the same station guard before validation and write. Working-hours changes and reservations similarly coordinate through a club guard. ECR-8 must preserve this protocol when administrative endpoints are designed.

4. Synthesis: tests need histories and diverse conflicts

Recent work S04-S08 demonstrates several ways of checking transactional histories or generating revealing workloads. Their guarantees differ, so citing a successful weak-isolation checker would not prove that CyberSeat never double-books a station. Nor is it proportionate to implement a new general-purpose isolation checker for this diploma.

The adopted scope is domain-specific: release requests through a start barrier, preserve their input/response/operation identity, then query the committed state using an overlap oracle. Add a forced network interruption after commit, restart/reconciliation checks and booking-versus-maintenance races. A small workload that produces a comprehensible failing history is more useful here than an unsupported claim of production-scale reliability.

5. Synthesis: user quality and access are separate questions

S09 supports systematic calculation and interpretation of a perceived-usability measure. S13 gives concrete accessibility checks. Neither establishes demand for an esports booking service; that remains ECR-3. Prototype task observations should therefore be reported alongside questionnaire results, with participant recruitment and limitations disclosed.

S12 requires a server authorisation decision for each resource operation. UI hiding is insufficient: tests should change reservation IDs and club IDs directly. Ownership and assigned-club boundaries must remain effective even when a visitor understands the API. Reports must distinguish scheduled utilisation from actual attendance, which the proposed system cannot observe without additional evidence or hardware.

6. Research-to-requirement decisions and remaining gaps

Integrity cannot be inferred from labels or ORM validation

Design decision: Database overlap constraint and common transactional write path

Requirement / planned verification: BR-04/09/10, NFR-01; real-DB race tests

Cancel/retry/network failure alters the observable history

Design decision: Durable operation identity and result reconciliation

Requirement / planned verification: BR-06/11, NFR-03; response-loss and retry tests

Scheduling uses interval rather than scalar uniqueness

Design decision: `[start,end)` and club-zone conversion

Requirement / planned verification: BR-05/14; adjacency, containment and overnight fixtures

Testing only one conflict pattern misses behaviours

Design decision: Controlled family of cross-operation histories

Requirement / planned verification: NFR-01; final-state oracle after each run

Resource access depends on actor and object

Design decision: Server owner/assignment checks

Requirement / planned verification: BR-07/08, NFR-02; foreign-resource negative tests

Perceived usability is only one quality dimension

Design decision: Task metrics plus questionnaire and accessibility checks

Requirement / planned verification: NFR-06/07; formative prototype study

The ECR-3 evidence record now contains three owner-supplied anonymised visitor discussions, and ECR-5 supplies the product comparison. These are exploratory visitor findings, not independently authenticated exchanges or representative market evidence. Gaps remain: no administrator/club-workflow validation, executed CyberSeat benchmark, final authentication mechanism or complete accessibility audit, and limited domain-specific empirical research in this focused review. Mansur is requested to review the UX interpretation; Nurali is requested to review authorisation and reporting implications. Alibi confirmed his teammates' agreement with the prepared materials in the working chat on October 23, 2026.

4. Existing systems comparison

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/research/existing-systems-comparison.md

Version: 1.1 **Date:** October 23, 2026 **Related issue:** ECR-5 **Scope:** systems that support gaming-club or esports-venue operations and station booking

Purpose and method

This comparison evaluates three directly relevant systems against the same nine criteria. Three systems are sufficient because all three are specialised products for gaming or esports venues, expose visitor booking functions, and provide venue-management functions. General-purpose appointment products were excluded because they do not model gaming stations, venue maps, device state, or gaming-centre operations.

The analysis uses publicly available official product pages and official support documentation accessed on October 23, 2026. A capability is marked **Confirmed** only when an official source explicitly describes it. **Not confirmed** means that the reviewed public sources do not establish the capability; it does not mean that the product lacks it.

Systems

1. **SENET** - cloud software for managing gaming centres, with a player mobile application and web portal.
2. **ggLeap** - esports-venue management software with a player web portal and integrated booking.
3. **LANGAME** - gaming-club software and a consumer application/aggregator for finding clubs and booking stations.

Comparison matrix

1. Visitor self-service booking

SENET: Confirmed. Players can book PCs and consoles through the SENET ID portal or mobile app.

ggLeap: Confirmed. Players can reserve PCs online through the player web portal; the portal can be linked or embedded.

LANGAME: Confirmed. Visitors can book gaming-club stations through the website or mobile application.

2. Specific station selection

SENET: Confirmed. The booking module and venue map expose PCs/consoles and booking state.

ggLeap: Confirmed. The system supports selecting one or several devices during booking.

LANGAME: Confirmed. The application opens a club map and allows selection of a specific PC.

3. Group booking / adjacent stations

SENET: Confirmed. Official SENET documentation states that group booking of multiple PCs or consoles is supported. Adjacency rules are not confirmed.

ggLeap: Confirmed. Customers can book multiple PCs; explicit automatic adjacency rules are not confirmed.

LANGAME: Confirmed. Users can reserve several stations for themselves and friends. Automatic adjacency rules are not confirmed.

4. Mobile visitor access

SENET: Confirmed. SENET ID is available as a mobile app and web portal.

ggLeap: Confirmed for web. A player web portal is documented. A dedicated mobile application for booking was not confirmed in the reviewed sources.

LANGAME: Confirmed. The consumer mobile app supports club search and booking.

5. Admin booking calendar and resource state

SENET: Confirmed. Admin map, booking list/timeline, device status, pricing, cancellation, reminders, and reports are documented.

ggLeap: Confirmed. Web Admin provides agenda/grid views, calendar scheduling, device selection, locking rules, and booking management.

LANGAME: Partially confirmed. Management software and online booking are documented, but the reviewed public pages do not describe the administrator calendar in enough detail.

6. Cancellation and modification by visitors

SENET: Confirmed for cancellation/status. Official material describes booking cancellation and Active/Cancelled/Expired states; detailed visitor-side modification rules depend on venue settings.

ggLeap: Partially confirmed. Admin booking editing and booking settings are documented; a visitor self-service cancellation flow was not confirmed in the reviewed sources.

LANGAME: Not confirmed. The reviewed visitor pages do not establish a self-service rescheduling/cancellation policy. Obtain a direct support article or vendor confirmation before claiming a limitation. L1

7. Events and tournaments

SENET: Not confirmed as a visitor calendar in the reviewed SENET booking sources.

ggLeap: Confirmed. Event Builder, competitions, challenges, quests, and event-related bookings are documented.

LANGAME: Not confirmed as an event calendar or management workflow. Esports articles and videos do not establish either capability. L2

8. Reporting and analytics

SENET: Confirmed. Real-time reports, analytics, booking reports, activity logs, and device/session information are documented.

ggLeap: Not confirmed for defined report outputs. Web administration and analytics integrations are described, but the cited sources do not establish native reservation/utilisation reports and their calculation rules. G1

LANGAME: Partially confirmed. Business control and management are advertised, but the reviewed public visitor pages do not specify report structure.

9. Multi-venue discovery and comparison

SENET: Confirmed for connected venues. A SENET ID account can connect to venues and switch between them; the sources do not establish a broad public aggregator with comparable station specifications.

ggLeap: Confirmed for an operator's branches. Visitors can check availability at multiple locations belonging to a venue organisation. A cross-provider aggregator is not confirmed.

LANGAME: Confirmed. LANGAME operates a catalogue/aggregator with club pages, prices, equipment information, reviews, maps, and online booking.

Source traceability

Each matrix assessment is limited to the specific capability described, not a claim of complete feature parity. Sources were checked on October 23, 2026. Product marketing establishes an advertised feature, not independently tested correctness.

1-3 Booking, station selection, groups

SENET evidence: S1 booking changelog, sections User booking and Admin booking

ggLeap evidence: G2 booking guide and G3 booking overview, Website and Web Admin

LANGAME evidence: L1 application, club selection and booking for friends; automatic adjacency is not established

4 Mobile access

SENET evidence: S1, User booking

ggLeap evidence: G3, web portal only

LANGAME evidence: L1, mobile application

5 Admin booking and resources

SENET evidence: S1, Admin booking

ggLeap evidence: G2, booking settings and Web Admin

LANGAME evidence: L3 business software; calendar detail remains unconfirmed

6 Cancellation and modification

SENET evidence: S1, October 14 cancellation control

ggLeap evidence: G3, staff booking changes only

LANGAME evidence: L1; visitor policy not established

7 Events

SENET evidence: Booking sources do not establish a visitor calendar

ggLeap evidence: G1 product, Event Builder; G3, event bookings

LANGAME evidence: L2 portal, editorial content only; calendar not established

8 Reports

SENET evidence: S2 product, real-time reports; S1, booking reports

ggLeap evidence: G1, integrations only; specific reports unconfirmed

LANGAME evidence: L3, management claims; report structure unconfirmed

9 Multiple venues

SENET evidence: S1, SENET ID venue access

ggLeap evidence: G3, multiple branches

LANGAME evidence: L2, catalogue with prices, addresses and PC specifications

Findings

Common capabilities

All three systems confirm the market need for advance gaming-station booking. Each connects booking to a venue-specific station or device model rather than treating a booking as a generic appointment. Each also connects the visitor experience with operational software used by the club.

Relevant strengths

- **SENET** provides the strongest publicly documented integration between visitor booking, device state, venue operations, payments, and reporting.

- **ggLeap** provides detailed booking controls for employees and customers, including multiple-device bookings, resource-lock behaviour, web embedding, payments, and competition features.

- **LANGAME** provides the clearest consumer discovery experience: users can find multiple clubs, compare club information, open a venue map, and reserve particular stations or several stations for a group.

Limitations identified from public evidence

- Public product descriptions do not consistently document conflict-handling behaviour when availability changes between selection and confirmation.

- None of the reviewed sources clearly confirms automatic selection of adjacent stations using a formal adjacency rule.

- Visitor-side cancellation and rescheduling evidence is incomplete. SENET documents a venue-level cancellation setting; the reviewed LANGAME sources do not establish the claimed rescheduling restriction.

- Cross-product comparisons of exact prices, regional payment support, data portability, and API limits would require quotations, trial accounts, or vendor confirmation and are therefore outside this evidence set.

Why CyberSeat is still justified

CyberSeat is not intended to reproduce every venue-management function of mature commercial platforms. Its academic and practical focus is a transparent reservation workflow for local esports centres, with the following design priorities derived from the user discussions and comparison:

1. show station specifications, prices, location, and real-time availability before confirmation;

2. support individual station selection with visible adjacency information; multi-station transactions remain a separate scope decision;

3. revalidate availability immediately before confirmation and explain conflicts;

4. allow visitors to view and cancel their own reservations with explicit rules;

5. expose loading, empty, validation, server-error, and connection-loss states;

6. combine a visitor event calendar with club-level administrative scheduling and reports;

7. enforce role-based access and consistent booking rules in one documented system;

8. evaluate the proposed workflow against measurable usability and correctness criteria.

The project therefore has a distinct evaluation target: whether a focused, clearly documented prototype and later implementation can reduce the communication delays and uncertainty reported by the interviewed visitors while preserving reliable administrative control.

Sources

1. SENET, *Gaming Centers - Venue Management Software*. https://senet.cloud/gaming-centers

2. ENESTECH Software, *SENET Changelog* (booking timeline, booking cancellation, statuses, SENET ID, group booking, admin booking and reports).
https://enestech.zendesk.com/hc/en-gb/articles/26052144991516-Changelog-Last-updated-Oct-14-2026

3. SENET, *Mobile App* (supplementary link, unavailable during the correction check; mobile claims above rely on source 2).
https://senet.cloud/mobile-app

4. ggCircuit, *ggLeap - Esports Venue Management Software*. https://www.ggcircuit.com/ggleap

5. ggCircuit Knowledge Base, *ggLeap Booking System*.
https://helpdesk.ggcircuit.com/en/article/ggleap-booking-system-18m36ss/

6. ggCircuit, *Introducing the New Esports Booking System*.
https://blog.ggcircuit.com/2023/05/02/ggcircuit-introduces-its-new-esports-booking-system/

7. LANGAME, *Computer Club Software*. https://langame.ru/en/software

8. LANGAME, *Gaming Clubs and Online Booking*. https://langame.ru/

9. LANGAME, *Booking Application*. https://langame.ru/app

Review checklist

Recorded in source: Three directly relevant systems compared.

Recorded in source: Nine common criteria applied consistently.

Recorded in source: Claims tied to official sources.

Recorded in source: Unknown capabilities described as not confirmed rather than absent.

Recorded in source: Need for CyberSeat explained using comparison and user evidence.

Recorded in source: Comparison reviewed by another team member. See the review record below.

Publication and review status

The initial version was published through PR #4. The applicable revision has a recorded review: Alibi-stack's approval of PR #7 (commit `d7bae8ea9931bbdb8e4caec01c243f33c8cea493`). See the October 24 review and agreement record (new package record; see chapter 11) for the distinction between GitHub approval, reported team agreement and outstanding execution checks.

5. Technology choices

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/research/technology-comparison.md

Version 1.1 | October 23, 2026 | Owner: Aibekuly Alibi | ECR-6

Status: proposed decisions, team agreement confirmed by Alibi on October 23, 2026. No technology benchmark or implementation is claimed; team agreement is recorded from Alibi's report. The baseline matches the current README.

1. Evaluation basis

Priorities are reservation integrity (BR-04/09), explicit authorisation (NFR-02), reproducible testing (NFR-01/08), a browser-based interface and manageable integration for a three-person diploma team. Ratings below are qualitative engineering judgements, not measurements. Team expertise has not been independently measured and is not used to invent numerical scores. All candidates can support a correct design; none makes the reservation transaction correct automatically.

2. Server platform: FastAPI, Django and NestJS

API contract and validation

FastAPI / Python: Type-based request/response models and OpenAPI integration

Django / Python: Structured web framework; an API layer can be added, with serializer and contract choices to make

NestJS / TypeScript: Controllers, DTO validation, modules and OpenAPI integration available

Administrative foundation

FastAPI / Python: Custom admin screens, authorisation and session integration required

Django / Python: Built-in admin, authentication and ORM reduce common administrative setup

NestJS / TypeScript: Guards and modules support access policies; persistence and admin UI still chosen separately

Booking consistency

FastAPI / Python: SQLAlchemy/driver transaction and PostgreSQL constraints required

Django / Python: ORM transactions available; domain rules and appropriate constraints still required

NestJS / TypeScript: ORM/driver transactions required; JavaScript execution does not solve cross-worker races

Team integration

FastAPI / Python: Clear typed HTTP contract with React; two languages across client/server

Django / Python: Strong fit if conventional data management dominates

NestJS / TypeScript: Shared TypeScript language across client/server; language familiarity remains to be confirmed

Main tradeoff

FastAPI / Python: Lean API construction, but more security/admin composition work

Django / Python: More integrated conventions; custom booking workflow still needs application-specific design

NestJS / TypeScript: More module/DTO setup and a different backend toolchain from the current baseline

Proposed decision D1: retain FastAPI because a separately developed React client and explicit reservation API are central to the current design. Use request validation and generated API documentation to reduce integration ambiguity. Do not infer a latency advantage from vendor feature descriptions. Django is a credible fallback if the team prefers its integrated administrative foundation. NestJS becomes preferable if confirmed backend TypeScript expertise outweighs switching cost. Revisit D1 before implementation if Nurali cannot support the proposed security/admin composition.

Primary sources: FastAPI features, Django transactions, Django admin, NestJS documentation. Consulted October 23, 2026; versioned pages are reference editions, not claims of the newest available release.

3. Storage: PostgreSQL, MySQL and MongoDB

Time-interval model

PostgreSQL: Native timestamp ranges and overlap operators

MySQL / InnoDB: Start/end columns with explicit overlap predicates

MongoDB: Start/end fields and application-defined predicates

Declarative no-overlap rule

PostgreSQL: GiST exclusion constraint can combine station equality and range overlap

MySQL / InnoDB: Scalar unique keys do not express arbitrary interval overlap; use an explicit transactional locking design

MongoDB: Unique indexes likewise do not express arbitrary interval overlap; schema and transaction protocol must prevent conflicts

Transactions and coordination

PostgreSQL: Row locks and serialisable isolation; range constraint complements the application protocol

MySQL / InnoDB: InnoDB transactions and locking support a correct solution when consistently applied

MongoDB: Single-document atomicity and multi-document transactions supported; deployment and data modelling choices matter

Relationships/reporting

PostgreSQL: Foreign keys and SQL fit clubs, stations, owners, assignments and utilisation queries

MySQL / InnoDB: Relational model also fits the domain well

MongoDB: Flexible documents, but related booking/schedule/reporting records require explicit consistency design

Main tradeoff

PostgreSQL: PostgreSQL-specific range features and operational setup

MySQL / InnoDB: More custom interval-integrity logic to demonstrate

MongoDB: Less direct fit for this relational scheduling invariant; not excluded because of a false lack-of-transactions claim

Proposed decision D2: PostgreSQL. Its interval constraint makes the central invariant visible and independently enforceable. `tstzrange` values use `)` bounds; `btree_gist` may be needed for station equality in the exclusion index. A partial constraint applies only to `CONFIRMED` reservations. This covers booking-versus-booking overlap, not maintenance in another table or working-hours containment. Those rules require the shared guards described in the [architecture. MySQL and MongoDB remain viable with an alternative proven protocol; no unsupported performance ranking is asserted.

Primary sources: PostgreSQL 18 range types, PostgreSQL 18 locking, MySQL 8.4 InnoDB locking, MongoDB transactions. Consulted October 23, 2026.

4. Component interaction: REST, GraphQL and gRPC

Browser integration

REST over HTTP with JSON: Direct fetch/client calls with explicit endpoints

GraphQL: Browser clients can request shaped data through a typed schema

gRPC: Browser use needs a compatible web integration; native gRPC targets service RPC

Contract

REST over HTTP with JSON: OpenAPI documents operations, bodies and status codes

GraphQL: Schema plus query/mutation validation

gRPC: Protocol Buffers and generated service/client contracts

Core workflow fit

REST over HTTP with JSON: Catalogue reads and create/cancel commands are straightforward

GraphQL: Flexible catalogue composition, but per-field authorisation and query limits need attention

gRPC: Strong typed service interaction; little current need for internal service RPC in a monolith

Failure/retry semantics

REST over HTTP with JSON: Explicit HTTP statuses and application operation key

GraphQL: Domain errors and retry identity still application responsibilities

gRPC: Deadlines and status model help RPC; retries still need business-level deduplication

Complexity for this scope

REST over HTTP with JSON: One browser API and generated documentation

GraphQL: Additional resolver/query design without an established need

gRPC: Additional browser transport/tooling without planned independent backend services

Proposed decision D3: REST with a versioned `/api/v1` contract and OpenAPI. Use 409 for a valid request conflicting with current state, 422 for validation and 401 for missing/expired authentication. Foreign own-reservation IDs return 404. POST creation is made replay-safe by the application protocol, not by the HTTP method itself. Availability refresh is sufficient for the core scope; WebSockets or server push can be reconsidered if user evidence establishes a real need.

Sources: HTTP semantics, RFC 9110, GraphQL specification, October 2021 reference edition, gRPC introduction. Consulted October 23, 2026. These sources describe mechanisms; the suitability judgement is this project's decision.

5. Client and deployment alignment

D4, existing baseline: React, TypeScript and Vite for visitor/admin interfaces. React's component model and TypeScript's static checks fit reuse across catalogue, forms and admin views, but do not validate incoming data or authorise requests. Vue with TypeScript is a credible component-based alternative; server-rendered templates reduce a separate client build but fit less directly with the existing interface ownership and API plan. Keeping the documented baseline avoids a speculative redesign. Sources: React documentation, TypeScript handbook, Vue guide.

React/TypeScript appears in the current README and ECR-1 interface baseline. **Alibi confirmed on October 23, 2026 that his teammates agree with the prepared materials**, including this comparison. The agreement is recorded from Alibi's report rather than a separate GitHub review. Before implementation, request review of forms, error handling, API models and accessibility behaviour.

D5, provisional tooling: SQLAlchemy with Alembic migrations, Pytest for real-DB integration tests, Playwright for browser scenarios and k6 for load generation, consistent with the README. Pin compatible versions only after the first reproducible environment is validated. Docker Compose will describe the local client/API/database environment. It is sufficient for a demo, but does not establish production availability, backups or disaster recovery.

6. Cross-cutting decisions and reconsideration triggers

Modular monolith with one transactional database

Requirement served: BR-04/09, NFR-08

Cost / risk: Coupled deployment and shared schema

Revisit when: Independently deployable teams or measured scale requirements appear

Club guard then station guard for state-changing transactions

Requirement served: BR-09/10

Cost / risk: Contention; every writer must comply

Revisit when: NFR-04 benchmark fails or administrative batch writes become common

PostgreSQL overlap constraint as defence in depth

Requirement served: BR-04/05

Cost / risk: Database-specific migration

Revisit when: A different DB becomes mandatory and equivalent safety is demonstrated

Durable owner + operation-key deduplication

Requirement served: BR-06/11

Cost / risk: Storage and reconciliation contract

Revisit when: Retention/deletion policy is introduced

REST/OpenAPI

Requirement served: BR-03/07/08, NFR-08

Cost / risk: Some view-specific request composition

Revisit when: Measured over-fetching or integration requirements justify GraphQL or push

The first implementation spike must prove migration support, concurrent creation, maintenance coordination, cancellation, and replay after response loss on actual PostgreSQL. Framework validation and an in-memory SQLite test cannot substitute for that evidence. Authentication/session choice, deployment host, exact versions and measured resource requirements remain open implementation decisions. Reviewer requested: Mansur for D4/API interaction; Nurali for D1 and administrative transaction use. Team agreement confirmed through Alibi's report on October 23, 2026.

6.1 User and administrator scenarios

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/requirements/user-scenarios.md

Version 1.1 · October 23, 2026 · ECR-1

Document owner: Mansur Urynbassar. Status: revised design baseline with Alibi's approval in PR #7; see the October 24 review and agreement record (new package record; see chapter 11). The previous version reported team agreement on October 23, 2026; this statement does not constitute a recorded GitHub review.

Purpose and Scope

This document defines the expected behavior of the proposed CyberSeat system and provides a basis for the functional requirements, use-case model, UX mockups, and subsequent system evaluation. It is part of the first submission due by **6:00 PM on October 24, 2026**. This deadline applies to the project documentation and design materials; it does not require every described feature to be implemented by that date.

The document is based on the CyberSeat README and the project justifications prepared by the three team members. Visitor needs are supported by the three discussions documented in user-needs evidence. Administrative needs remain assumptions pending administrator research. Prototype evaluation remains a future activity. Online payment, a loyalty programme, and hardware management are outside the mandatory scope.

Roles and General Rules

Role	Access scope
Guest	Browse clubs, gaming-station specifications, availability, and published events
Registered visitor	Create, view, and cancel only their own reservations; view the event calendar
Club administrator	Manage stations, schedules, reservations, events, and reports for assigned clubs
System administrator	Manage clubs and administrator assignments

Dates and times shall be displayed in the selected club's time zone, which shall be clearly identified in the interface. Availability depends on the club's working schedule, active reservations, and unavailable intervals. A station shown as available may become unavailable before confirmation; therefore, the server shall perform the final availability check. A missing server response shall not be interpreted as either a successful or a failed reservation. Cancelled reservations shall remain in the history.

The following scenarios define team-approved behavior for the proposed system. They are design and acceptance criteria, not claims that the application has already been implemented or tested.

SC-01. Browse Clubs and Gaming Stations

Actor: Guest or registered visitor **Preconditions:** The catalogue is available. **Trigger:** The user opens the club catalogue.

1. The user browses the available clubs and applies station-related filters.

2. The user selects a club and opens its details page.

3. The user reviews the address, equipment, working hours, and gaming-station layout.

Expected result: A club is selected for an availability search; no reservation is created.

Alternative and error flows: An empty catalogue or no matching results shall be explained and accompanied by an option to change the filters. A loading failure shall display an error message and a retry action.

Acceptance criteria: Applied filters remain visible; an empty result is distinguishable from a loading error; the selected club is clearly identified. At the design stage, the floor layout may be represented by a mockup.

SC-02. Select a Date, Time, and Available Station

Actor: Guest or registered visitor **Preconditions:** A club has been selected. **Trigger:** The user starts selecting a visit interval.

1. The user enters a date, start time, and end time.
2. The system validates the interval and requests current availability.
3. The system displays available, occupied, and unavailable stations together with their specifications.
4. The user selects an available station and proceeds to confirmation.

Expected result: The club, station, and time interval are selected, but availability is not yet guaranteed.

Alternative and error flows: Missing values, an end time earlier than the start time, and invalid times shall be identified next to the relevant fields. If the interval is outside working hours, the system shall request another interval. If no station is available, the user may change the time or club. Availability data from a previous interval shall not be used after the interval changes.

Acceptance criteria: Every station status is communicated through both text and colour; the time zone is visible; a blocked station cannot be selected as available; a loading state is distinguishable from an empty result.

SC-03. Confirm a Reservation

Actor: Registered visitor **Preconditions:** The user is authenticated and has selected a club, station, and interval. A guest must sign in before confirmation. **Trigger:** The user opens the reservation confirmation screen.

1. The user reviews the club, station, date, start time, end time, and time zone.
2. The user selects **Confirm reservation**.
3. The server rechecks access rights, working hours, unavailable intervals, and overlaps with active reservations.
4. If the reservation is saved, the system displays its identifier and details.
5. The user opens the new reservation in the personal account.

Expected result: Exactly one confirmed reservation belongs to the current user and occupies the selected interval.

Alternative and error flows: If the station becomes unavailable before confirmation, the system shall not create a conflicting reservation and shall offer another station or time. An expired session shall require the user to sign in again. If the connection is lost after submission, the interface shall display an uncertain outcome and request an updated reservation list; it shall not claim success without server confirmation.

Acceptance criteria: Concurrent overlapping requests for the same station shall result in no more than one active reservation. Client-side validation shall not replace the server-side check. Repeated submission shall be disabled while a request is pending, and the API shall prevent duplicate reservations when a request is retried after a connection failure.

SC-04. View Personal Reservations

Actor: Registered visitor **Preconditions:** The user is authenticated. **Trigger:** The user opens the personal account.

1. The user opens active reservations or reservation history.
2. The user selects a record and reviews the club, station, interval, and status.
3. If permitted, the user proceeds to reservation cancellation.

Expected result: The user receives current information about their own reservations.

Alternative and error flows: If no records exist, the interface shall show an empty state. A loading error shall not be presented as an empty list. A request for another user's reservation shall be rejected without disclosing its data.

Acceptance criteria: Other users' reservations are absent from the list and inaccessible by identifier; cancelled records remain in history; dates and statuses remain readable on mobile screens.

SC-05. Cancel a Personal Reservation

Actor: Registered visitor **Preconditions:** The user owns an active reservation and its start time has not passed. **Trigger:** The user selects the cancellation action.

1. The user reviews the selected reservation.

2. The user confirms the cancellation.

3. The server verifies ownership and the current reservation status and records the cancellation.

4. The interface displays the updated status, while the record remains in history.

Expected result: The cancelled reservation no longer occupies the interval. The station becomes available unless another restriction applies.

Alternative and error flows: Closing the confirmation dialog shall make no change. An already cancelled reservation shall display its current status. After a connection failure, the client shall request the latest status before repeating an action. A user shall not be able to cancel another user's reservation.

Acceptance criteria: Cancellation does not delete the history record; availability changes only after server confirmation. At or after the reservation start, cancellation/status changes are not permitted in this baseline, including for administrators (ADM-09). Any future post-start exception requires separately approved club rules and an audit record.

SC-06. View the Event Calendar

Actor: Guest or registered visitor **Preconditions:** The event calendar is available. **Trigger:** The user opens the events section.

1. The user selects a period and, where supported by the interface, a club.

2. The user browses events and opens a selected event.

3. The user reviews its title, description, location, date, time, and time zone.

Expected result: The user receives current event information. Tournament registration and payment are outside this scenario.

Alternative and error flows: A period with no events shall be distinguished from a loading error. A cancelled event shall be retained with a clear cancelled status rather than displayed as active. Updated details shall be shown when the event is reopened.

Acceptance criteria: The calendar and event details display consistent dates; the user can return to the previously selected period; published club events are available to guests and registered visitors.

SC-07. Access the Club Administration Area

Actor: Club administrator **Preconditions:** The user is authenticated and assigned to at least one club. **Trigger:** The user opens the administration area.

1. The system verifies the administrative role.

2. The system displays the clubs assigned to the administrator.

3. The administrator selects a club and opens its management sections.

Expected result: Management functions are available only within the administrator's authorised scope.

Alternative and error flows: A visitor shall not receive administrative data. A direct request for an unassigned club shall be rejected. After an assignment is revoked, the next operation shall require a new authorisation check.

Acceptance criteria: Access restrictions are enforced by the server, including direct requests using resource identifiers; the interface clearly identifies the currently selected club.

SC-08. Manage Gaming Stations

Actor: Club administrator **Preconditions:** An assigned club is open in the administration area. **Trigger:** The administrator opens the gaming-station section.

1. The administrator creates a station or selects an existing station.
2. The administrator enters or edits its label and specifications.
3. The administrator reviews the data and saves the changes.
4. The system updates the station representation.

Expected result: The club catalogue contains current station information.

Alternative and error flows: Required fields and duplicate labels shall be validated. A saving failure shall not be presented as success. A station with existing reservations may be deactivated, but its reservations and history shall not be silently deleted.

Acceptance criteria: An administrator cannot edit a station belonging to an unassigned club; saved specifications match the information shown to visitors; existing reservations remain linked to the station.

SC-09. Update the Working Schedule

Actor: Club administrator **Preconditions:** The administrator has access to the selected club's schedule. **Trigger:** The administrator edits the working intervals.

1. The administrator defines working intervals and verifies the club's time zone.
2. The system validates the data and identifies affected active reservations.
3. If no conflict exists, the system saves the schedule.
4. Subsequent availability searches use the updated schedule.

Expected result: New reservations comply with the current working schedule.

Alternative and error flows: Invalid intervals shall not be saved. If a change conflicts with an active reservation, the system shall reject the change and identify the conflict. An existing confirmed reservation shall never be cancelled silently.

Acceptance criteria: A conflict shall not leave a partially updated schedule; saved changes are applied by the server-side reservation check.

SC-10. Block a Station for Maintenance

Actor: Club administrator **Preconditions:** A station belonging to an assigned club has been selected. **Trigger:** The administrator creates an unavailable interval.

1. The administrator enters the start, end, and reason for the block.
2. The server validates the interval and checks active reservations.
3. If the interval is valid and conflict-free, the server saves the block.
4. Visitors see the station as unavailable during the specified interval.

Expected result: A conflicting reservation cannot be created during the maintenance interval.

Alternative and error flows: If an active reservation conflicts with the proposed block, the system shall reject the block and preserve the reservation. Concurrent reservation and blocking requests shall not both receive contradictory confirmations. Removing a block shall not remove other availability restrictions.

Acceptance criteria: The unavailable interval is enforced by the server for direct reservation requests, not only by the floor-layout interface; failed operations are clearly reported.

SC-11. View Club Reservations

Actor: Club administrator **Preconditions:** The administrator has access to the selected club. **Trigger:** The administrator opens the schedule or reservation section.

1. The administrator selects a date or period and, if necessary, a station.

2. The administrator reviews reservations and their statuses.

3. The administrator opens a record to verify its interval and station.

Expected result: The administrator can assess the workload of the assigned club.

Alternative and error flows: No reservations shall be distinguished from a loading failure. Data from unassigned clubs shall be inaccessible. Only personal data required to service a reservation shall be displayed.

Acceptance criteria: Cancelled reservations shall not appear as occupied intervals; the server shall enforce the selected-club scope. Administrative changes to a visitor's reservation shall follow club rules and be recorded in the audit log.

SC-12. View a Club Report

Actor: Club administrator **Preconditions:** The administrator has access to reports for the selected club. **Trigger:** The administrator selects a report.

1. The administrator selects a club and reporting period in the club's time zone.

2. The administrator requests reservation and utilisation indicators.

3. The system displays the results, reporting period, and calculation definitions.

Expected result: The administrator receives approved indicators for the selected period.

Alternative and error flows: An invalid period shall require correction. A lack of data shall be explained. A calculation error shall not be represented as a valid zero value.

Acceptance criteria: Calculations shall be compared with a controlled data set. Cancelled reservations shall be reported separately and excluded from occupied intervals. The utilisation denominator shall be the station time available within the club's working schedule after approved unavailable intervals are excluded. Financial reporting is outside scope unless a pricing model is approved separately.

SC-13. Manage Club Events

Actor: Club administrator **Preconditions:** The administrator has access to events for the selected club. **Trigger:** The administrator creates or edits an event.

1. The administrator enters the title, description, location, date, and time.

2. The administrator reviews the data and saves the event.

3. The administrator verifies its appearance in the calendar.

4. When an event changes or is cancelled, the administrator updates its published record.

Expected result: Visitors receive current event information.

Alternative and error flows: Required fields and time intervals shall be validated. An administrator shall not modify an event belonging to an unassigned club. A saving failure shall not be presented as a published update.

Acceptance criteria: The calendar and event details show the saved information; a cancelled event remains in history with a cancelled status; creating an event does not automatically block gaming stations.

SC-14. Manage Clubs and Administrator Assignments

Actor: System administrator **Preconditions:** The user has system-administrator privileges. **Trigger:** The user opens club management.

1. The system administrator creates or opens a club and reviews its primary details, including its time zone.
2. The system administrator selects a user for assignment to a specific club.
3. The system administrator saves a new assignment or revokes an existing assignment.
4. The system displays the updated assignment list.

Expected result: Administrative permissions are associated with specific clubs.

Alternative and error flows: A club administrator shall not grant themselves additional privileges. Validation and saving errors shall be clearly displayed. Updating club information shall not damage reservation history.

Acceptance criteria: An assignment grants access only to the specified club; revocation applies to subsequent requests. Club deletion and bulk permission changes require separate requirements and are outside this scenario.

Common Interface States

State	Expected behavior
Loading	Display progress and do not present it as an empty result
No results	Explain the state and offer changes to filters or the selected period
Invalid input	Display a message next to the relevant field and preserve other valid values
Expired session	Request sign-in and do not expose protected data
Insufficient permissions	Reject the action without disclosing another user's or club's data
Server error	Clearly state that the operation failed and provide a safe retry action
No connection before submission	Do not show the operation as saved; preserve entered data where possible
No response after submission	Display an uncertain outcome and verify the result before allowing a retry
Availability conflict	Refresh availability and offer another station or time

The UX mockups shall cover desktop and mobile layouts, use clear labels, and communicate statuses through more than colour alone. Validation of these states in a working application belongs to later implementation and evaluation stages.

Traceability to ECR-1 and Subsequent Deliverables

Club, station, date, and time selection

Coverage: SC-01-SC-02

UX mockup basis: Catalogue, club details, interval and station selection

Reservation creation and conflict handling

Coverage: SC-03

UX mockup basis: Confirmation, success, conflict, and uncertain-response states

Personal reservation viewing and cancellation

Coverage: SC-04-SC-05

UX mockup basis: Personal account, reservation details, cancellation confirmation

Event calendar

Coverage: SC-06

UX mockup basis: Calendar and event details

Administrative operations

Coverage: SC-07-SC-14

UX mockup basis: Club selection, stations, schedules, blocks, reports, and events

Loading, errors, and connection loss

Coverage: All scenarios and the common-state table

UX mockup basis: Relevant states for each principal screen

Team review of reservation and administrative behavior

Coverage: Team Review Results below

UX mockup basis: Completed on October 23, 2026

SC-01-SC-05 form the principal visitor flow for the first UX mockup sequence. SC-07, SC-10, and SC-11 form the principal administrative flow. This document provides input for the requirements specification, use-case model, and prototype; it does not replace those separate deliverables.

Agreed Team Decisions

Topic	Approved decision
Interval boundaries	Reservations use the half-open interval `[start, end)`, so one reservation may end when the next begins. Minimum duration shall be configurable by the club.
Cancellation	A visitor may cancel their own reservation before its start time. At or after the start, status changes are deferred and prohibited in this baseline (ADM-09); future exceptions require separately approved club rules and an audit-log entry.
Lost response during creation	The client shall request the current reservation list after reconnection. The API and database shall prevent duplicate reservations during retries.
Schedule or maintenance conflict	A proposed change that conflicts with an active reservation shall be rejected; the confirmed reservation shall be preserved.
Administrator actions on visitor reservations	Administrators may view reservations for assigned clubs. Any permitted status change shall follow club rules and be recorded in the audit log.
Reports	Indicators shall use an explicitly selected period in the club's time zone. Cancelled reservations shall be shown separately and excluded from occupied intervals.
Events	Administrators manage events for assigned clubs. Cancelled events remain in history with a clear status.

Topic	Approved decision
Authentication and personal data	Authentication is required to create and manage reservations. Administrators see only the personal data required to service a reservation.

Review history and current revision

Recorded in source: Mansur Urynbassar reviewed the scenarios for consistency with the planned user and administrator interfaces.

Recorded in source: Aibekuly Alibi reviewed SC-02-SC-05 and SC-09-SC-10, including reservation and conflict-prevention rules.

Recorded in source: Akhmetov Nurali reviewed SC-07-SC-14, including access control, schedules, events, and reporting.

Recorded in source: The team reviewed and approved the decisions concerning error handling, connection loss, and availability conflicts.

Recorded in source: The document satisfies the expected result and acceptance criteria of ECR-1.

Recorded in source: The document was published on GitHub through a dedicated branch and Pull Request.

Recorded in source: Links to the document, commit, and Pull Request were added to ECR-1.

The checked entries above preserve the previous document's report of team agreement. Review of the correction revision is separately established by Alibi-stack's approval of PR #7 (commit `d7bae8ea9931bbdb8e4caec01c243f33c8cea493`). See the October 24 review and agreement record (new package record; see chapter 11); historical reported agreement and submitted GitHub review remain distinct. This document remains an input to the separate requirements, UML and UX deliverables.

6.2 Booking and non-functional requirements

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/requirements/booking-and-nfr.md

Version 1.1 | October 23, 2026 | Owner: Aibekuly Alibi | ECR-7

Status: design baseline; team agreement confirmed by Alibi on October 23, 2026. Must = mandatory for the core system; Should = desirable for the diploma version. Acceptance statements describe future verification, not tests already passed. Sources: ECR-1 scenarios, proposal, research.

Domain semantics

A station belongs to exactly one club. A reservation identifies its owner, station, start and end instants and persisted status `CONFIRMED` or `CANCELLED`. A past visit is determined from the end instant; it is not a third stored booking status. Sending, failed and unknown are interface states. An event is informational and creates no reservation automatically.

The club-configured minimum duration is validated for new reservations (as agreed in ECR-1); its initial value requires club/user confirmation. All intervals are finite and half-open `[s,e)` with `s < e`. Two intervals overlap exactly when `s1 < e2 AND s2 < e1`. Times are transmitted with an offset, validated against the club's IANA zone and stored as UTC instants; the zone identifier remains part of club configuration. Ambiguous or nonexistent local times require explicit resolution, never silent guessing. Future working intervals may cross midnight and must be converted to a continuous instant interval before containment is checked.

Availability requires an active club and station, full containment within an open working interval, and no overlap with a confirmed reservation or maintenance interval. Displayed availability is advisory. At commit, the server clock must still place the start in the future.

Functional requirements and acceptance criteria

BR-01 Must

Requirement: Search availability for the selected club, station type and interval

Verifiable acceptance criterion: Seed one free PC, one reserved PC and one blocked console; search returns the expected station statuses for that interval, with visible club zone

Trace: SC-01, SC-02

BR-02 Must

Requirement: Validate interval and station/club identity on the server

Verifiable acceptance criterion: Missing fields, `end <= start`, duration below the configured club minimum, past start, inactive station, wrong club/station pairing or out-of-hours interval are rejected without a new reservation

Trace: SC-02, SC-03

BR-03 Must

Requirement: Create one reservation for the authenticated visitor after revalidation

Verifiable acceptance criterion: For a valid free interval, committed record has the authenticated owner, selected station, exact instants, unique ID and `CONFIRMED`; success is shown only after committed response

Trace: SC-03

BR-04 Must

Requirement: Prevent overlapping confirmed bookings, including concurrent requests

Verifiable acceptance criterion: Release 20 different visitors simultaneously against one free station and identical interval: exactly one success absent infrastructure faults; DB contains exactly one confirmed record. Repeat for partial overlap and containment

Trace: SC-03

BR-05 Must

Requirement: Permit adjacent intervals and independent stations

Verifiable acceptance criterion: Book `[14:00,15:00)` then `[15:00,16:00)` on the same station: both succeed; same interval on two stations also succeeds

Trace: SC-02, SC-03

BR-06 Must

Requirement: Deduplicate retries of one logical creation operation

Verifiable acceptance criterion: Two concurrent requests with the same authenticated owner and operation key and identical body return the same reservation ID; one record exists. Reusing the key for a different body returns a conflict

Trace: SC-03; common states

BR-07 Must

Requirement: List and show only the visitor's own reservations

Verifiable acceptance criterion: Visitor A cannot obtain B's details by changing an ID or list filter; authenticated ownership comes from the session, not a client owner field

Trace: SC-04; common states

BR-08 Must

Requirement: Cancel an own confirmed booking only before its start

Verifiable acceptance criterion: A valid cancellation records `CANCELLED`, preserves history and releases only that booking's occupancy. At/after start or for another owner it is rejected. A repeat of a completed cancellation is harmless

Trace: SC-05

BR-09 Must

Requirement: Coordinate bookings with maintenance creation and changes

Verifiable acceptance criterion: Race a valid reservation and overlapping maintenance for the same station: no committed state contains both. If a booking wins, maintenance is rejected; if maintenance wins, booking is rejected

Trace: SC-03, SC-10

BR-10 Must

Requirement: Preserve reservations when working hours or station status change

Verifiable acceptance criterion: A change that would invalidate a future confirmed booking is rejected with a conflict; it does not silently cancel or delete the booking

Trace: SC-08, SC-09

BR-11 Must

Requirement: Reconcile an unknown network outcome

Verifiable acceptance criterion: Drop the response after commit; refresh own reservations and retry/query using the same operation key resolves to the original ID. UI says result is being checked until reconciliation, and does not create a new operation automatically

Trace: SC-03; common states

BR-12 Must

Requirement: Revalidate after stale availability or session expiry

Verifiable acceptance criterion: A second client takes the displayed slot; first client receives a conflict and refreshes choices. An expired session cannot commit; after login the interval is checked again

Trace: SC-02, SC-03; common states

BR-13 Must

Requirement: Distinguish loading, empty, validation, conflict and server failure

Verifiable acceptance criterion: Each state has a textual message and an appropriate next action. A server error is never displayed as an empty catalogue or confirmed booking

Trace: All scenarios; common states

BR-14 Must

Requirement: Show club-local time consistently

Verifiable acceptance criterion: Confirmation, own history, administration and calendar show the same instant in the selected club's zone with zone label; UTC/local conversion round-trips in a controlled fixture

Trace: SC-02 through SC-06

BR-15 Must

Requirement: Retain an audit trail for changes

Verifiable acceptance criterion: Creation, cancellation and rejected administrative conflicts can be correlated through operation/request IDs; audit records contain actor, action, target, time and result, without credentials

Trace: SC-03, SC-05, SC-09

BR-16 Should

Requirement: Offer another interval or station after a conflict

Verifiable acceptance criterion: Conflict view preserves the selected club/time and allows another availability search; alternatives are not presented as already held

Trace: SC-02, SC-03

Invariants and state transitions

I1: no two confirmed reservations for one station overlap. I2: confirmed reservations and maintenance for one station do not overlap. I3: a new or changed reservation fits an open schedule at commit. I4: a logical creation operation has at most one reservation. I5: owner and assigned-club checks are server-side. I6: cancellation cannot erase history or another occupancy source.

Only `CONFIRMED -> CANCELLED` is a persisted user transition. Creation enters `CONFIRMED` atomically; rollback creates no booking. Repeated cancellation returns the existing cancelled result. Creating a new reservation after cancellation uses a new operation key; retrying the old key returns the original, now cancelled booking.

Proposed API contract (design, not implemented endpoints)

Availability

Proposed contract: `GET /api/v1/clubs/{id}/availability?start=...&end=...`

Principal result/error: 200 advisory statuses; 422 invalid interval

Create

Proposed contract: `POST /api/v1/reservations`, operation key plus station and offset-aware interval

Principal result/error: 201 first commit; 200 replay; 409 conflict; 401 session absent/expired; 422 invalid input

Own list/details

Proposed contract: `GET /api/v1/me/reservations`, `GET /api/v1/me/reservations/{id}`

Principal result/error: 200 own data; 404 absent or foreign resource

Cancel

Proposed contract: `POST /api/v1/me/reservations/{id}/cancellation`

Principal result/error: 200 cancelled/replayed result; 409 started; 404 absent/foreign

Reconcile creation

Proposed contract: `GET /api/v1/me/reservation-operations/{key}`

Principal result/error: 200 committed result; 202 in progress; 404 no durable result yet, which alone is not proof that an in-flight request failed

Error bodies carry a stable machine code, safe message and correlation ID. The client cannot supply a trusted owner or role. Proposed idempotency records are unique by owner and operation key, include a normalised-body fingerprint and are committed with the booking. Retain the operation mapping for the lifetime of the booking in the MVP. A future retention limit requires an explicit protocol change. Standard HTTP POST is not assumed to be inherently idempotent.

Non-functional requirements

NFR-01 Must

Requirement and target: Zero invariant violations in the defined concurrency suite

Verification conditions: Real PostgreSQL, independent connections/API workers, synchronised race starts; 100 runs each for same-slot, partial overlap, booking vs maintenance, booking vs closure and cancel vs create; query DB state after each run

NFR-02 Must

Requirement and target: No unauthorised own-record or cross-club operation

Verification conditions: Negative tests for unauthenticated, visitor A/B, assigned/unassigned administrator and system administrator; direct HTTP requests as well as UI checks. ECR-8 defines complete administrative permissions

NFR-03 Must

Requirement and target: Committed data survives API restart

Verification conditions: Commit a booking, restart API, reconcile operation and read history; ID/status unchanged. Force failure before commit: neither booking nor success mapping remains

NFR-04 Must

Requirement and target: Draft performance target: p95 availability <=500 ms, creation/cancellation <=1000 ms; unexpected HTTP 5xx <1%

Verification conditions: Dedicated 4-vCPU/8-GB host, local Docker network, one API service and PostgreSQL; 10 clubs, 500 stations, 100,000 reservations, 30-day horizon; 50 virtual users, 80% availability/10% creates/10% cancels, 2-min warmup and 10-min measured run, three runs. Report achieved request rate and raw results; classify expected 409 conflicts separately. Targets were accepted as proposed design targets according to Alibi's team-agreement report

NFR-05 Must

Requirement and target: Bounded contention and errors

Verification conditions: Draft lock timeout 2 s, maximum 3 full-transaction retries for retryable deadlock/serialization failure with bounded backoff; exhaustion returns explicit retryable failure, never success. Timeouts and retry counts must be recorded with load results

NFR-06 Must

Requirement and target: Usable core flow at 360px and desktop width, with keyboard access

Verification conditions: Check visible focus, meaningful labels and text statuses, field errors, no keyboard trap and no horizontal page overflow for core screens; use WCAG 2.2 AA as the design target, not a claim of certified conformance

NFR-07 Should

Requirement and target: Draft usability goal: at least 80% unassisted task completion

Verification conditions: Recruit at least 5 relevant visitors; each finds a free station, books and cancels. Report per-task counts, time and errors plus SUS where appropriate. Small convenience samples are formative and cannot establish population-wide benefit

NFR-08 Must

Requirement and target: Reproducible environment and migrations

Verification conditions: A teammate uses documented pinned dependencies, synthetic seed and migrations in a fresh environment; no undocumented manual schema edits. This check belongs to implementation, not current documentation

NFR-09 Must

Requirement and target: Safe operational data handling

Verification conditions: HTTPS in deployment; secrets excluded from source; credential/token values absent from error bodies and logs; server-side authorisation repeated on every operation. Authentication mechanism to be agreed in ECR-8

NFR-10 Must

Requirement and target: Reports use defined denominators and consistent snapshots

Verification conditions: Utilisation definition specifies booking statuses, time clipping and available station-minutes; compare with hand-calculated synthetic fixtures. Past end time alone does not prove attendance; occupancy reporting is scheduled utilisation, not measured actual usage

These numerical targets are explicit proposed evaluation conditions, not derived user evidence or measured performance. NFR-04 and NFR-07 were included in Alibi's report of team agreement; ECR-13 should retain or explicitly revise their measurement conditions.

Boundary examples for later tests

Existing / requested data	Expected result
Confirmed `[14:00,15:00)` / request `[14:30,15:30)`	Reject overlap

Existing / requested data	Expected result
Confirmed `[14:00,15:00)` / request `[13:00,16:00)`	Reject containment
Confirmed `[14:00,15:00)` / request `[15:00,16:00)`	Allow if other rules pass
Cancelled `[14:00,15:00)` / same interval new operation	Allow if other rules pass
Maintenance `[14:00,15:00)` / request same interval	Reject regardless of cancelled booking history
Open interval 22:00-02:00 next day / request 23:00-01:00	Allow after club-zone conversion if other rules pass
Booking vs concurrent closure of the same club	Serialise at shared club guard; whichever commits second sees the first
Same operation key / changed station or interval	Reject key reuse; never create second booking

Review and dependencies

Review requested: Nurali for BR-09/10, administrative write protocol and NFR-02/10; Mansur for BR-11/13/14 and interface feasibility. Alibi confirmed his teammates' agreement in the working chat on October 23, 2026. ECR-8 must use the same guards for schedule/maintenance writes; ECR-13 must record agreed targets and executed test evidence later; ECR-12 must translate invariants into schema/migrations. The October deliverable is this specification with its acceptance criteria.

6.3 Roles and access-control matrix

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/requirements/roles-and-permissions.md

Version 1.0 · October 23, 2026 · Owner: Akhmetov Nurali · ECR-8

Status: design specification with GitHub approval and reported team agreement; see the October 24 review and agreement record (new package record; see chapter 11). This is planned behaviour, not implemented authorisation or completed tests. Sources: scenarios, booking baseline, administrative specification. System administrator in existing documents means superadministrator here.

Role definitions

Role	Identity and scope
Guest	Unauthenticated person; public information only.
Registered visitor	Authenticated person; manages only reservations owned by their authenticated user ID.
Club administrator	Authenticated person with active assignments to specific clubs; operational authority is limited to those clubs.
Superadministrator	System-level authority to create clubs and grant/revoke club assignments; no automatic operational access to every club.

Administrators may use visitor functions for their own reservations. That personal capability never expands administrative scope. A superadministrator needing operational access must also have an explicit club assignment.

Permission matrix

Own = ownership verified by the server. Assigned = a current server-side club assignment. No = denied even when a client displays a button or supplies another ID. Permissions are Must requirements unless deferred explicitly.

Browse active clubs, public station details and availability

Guest: Public

Visitor: Public

Club administrator: Public

Superadministrator: Public

View published events and clearly cancelled published events

Guest: Public

Visitor: Public

Club administrator: Public

Superadministrator: Public

Create/list/view personal reservations

Guest: No

Visitor: Own

Club administrator: Own via visitor flow

Superadministrator: Own via visitor flow

Cancel personal reservation before its start

Guest: No

Visitor: Own

Club administrator: Own via visitor flow

Superadministrator: Own via visitor flow

Open operational administration

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

Edit club name/address and minimum booking duration

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

Create/edit/deactivate stations

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

Change working schedules and maintenance intervals

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

List/view club reservations and minimum service data

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

Cancel club reservation before start, with reason

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

Change reservation after its start

Guest: No

Visitor: No

Club administrator: Deferred pending approved club rules

Superadministrator: Deferred pending approved club rules

Create/edit/publish/cancel club events

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

View reservation and scheduled-utilisation reports

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

Inspect operational audit entries

Guest: No

Visitor: No

Club administrator: Assigned

Superadministrator: Explicit club assignment required

Create clubs; manage activation and IANA time zone

Guest: No

Visitor: No

Club administrator: No

Superadministrator: System-wide; booking safeguards apply

Grant/revoke administrator-to-club assignments

Guest: No

Visitor: No

Club administrator: No

Superadministrator: System-wide

Inspect club/assignment administration audit

Guest: No

Visitor: No

Club administrator: No

Superadministrator: Limited to these system actions

Grant superadministrator privileges via application

Guest: No

Visitor: No

Club administrator: No

Superadministrator: Outside this specification

Permanently delete clubs, stations, bookings or audit history

Guest: No

Visitor: No

Club administrator: No

Superadministrator: No

Server access rules and planned verification

RBAC-01 / Must

Requirement: Authenticate and authorise every protected request; never trust a client role/owner.

Verifiable acceptance criterion: Guest receives 401 for protected operation; visitor receives 403 for administrative route. Supplying another role/user ID grants no capability.

RBAC-02 / Must

Requirement: Resolve the actual owning club of every station, block, booking, event and report.

Verifiable acceptance criterion: Admin assigned only to A cannot read/mutate B by changing path, body, nested resource IDs or filters; no protected details are disclosed.

RBAC-03 / Must

Requirement: Apply club scope to lists, details, aggregates, writes and any later exports.

Verifiable acceptance criterion: Two-club fixture returns only authorised data; explicitly requesting an unauthorised club or mixed scope is rejected.

RBAC-04 / Must

Requirement: Personal routes enforce authenticated ownership independently of admin authority.

Verifiable acceptance criterion: Visitor A cannot read/cancel B's record. Return 404 for absent/foreign records as in the booking baseline.

RBAC-05 / Must

Requirement: Revocation removes access on subsequent requests and is ordered with protected writes.

Verifiable acceptance criterion: After revocation commits, next request is denied. An in-flight write either commits before revocation or observes revoked access and fails; stale session cannot restore assignment.

RBAC-06 / Must

Requirement: Administrators cannot alter assignments, club identity/time zone or their privileges.

Verifiable acceptance criterion: Direct system/assignment requests receive 403. Station/event updates cannot reparent a resource to another club.

RBAC-07 / Must

Requirement: Default deny; separate system and operational capabilities.

Verifiable acceptance criterion: Unassigned superadministrator can manage assignments but cannot read club reservation reports. Grant/revoke explicitly enables/removes operational access.

RBAC-08 / Must

Requirement: Minimise reservation service data.

Verifiable acceptance criterion: Show booking ID, station, interval, status, user reference/display name and relevant cancellation reason; exclude credentials and unrelated profile/contact data. Reports contain aggregates.

RBAC-09 / Must

Requirement: Audit protected changes and failures safely.

Verifiable acceptance criterion: Records contain actor/action/target/club/time/result/reason/correlation ID without credentials; operational audit queries obey club scope.

An authenticated user lacking an administrative capability receives 403. For an administrator requesting an absent/out-of-scope individual resource, consistently use 404 without revealing existence. Invalid input uses 422 and conflicting valid changes use 409. Authorisation precedes disclosure of protected validation/conflict details.

Review required before task closure

Recorded in source: Team agreement with the proposed assignment and transaction/revocation design is recorded from Mansur's October 24 confirmation on behalf of the team. See the October 24 review and agreement record (new package record; see chapter 11); no separate execution of a technical review by Alibi is asserted.

Recorded in source: Mansur's approval of the role/access specification is recorded in PR #5. See the October 24 review and agreement record (new package record; see chapter 11); this is design acceptance, not a live-interface test.

Recorded in source: Named reviewer, reviewed commit and approval outcome are recorded in NeryBR's approval of PR #5 (commit `834f87e1a78a5ebc1b5c29097b856427dd43f993`).

The approval and reported agreement above are recorded; external task-evidence actions remain separate. See the full acceptance checklist.

6.4 Administrative requirements

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/requirements/administrative-requirements.md

Version 1.0 · October 23, 2026 · Owner: Akhmetov Nurali · ECR-8

Status: design specification approved by Mansur in PR #5; team agreement with the documented model was reported by Mansur on October 24. See the October 24 review and agreement record (new package record; see chapter 11). Acceptance examples describe future verification, not executed application tests.

Purpose and submission scope

This specification and the role matrix cover the administrative part of university requirement 6, Requirements specification. They complement Alibi's booking and non-functional baseline and Mansur's SC-01-SC-14 scenarios.

October covers analysis, design and planning. Mandatory submission: **October 24, 2026 before 18:00 in the university-specified time zone**. The team's October 21, 2026 23:00 target is an internal milestone. A completed frontend/backend and final test results are not required for this submission.

Must = required product behaviour; Should = desirable capability that may be deferred explicitly. Documenting a Must feature does not assert implementation in October. Payments, revenue reports, loyalty, hardware control, event registration, permanent deletion and superadministrator provisioning are outside this task.

Shared domain contract

Entity	Required meaning and relationships
User	Stable authenticated identity; personal booking ownership comes from this identity.
Club	Stable ID, name, address, IANA time zone, active state and configurable positive minimum booking duration.
Administrator assignment	User-to-club relationship with active/revoked state or validity interval. Many clubs per administrator and many administrators per club; no duplicate active pair.
Gaming station	Stable ID, exactly one club, club-unique label, PC/console type, specifications and effective active intervals. Referenced stations cannot be moved or deleted.
Working schedule	Club-local weekly intervals plus dated exceptions. An exception replaces the regular schedule for its local date; a closed exception supplies no opening interval. Merge overlapping opening intervals before calculations.
Maintenance interval	Station, start/end instants, reason, effective/cancelled state and audit metadata; restricts availability independently of reservation history.
Reservation	Owner, station, start/end instants and persisted CONFIRMED or CANCELLED status, matching the booking baseline. Retain cancellation actor/time/reason. Past end time is derived, not proof of attendance.
Event	Exactly one club; title, description, location, start/end instants and DRAFT/PUBLISHED/CANCELLED state. No automatic station occupancy.
Audit entry	Actor, action, target, club where applicable, UTC instant, result, change/reason summary and correlation ID; no credentials.

All booking, maintenance and report intervals are finite and half-open: [start,end), start < end. Overlap is start1 < end2 AND start2 < end1; adjacent intervals do not overlap. Store instants in UTC, retain the club's IANA zone and display local time with the zone. Ambiguous/nonexistent local times require explicit resolution. Overnight hours become continuous intervals

after conversion.

Retain effective-dated club/station activity, schedule and maintenance history to reproduce historical capacity. Later deactivation, block removal or schedule revision must not overwrite past capacity. Exact tables, revision fields and migrations require Alibi's shared-model review; they are proposed requirements here, not an assertion about the current ER model.

Functional requirements and acceptance criteria

Every operation obeys the permission matrix; operational permissions never extend beyond assigned clubs.

ADM-01 / Must

Requirement: Superadministrator creates clubs and manages activation/time zone. Assigned admin edits only club name/address and operational settings.

Verifiable acceptance criterion: Missing/invalid zone or non-positive minimum duration is rejected. Admin cannot create clubs, change club identity/zone or access another club. System changes obey ADM-08.

Trace: SC-07, SC-14

ADM-02 / Must

Requirement: Superadministrator grants/revokes administrator assignments.

Verifiable acceptance criterion: Grant enables exactly the selected existing club/user pair; duplicate active pair is rejected; revocation denies subsequent operations and is audited.

Trace: SC-14; RBAC-05

ADM-03 / Must

Requirement: Assigned admin creates/edits stations.

Verifiable acceptance criterion: Validate required label/type; reject duplicate label in one club, allow same label in different clubs. Public specifications match saved data. Reparenting is rejected.

Trace: SC-08

ADM-04 / Must

Requirement: Deactivate stations without deleting bookings/history.

Verifiable acceptance criterion: Immediate deactivation with a future or in-progress confirmed booking is rejected. Otherwise it succeeds and blocks new bookings; past records remain resolvable.

Trace: SC-08; BR-10

ADM-05 / Must

Requirement: Define weekly hours and dated exceptions.

Verifiable acceptance criterion: Reject invalid instant intervals; accept valid overnight hours; apply dated exception instead of regular hours. Reject any change invalidating a protected booking atomically.

Trace: SC-09; BR-10

ADM-06 / Must

Requirement: Create/update/remove maintenance with reason.

Verifiable acceptance criterion: Booking [14:00,15:00) prevents block [14:30,15:30); adjacent [15:00,16:00) is permitted if otherwise valid. Removal retains history and never removes another block/booking.

Trace: SC-10; BR-09

ADM-07 / Must

Requirement: List club bookings by period/station/status with minimum service data.

Verifiable acceptance criterion: Return only bookings overlapping selected period in the assigned club. Cancelled records never appear occupied; foreign nested station IDs are rejected.

Trace: SC-11

ADM-08 / Must

Requirement: Preserve protected bookings during club/station/schedule/block/zone changes.

Verifiable acceptance criterion: Reject the whole change if any CONFIRMED booking with end > server_now becomes invalid; show a safe conflict summary within authorised scope. Never automatically cancel, shorten, move or delete bookings.

Trace: SC-08-10, SC-14; BR-09/10

ADM-09 / Must

Requirement: Allow explicit admin cancellation before booking start, with a reason.

Verifiable acceptance criterion: Recheck assignment, reason and server_now < start; retain record and actor/time/reason; release only its own occupancy. Replay is harmless. At/after start return 409 until separate club rules are approved.

Trace: SC-05, SC-11

ADM-10 / Must

Requirement: Create/edit drafts and publish valid club events.

Verifiable acceptance criterion: Require nonempty title/description/location and a valid interval. Public queries exclude drafts and show saved published details in club-local time.

Trace: SC-06, SC-13

ADM-11 / Must

Requirement: Cancel published events while retaining history.

Verifiable acceptance criterion: Public details clearly show CANCELLED; retries preserve one event record. Publication/cancellation never changes station occupancy.

Trace: SC-06, SC-13

ADM-12 / Must

Requirement: Produce defined reservation and scheduled-utilisation reports.

Verifiable acceptance criterion: Show club, zone, [from,to), generation instant, snapshot cutoff and formula version; reject invalid/unauthorised scope. Controlled fixtures below give stated results.

Trace: SC-12; NFR-10

ADM-13 / Must

Requirement: Audit administrative changes and rejected conflicts.

Verifiable acceptance criterion: Successful mutations have durable audit records; failed transactions leave no partial mutation. Safe failure records correlate actor/action/target/time/result/reason/request.

Trace: SC-09-14; BR-15

ADM-14 / Must

Requirement: Detect stale updates and reconcile uncertain writes.

Verifiable acceptance criterion: Obsolete resource version yields 409. Creation retries with same actor/key/body return original result; changed body conflicts. Lost response never appears as confirmed success before reconciliation.

Trace: Common states; BR-06/11

ADM-15 / Should

Requirement: Show a change summary before saving.

Verifiable acceptance criterion: Summary identifies club/station, effective interval and change. A later concurrent conflict still rejects the write at commit.

Trace: SC-08-10

Transactional booking protection

Booking creation and administrative writes must use the same protection protocol, not only an advisory UI check. Reuse Alibi's shared club guard and deterministic station lock ordering, or an equivalent agreed transactional mechanism. Validate current assignments, versions, schedule, effective states and conflicts atomically with the write. Order assignment revocation with protected mutations as required by RBAC-05.

Race/boundary	Required committed outcome
Booking vs overlapping maintenance	One may commit; the other observes the conflict. Never commit both.
Booking vs closure/reduced hours/deactivation	Booking first means conflicting admin change fails. Admin change first means invalid booking fails.
Two edits from one version	First succeeds; stale second edit fails without overwriting it.
Cancellation vs block	Serialize and revalidate; block may follow cancellation if all remaining constraints permit it.
Multi-station change	All valid changes commit together or none.
Increased minimum duration	Applies to new requests only; existing bookings are not invalidated retroactively.
Zone change with future/in-progress confirmed bookings	Reject in this baseline. Without protected bookings, change prospectively and preserve historical zone/schedule revisions.

SC-08 permits deactivation while preserving records; BR-10 rejects changes invalidating future bookings. Together they mean that immediate conflicting deactivation must fail. Future-effective deactivation may be specified separately.

SC-05/SC-11 leave post-start status changes dependent on approved club rules. This baseline enables explicit pre-start cancellation only. Post-start cancellation, attendance, refunds, relocation and additional statuses remain deferred; an administrative role alone does not define those business rules.

Report calculation rules

Scope, period and snapshot

Select one authorised club and finite nonempty P=[from,to) in its zone. Convert boundaries to UTC; measure elapsed durations between instants in minutes. A date-only inclusive last day converts to midnight of the following local day and is displayed as the resulting exclusive end. Reject empty/reversed/malformed or unresolved ambiguous periods.

Read all inputs from one consistent committed snapshot T; show T and definition version with results. Apply historical schedules, club/station activity and maintenance effective during P, not today's configuration projected backwards. Later cancellation may restate a newly generated report; an earlier saved result retains its snapshot identity. These indicators measure scheduled bookings, not actual attendance or revenue.

For each station s existing during some part of P:

- W_s is the union of effective club opening intervals, intersected with P and effective active club/station intervals.

- M_s is the union of effective maintenance intersected with W_s. Merge overlaps before subtraction; never subtract a minute twice.

- A_s = W_s minus M_s: available capacity, including time occupied by bookings.

- B_s is the union of booking intervals whose status is CONFIRMED at T, intersected with P.

- O_s = B_s intersect A_s: scheduled occupied time. CANCELLED contributes no occupied minutes.

Indicator	Exact definition
Confirmed count	Distinct CONFIRMED IDs at T whose visit intervals overlap P, including bookings starting outside P.
Cancelled count	Distinct CANCELLED IDs at T whose original visit intervals overlap P; not a count by cancellation timestamp.
Total count	Confirmed + cancelled count; no additional persisted status assumed.
Available station-minutes	A = sum duration(A_s). Subtract maintenance and closed/inactive time, not bookings.
Scheduled occupied station-minutes	O = sum duration(O_s), clipped to P and capacity.
Free station-minutes	A - O.
Scheduled utilisation	100 * O / A when A > 0; otherwise N/A with no-capacity explanation.
Cancellation share	100 * cancelled_count / total_count when total_count > 0; otherwise N/A.

Aggregate exact durations before formatting; display minutes and percentages to two decimal places, rounding half up. Compute club utilisation from total O/A, not an average of station percentages. A spanning booking can count in several overlapping periods; do not sum those distinct counts into a unique total. Durations over disjoint periods are additive under the same snapshot and definitions.

If historical inputs are missing, show insufficient historical data rather than fabricated zero. If confirmed bookings overlap each other or fall outside effective capacity, report a data-integrity error and withhold utilisation/free-time results; do not silently hide invalid data through clipping or interval unions. An empty booking set with positive capacity gives O=0 and utilisation=0.00%. A calculation/loading error is never an empty result.

Controlled acceptance examples

These are proposed future fixtures, not executed test results. Times within each fixture use one explicitly labelled club zone.

Fixture	Expected result
P 10:00-12:00; two active/open stations; S1 maintenance 10:00-10:30, confirmed 10:30-11:30; S2 confirmed 11:00-12:00, cancelled 10:00-11:00	A=210 min; O=120; free=90; utilisation=57.14%; confirmed=2; cancelled=1; total=3; cancellation share=33.33%.
P 11:00-12:00; one station open 10:00-14:00; confirmed 10:30-11:30	A=60; O=30; free=30; utilisation=50.00%; confirmed count=1.
P 10:00-12:00; one station open throughout; blocks 10:00-10:45 and 10:30-11:00; no bookings	Maintenance union=60 min, not 75; A=60; O=0; free=60; utilisation=0.00%; counts=0; cancellation share=N/A.
All stations closed/blocked throughout P	A=0; O=0; free=0; utilisation=N/A, with no-capacity explanation.
P 10:00-11:00; cancelled booking ends exactly at 10:00 or starts exactly at 11:00	Neither boundary-only booking is counted.
Historical P has 120 open/active minutes; station deactivated after P	Historical A stays 120.
Historical schedule revision is missing	Insufficient historical data; no valid utilisation shown.
Invalid confirmed booking overlaps maintenance	Data-integrity error; no valid utilisation/free-time result shown.

Cross-cutting verification plan

Reuse NFR-01-NFR-10 from the booking baseline: concurrency, authorisation, persistence, performance and usability targets remain future evaluation conditions. No performance measurement or final test pass is asserted here.

Administrative forms distinguish loading, empty data, validation, insufficient permissions, conflicts, expired sessions, server errors and unknown write outcomes. Preserve valid input; use textual statuses alongside colour. Show a saved result only after server confirmation. Core flows support 360px and desktop widths, labelled controls and keyboard access according to NFR-06.

Later implementation checks use two clubs, a one-club admin, a multi-club admin, a visitor, an unassigned superadministrator, overnight hours and the report fixtures above. Check direct HTTP requests as well as UI flows. Reuse NFR-01's real-PostgreSQL concurrency conditions. October review verifies that cases and expected outcomes are documented, not that a finished application passes them.

Coordination and pending decisions

Shared relationships/statuses

Proposed decision: Stable IDs; many-to-many assignments; CONFIRMED/CANCELLED booking statuses.

Reviewer needed: Aibekuly Alibi

Transaction ordering

Proposed decision: Common club/station guards; revocation ordered with writes.

Reviewer needed: Aibekuly Alibi

Historical reports

Proposed decision: Effective activity/schedule/maintenance history and consistent snapshots.

Reviewer needed: Aibekuly Alibi

Admin cancellation

Proposed decision: Reason required, pre-start only; post-start policy deferred.

Reviewer needed: Aibekuly Alibi and Mansur Urynbassar

Navigation and states

Proposed decision: Guest events; separate personal/club scope; N/A/conflict/stale-data states.

Reviewer needed: Mansur Urynbassar

This specification has NeryBR's approval of PR #5 (commit `834f87e1a78a5ebc1b5c29097b856427dd43f993`). Mansur subsequently reported agreement with the audited materials on behalf of all three participants; see the October 24 review and agreement record (new package record; see chapter 11). Revise this document, the role matrix and affected scenarios/booking rules together if the team changes a proposed decision.

Open decisions: exact authentication/session design and superadministrator bootstrap; initial club minimum duration; post-start club policy; physical history schema/retention; final report response schema. A blocking disagreement on a Must rule must be resolved or explicitly scoped before closure.

October deliverable acceptance

Four separate roles

Evidence: Role definitions, permission matrix, RBAC-01-09

State: Documented; PR #5 approved

Clubs, stations, schedules, blocks, events, reports

Evidence: ADM-01-15

State: Documented; PR #5 approved

Admin limited to assigned clubs

Evidence: Matrix; RBAC-02/03/05/06

State: Documented; PR #5 approved

Existing bookings protected

Evidence: ADM-04/05/06/08; transaction table

State: Documented; team agreement reported by Mansur

Priorities and verifiable criteria

Evidence: Must/Should, IDs, acceptance columns

State: Documented

Explicit report calculations

Evidence: Formulas, boundaries, historical inputs, fixtures

State: Documented; team agreement reported by Mansur

Scenario and model coordination

Evidence: SC/BR/RBAC traces and review table

State: Traced; team agreement reported by Mansur

Result/commit/PR links and named reviewer

Evidence: GitHub files and PR; reviewer records commit/outcome

State: PR #5 approval recorded; ECR-8 result and Done status inspected October 24

Recorded in source: Both document permalinks, reviewed commit, PR #5 and actual NeryBR review link were published in ECR-8 on October 24 with Mansur’s explicit permission; the saved comment was verified.

Recorded in source: Agreement with the proposed shared data/transaction design is reported by Mansur on behalf of all three participants, including Alibi; see the October 24 review and agreement record (new package record; see chapter 11).

Recorded in source: Record the actual reviewing participant, reviewed commit and approval outcome: NeryBR's approval of PR #5 (commit `834f87e1a78a5ebc1b5c29097b856427dd43f993`). Any newly identified blocking issue still requires resolution.

Recorded in source: Matched the live ECR-8 criteria to the acceptance table above on October 24; ECR-8 was already Done. This verifies the design deliverable, not implementation or executed tests.

File creation or a PR alone does not complete team review. Do not claim approval or final application test results for this October specification.

7. Architecture and required diagrams

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/diagrams/architecture.md

Version 1.1 | October 23, 2026 | Owner: Aibekuly Alibi | ECR-10

Status: proposed architecture; team agreement confirmed by Alibi on October 23, 2026. These are design views, not deployed services. Detailed ER design is a later ECR-12 deliverable.

1. Views and editable sources

System context

Editable source: system-context.mmd

Readable export: system-context.svg

Purpose: People outside the CyberSeat boundary and their interactions

Component overview

Editable source: components.mmd

Readable export: components.svg

Purpose: Web client, API boundary, logical modules and PostgreSQL

Mermaid sources and SVG exports express the same nodes and relationships. SVG labels are editable text. The component view deliberately includes client and storage context; it is a practical component overview rather than a claim of strict UML notation. Solid arrows denote user requests or runtime calls; the dashed event-source arrow denotes manual information entry. A module is a logical responsibility inside one server deployment, not an independently deployed microservice.

2. System context

Guests browse clubs and availability. Registered visitors also create, view and cancel only their own reservations. Club administrators manage stations, working hours, maintenance, bookings, published events and reports within assigned clubs. System administrators manage clubs and assignments. A multi-club manager is represented through assignments rather than unrestricted access by default.

An organiser or public event notice is an information source outside the system. Administrators manually verify and enter event details; there is no required external-feed API. Hosting, the browser and the database are not additional business actors. GitHub and YouTrack manage development and are not runtime dependencies of the reservation workflow. Payment providers and hardware agents are absent because those capabilities are excluded.

3. Component overview

React/TypeScript web client

Responsibility and data: Catalogue, selected interval, reservation forms, own history, calendar and assigned-club admin views; transient loading/error states

Requirement / scenario trace: BR-01/11/13/14; SC-01 through SC-06; common states

REST API boundary / FastAPI

Responsibility and data: Request models, route contracts, safe errors, authenticated context, correlation/operation IDs; no trust in client owner or role

Requirement / scenario trace: BR-02/03/06/07; SC-03; common states

Identity and access

Responsibility and data: User/session identity, visitor ownership and current club assignments; checked for every operation

Requirement / scenario trace: BR-07/08, NFR-02; SC-07/14

Club and station catalogue

Responsibility and data: Clubs, station identity/specifications, active status and club zone

Requirement / scenario trace: BR-01/02/14; SC-01/02/08

Reservation and availability

Responsibility and data: Availability calculation, creation, own history, cancellation and durable replay semantics

Requirement / scenario trace: BR-03 through BR-12; SC-02 through SC-05

Club administration

Responsibility and data: Authorised station, schedule and maintenance changes; all booking-affecting writes use the same coordinator

Requirement / scenario trace: BR-09/10; SC-07 through SC-10

Events

Responsibility and data: Draft/published/cancelled event information, source, zone and club association; no automatic reservation

Requirement / scenario trace: SC-06/13

Reporting and audit

Responsibility and data: Defined scheduled-utilisation queries and actor/action/result records

Requirement / scenario trace: BR-15, NFR-10; SC-12

Transaction coordinator / data access

Responsibility and data: Shared lock order, revalidation, operation uniqueness, transaction and error mapping, migrations

Requirement / scenario trace: BR-04/06/09/10, NFR-01/03/05

PostgreSQL

Responsibility and data: Durable users, assignments, clubs, stations, schedules, maintenance, reservations, operation identities, events and audit data

Requirement / scenario trace: All persisted invariants; future ECR-12 schema

The API invokes identity/access and the applicable domain module. Modules use the shared data-access layer; administration must not implement an independent unsafe booking writer. Reporting reads committed data and never changes reservation status. Audit writes for successful mutations share the business transaction; logging a rejected request is a separate operational event and must not turn a rejection into a committed booking.

4. Transaction protocol (proposed implementation contract)

Database baseline: PostgreSQL `READ COMMITTED` with explicit guard locks and a reservation overlap constraint. This is a design to validate during implementation. All steps below belong to one transaction; reads that decide a write happen after the necessary guards are acquired. A process-local mutex cannot protect independent API workers.

1. Authenticate the actor. For creation, claim a unique `(owner_id, operation_key)` record and check its normalised payload fingerprint. A concurrent same-key request waits for the first transaction and then reuses its result. A different payload with the same key is a conflict. The successful operation mapping and reservation commit together.

2. Acquire the club guard row `FOR SHARE` for a station-level operation. Working-hours, club-zone or club-status writers acquire the same guard `FOR UPDATE`. Club configuration writers therefore conflict with in-progress station operations; ordinary reservations at different stations may proceed concurrently.

3. Acquire the station guard row `FOR UPDATE`. Reservation creation, cancellation, maintenance and station-status changes use this same guard. For any batch operation, acquire clubs and stations in ascending ID order; never acquire the club guard after a station guard.

4. Recheck actor/assignment, club and station state, server time, working-hours containment, overlapping confirmed reservations and maintenance. Assignment changes must also coordinate with the relevant authorisation guard or club guard; checking an assignment before waiting for locks is not sufficient for revocation-sensitive writes.

5. Insert or update the reservation and audit record. The database additionally rejects overlapping confirmed `tstzrange` intervals for the same station through a GiST exclusion constraint. Cancellation removes that reservation from active occupancy but preserves its history and operation mapping.

6. Commit before returning success. Map a known interval conflict to a stable 409 domain error. On deadlock/serialization failure, retry the complete transaction within NFR-05 bounds; on exhaustion or timeout return an explicit retryable error. Never retry a known business conflict as if it were a transient fault.

For a club schedule update, the writer takes the exclusive club guard, checks all affected confirmed future bookings and rejects a change that would invalidate them. No implicit cancellation occurs. If stations in several clubs are ever changed together, take all club guards before station guards in the universal order. Club-zone changes with future bookings require explicit migration policy or rejection; silent reinterpretation of instants is prohibited.

The overlap constraint protects reservations in one table. Maintenance in another table is safe only because all applicable writers take the same guards and revalidate; the constraint is not described as enforcing a cross-table invariant. Direct manual DB writes bypassing the protocol are outside authorised application operations and must be restricted operationally. An alternative future design could unify occupancy records or use serialisable transactions, but requires its own proof and tests.

5. Runtime flows and failure boundaries

Create: browser requests availability -> API returns advisory state -> visitor confirms with an operation key -> access check and transaction protocol -> commit -> response with reservation ID. If another writer wins the station interval, the loser gets a conflict and refreshes availability. After a lost response the browser refreshes its own reservation list and reconciles using the same key rather than inventing a second operation.

Cancel: browser requests cancellation -> server verifies owner under the shared protocol -> confirmed future reservation becomes cancelled -> commit -> refreshed own history. A second cancellation returns the cancelled state. Maintenance occupancy remains independent.

Maintenance: assigned administrator requests a block -> server checks club assignment and takes guards -> conflict check against confirmed bookings -> commit only if safe. A race with reservation creation has one order, so the second writer observes the first.

Read/report: apply access filters on the server -> query committed records -> return selected fields. A station's operational availability and report denominator need defined schedule/maintenance rules. A past reservation proves planned time, not attendance. Bulk reporting should use a defined snapshot when several queries contribute to one result.

The browser is untrusted; it has no database credentials and cannot commit directly. Only the API reaches PostgreSQL. TLS, secrets, session handling and resource access are implementation concerns to verify, not proven properties of these diagrams. No external payment, event feed or notification service can prevent core booking operation because none is mandatory in this scope.

6. Consistency and review checklist

- Roles and informational-calendar semantics match ECR-1 and the current README.

- Components use FastAPI, PostgreSQL and REST proposed in ECR-6.

- BR-04/09/10 map to one common transaction coordinator.

- BR-06/11 map to durable operation identity and network-outcome reconciliation.

- NFR-02 maps to resource/club checks in the API and domain operations.

- Sources and exports are provided for both required views; ER details remain in ECR-12.

Required human review: Mansur for client/API boundaries and failure states; Nurali for administrative writes, assignments and reporting. Alibi confirmed his teammates' agreement with the prepared architecture in the working chat on October 23, 2026. The diagrams do not replace Mansur's separate ECR-11 use-case diagram.

7.1 Use Case Diagram notes

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/diagrams/README.md

Use Case Diagram - ECR-11

- Editable source: `use-case-diagram.puml`

- Readable export: `use-case-diagram.svg`

- Related issue: https://nerybr.youtrack.cloud/issue/ECR-11

The diagram covers the four roles defined in the requirements: Guest, Registered Visitor, Club Administrator, and System Administrator. It includes browsing, availability, booking, cancellation, events, assigned-club administration, resource and schedule management, reporting, club management, and administrator assignment.

The `<<include>>` relationships identify required sub-behaviour of reservation creation. Independent administrator goals have direct actor associations; they are not compulsory substeps of a generic management operation. The `<<extend>>` relationship shows the exceptional conflict flow when a selected station becomes unavailable before confirmation.

Access boundaries are stated explicitly: visitors access only their own reservations, and club administrators manage only assigned clubs.

Review checklist

Recorded in source: All planned roles are shown.

Recorded in source: Booking, cancellation, events, and resource-management scenarios are shown.

Recorded in source: Relationships are aligned with the role/access matrix described in the requirements.

Recorded in source: Editable source is included.

Recorded in source: Readable SVG export is included.

Recorded in source: Mansur checked alignment with the interface scenarios.

Recorded in source: Another team member reviewed the diagram. See the review record below.

Publication and review status

The initial version was published through PR #4. The applicable revision has a recorded review: Alibi-stack's approval of PR #7 (commit `d7bae8ea9931bbdb8e4caec01c243f33c8cea493`). See the October 24 review and agreement record (new package record; see chapter 11) for the distinction between GitHub approval, reported team agreement and outstanding execution checks.

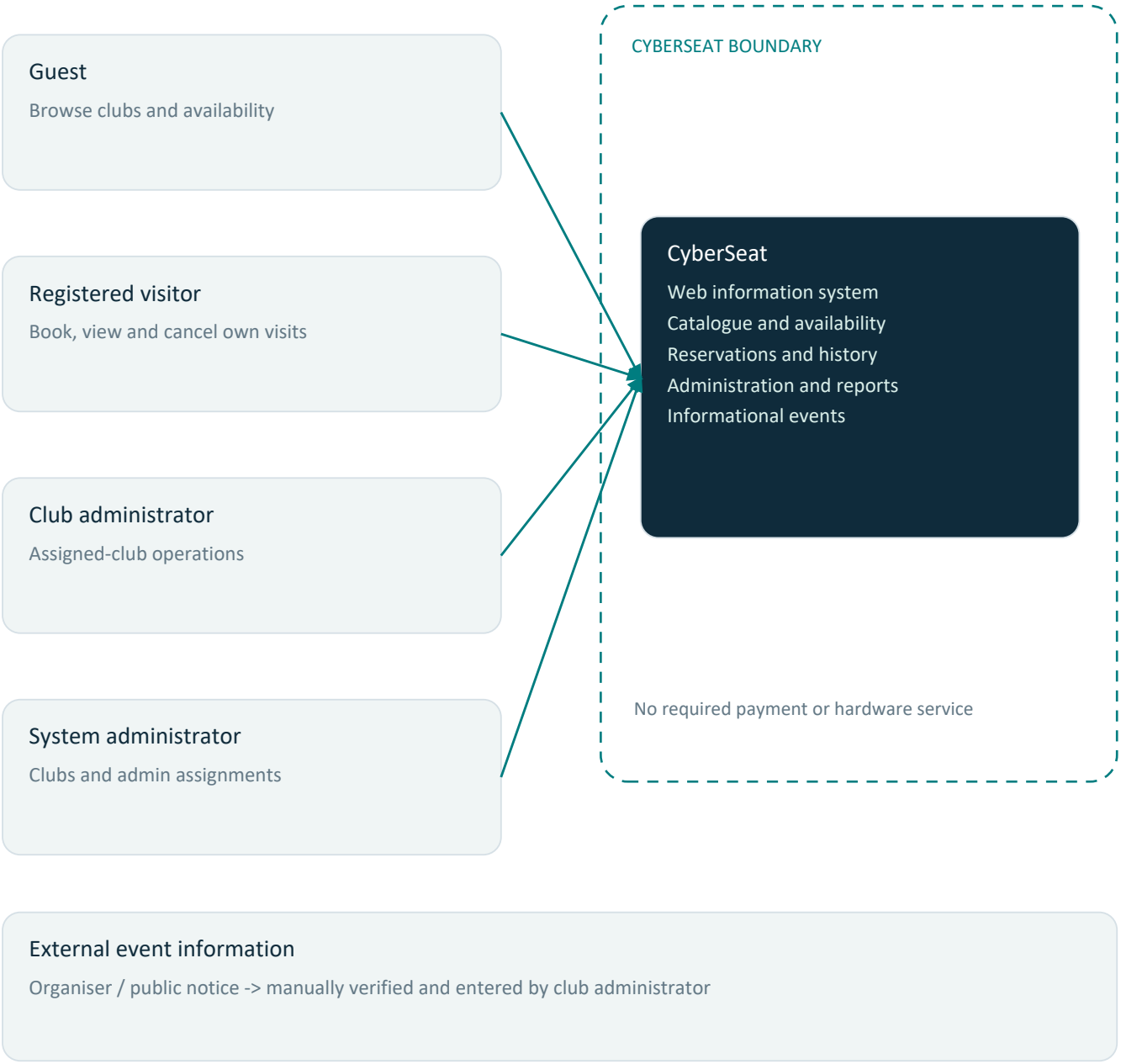
Reproduce the export

Use PlantUML with its bundled Smetana layout engine (selected in the source):

```
java -jar plantuml.jar -charset UTF-8 -tsvg docs/diagrams/use-case-diagram.puml
```

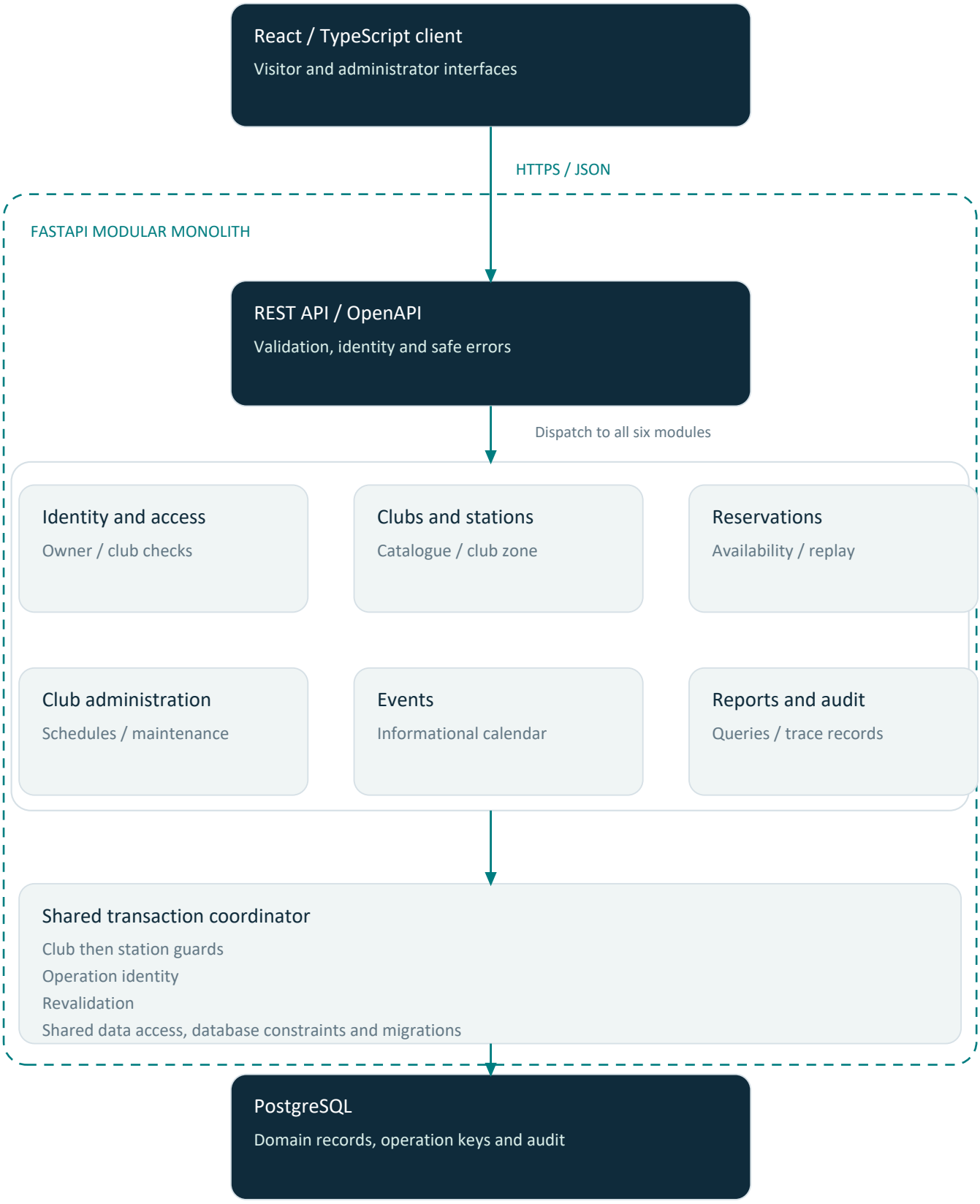
The SVG is generated from the `.puml` file. Do not manually redraw connectors in the export. Check that the conditional extension goes from Resolve booking conflict to Create reservation and that all six independent club-administrator goals remain connected to the actor.

Figure: System Context Diagram



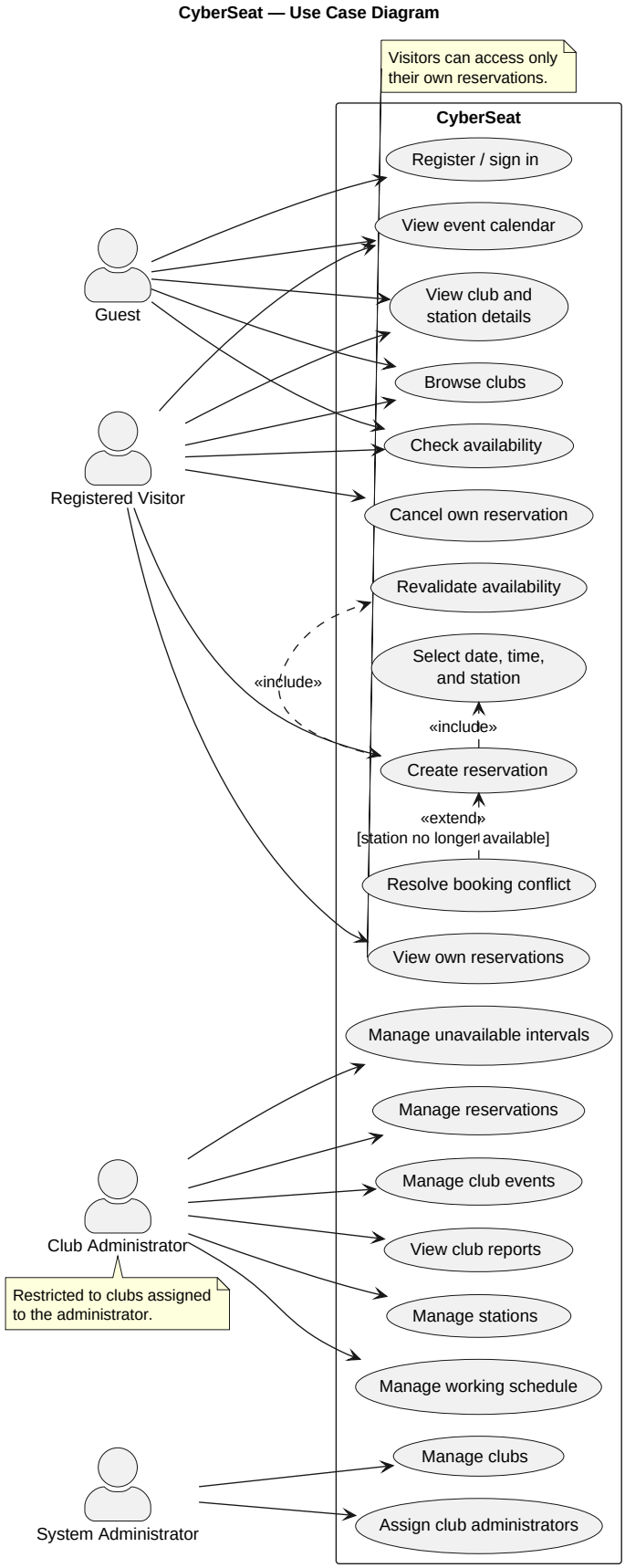
Design view from the pinned source baseline. Editable source and SVG export: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/diagrams/system-context.svg

Figure: Component Diagram



Design view from the pinned source baseline. Editable source and SVG export:
https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/diagrams/components.svg

Figure: Use Case Diagram



Design view from the pinned source baseline. Editable source and SVG export: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/diagrams/use-case-diagram.svg

8. UX prototype and scope

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/README.md

Version: 2.0

Date: October 24, 2026

Related issue: https://nerybr.youtrack.cloud/issue/ECR-9

Files

- `prototype.html` - interactive prototype entry point; open this file in a modern browser.

- `prototype.css` - graphite, lime and violet visual system, responsive layouts and reduced-motion support.

- `prototype.js` - page navigation, dialogs, filters and simulated response states.

- `prototype-model.js` - disposable in-memory example data and booking/maintenance simulation.

- `assets/gaming-lounge.png` - generated venue atmosphere art; see asset provenance.

- `cyberseat-ux-prototype.png` - preserved **version 1.1** static screen collection. It does not show the redesigned v2 interface.

- `tests/` - checks of the local simulation and DOM transitions, with their limitations.

Covered scenarios

1. club catalogue and filtering;

2. date, time, and single-station selection with nearby-seat information;

3. confirmation with server-side availability revalidation and conflict recovery;

4. personal account, history, cancellation, and cancellation rules;

5. event and tournament calendar;

6. administrative schedule and unavailable intervals;

7. administrative utilisation and cancellation reports;

8. loading, empty results, validation, connection-loss, server-error, and success states;

9. a mobile catalogue and booking entry variant.

Statuses use both text and colour. The booking path is consistent with the ECR-1 scenarios and explicitly avoids claiming that a working frontend has been implemented.

Primary prototype walkthrough

```
Catalogue -> filters -> station selection -> one station -> confirmation -> availability revalidation -> confirmed booking or conflict recovery -> personal account -> cancellation
```

Review checklist

Recorded in source: Key visitor screens included.

Recorded in source: Personal account and cancellation included.

Recorded in source: Event calendar included.

Recorded in source: Administrative schedule and reports included.

Recorded in source: Mobile variant included.

Recorded in source: Loading, empty, validation, server, and connection states included.

Recorded in source: Status is represented with text and colour.

Recorded in source: Prototype is aligned with the booking scenarios.

Recorded in source: Prototype reviewed by another team member. See the review record below.

Publication and review status

The initial version was published through PR #4. The applicable revision has a recorded review: NeryBR's approval of PR #9 (commit `723579d61ac025b50f59ff64cb0086cb03698a0d`). See the October 24 review and agreement record (new package record; see chapter 11) for the distinction between GitHub approval, reported team agreement and outstanding execution checks.

Revision 1.1 alignment

Screens 10-12 add the normal confirmation, pending/success, cancellation confirmation/history, pre-submission offline, uncertain response and maintenance-conflict states. Cancellation is permitted before the start time; no fee/refund policy is invented. All booking and event times identify the club zone. Each station has a textual status. A single reservation refers to one station, matching BR-03; adjacency is information, not an unapproved atomic group-booking API. Report figures are illustrative and identify the period and denominator.

The original version was a static set of screen designs. Version 2 replaces the primary HTML with a local interactive simulation, retaining the baseline scenarios described above. Future real API tests and user usability sessions are not claimed as completed.

Revision 2.0 - interactive visual redesign

The revision keeps Mansur's scenario baseline and initial artwork/export history. Alibi prepared the redesign on October 24, 2026 at the user's request. It does not imply that Mansur reviewed or authored the new code.

- A persistent desktop sidebar and compact mobile navigation lead to Discover, My bookings, Events, Club schedule, Insights and Interface states.

- The catalogue has a photographic venue hero, club cards and working local filters for name/neighbourhood, PC/console type, date, start and duration.

- The floor plan selects exactly one station. Reserved and maintenance stations are disabled and have textual statuses. The review step shows the club, station, interval, explicit time zone, estimated price and cancellation rule.

- Simulated confirmation is pending before success. Normal, stale-selection conflict, offline-before-send, uncertain response after saving, server-failure-before-save and session-expiry paths are available through Interface states or collapsed prototype controls.

- Unknown outcomes reconcile using the same local operation. Cancellation requires confirmation, retains history and releases only that booking's interval. An example maintenance block cannot overwrite a confirmed booking.

- Events remain club-published information; no event registration, payment, external-event curation or atomic group booking is introduced.

- Reports retain illustrative values and explicit period/denominator explanations. The period summary and example-week chart are separate fixtures.

- Semantic labels, visible focus, native modal dialogs, text statuses, a skip link and reduced-motion styles support accessible review. This is not a WCAG conformance claim.

Open and explore

Open `prototype.html` directly from this directory in a current browser. Keep its neighbouring CSS, JavaScript and `assets/` files in place. No installation or application server is required for this standalone design artifact. On GitHub, the HTML file page displays source; download the repository to run it locally.

Start with **Cyber Arena -> Choose a spot -> P02 -> Review session -> Confirm reservation -> My bookings -> Cancel booking**. Explore **Interface states** for recovery paths. Under **Club schedule**, blocking P04 at 18:00 conflicts with the initial demo booking; P02 is initially free.

The demo clock is fixed at **October 24, 2026, 17:00, Asia/Almaty (UTC+05:00)**. Default visits start at 18:00. Example club hours are 08:00-24:00 and the illustrative minimum is one hour; actual club configuration still requires confirmation. These fixtures do not change the requirements' general model, including overnight schedules.

All bookings, role views and responses are simulated in memory. There are no network requests, real authentication, database, payments or persistent data. Reloading resets the demo. The local model is not a security or concurrency guarantee for the future server.

Recorded validation and review

On October 24: JavaScript syntax checks passed; 12 local-model checks and 11 DOM integration checks passed. They cover one-station selection, overlap/maintenance behaviour in the simulation, cancellation/history, stable-operation replay, uncertain-response recovery, filters, navigation and error states. Native browser dialog behaviour, responsive rendering and human usability review remain unverified by these checks. See test instructions and limits.

The browser automation environment blocked opening a local `file:` URL. That action was not bypassed. The v2 interface has not been visually inspected in a real browser in this session, and no new v2 screenshot is presented as verified evidence. The preserved PNG remains labelled v1.1.

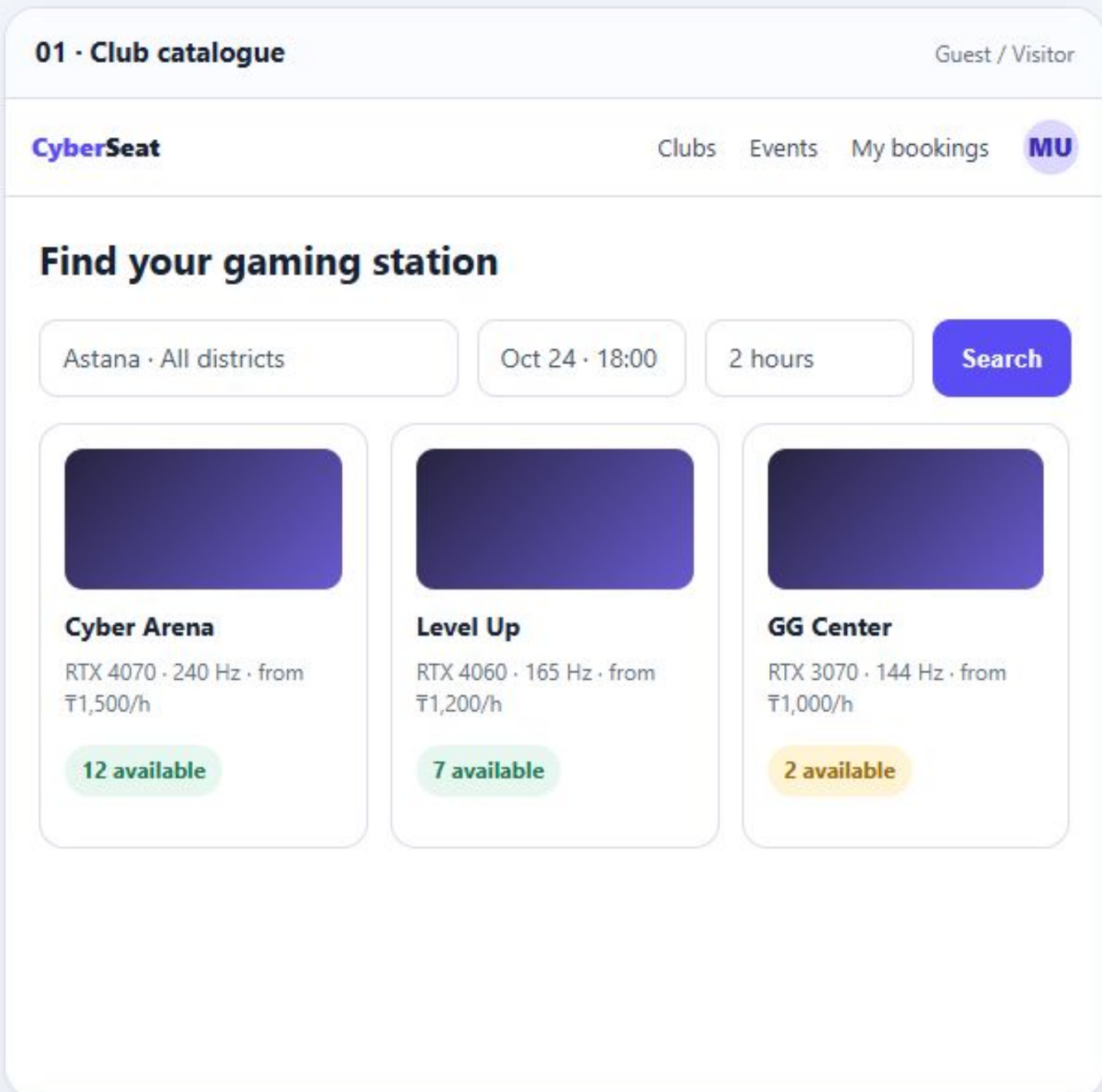
Independent review of version 2.0 is recorded in NeryBR's approval of PR #9 (commit `723579d61ac025b50f59ff64cb0086cb03698a0d`). It was submitted after merge. This design approval does not establish real-browser visual validation or participant usability results; those limitations above remain in force.

Scope note from the personal contribution statement

The September 30 contribution statement includes both club tournaments and major external esports events. The current approved scenarios and event model define club-published informational events. External-event curation, organiser verification and a model that does not require club ownership need a separate team/supervisor scope decision. This revision keeps the approved club-event baseline; it neither claims the broader calendar is complete nor introduces registration or payments. The personal statement itself is not published because it contains unnecessary personal information.

UX plate 01: historical screen design

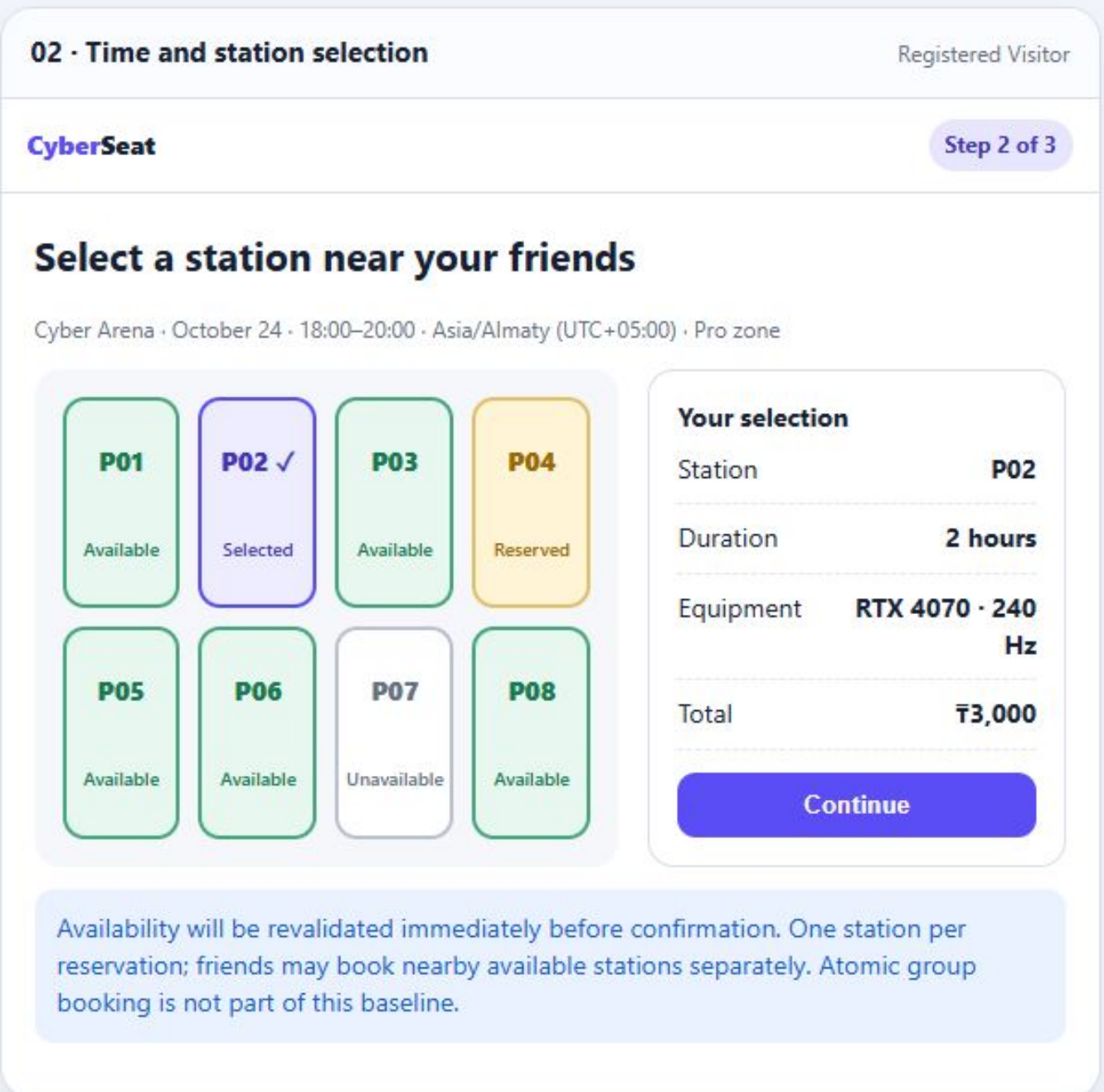
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 02: historical screen design

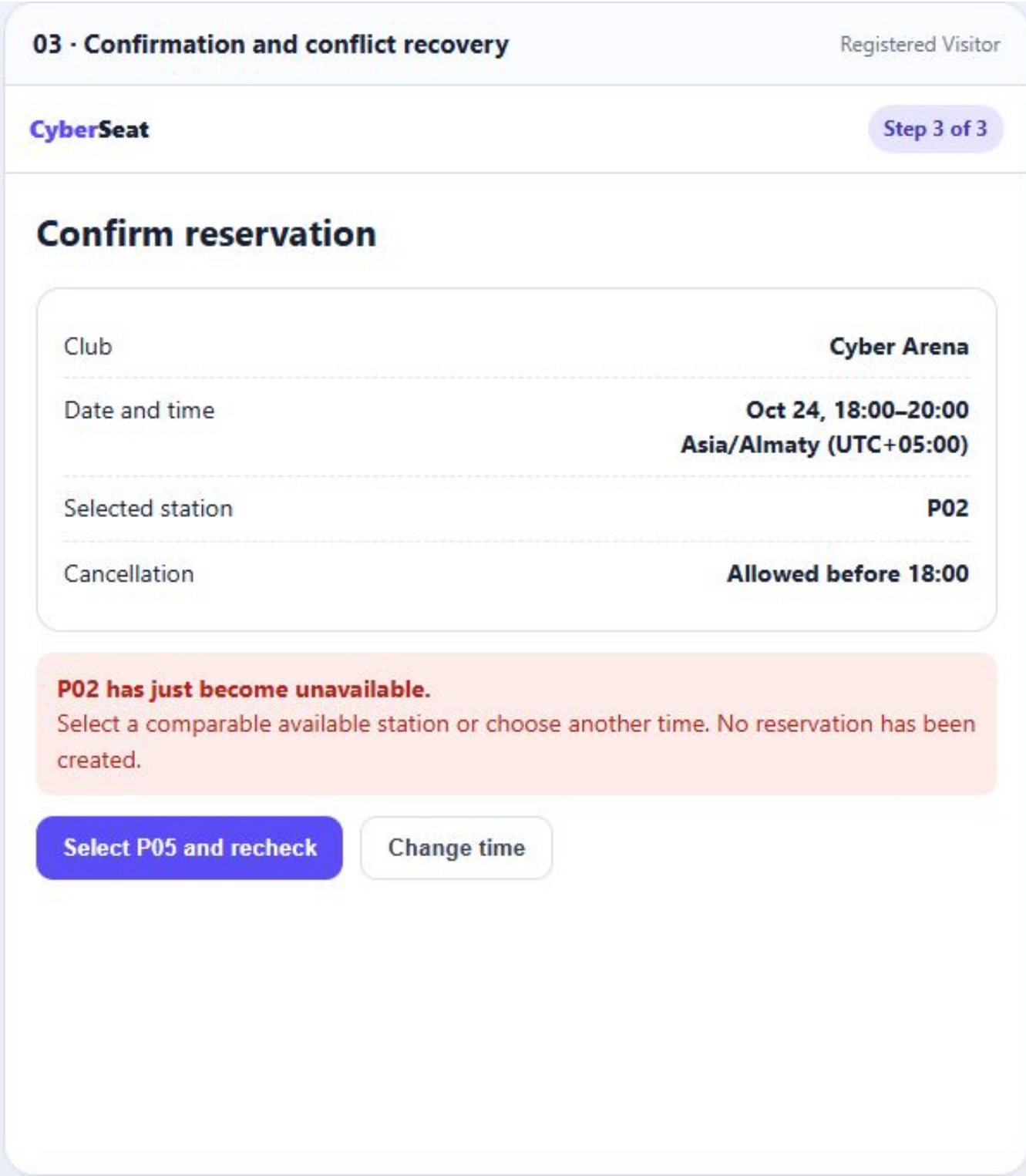
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 03: historical screen design

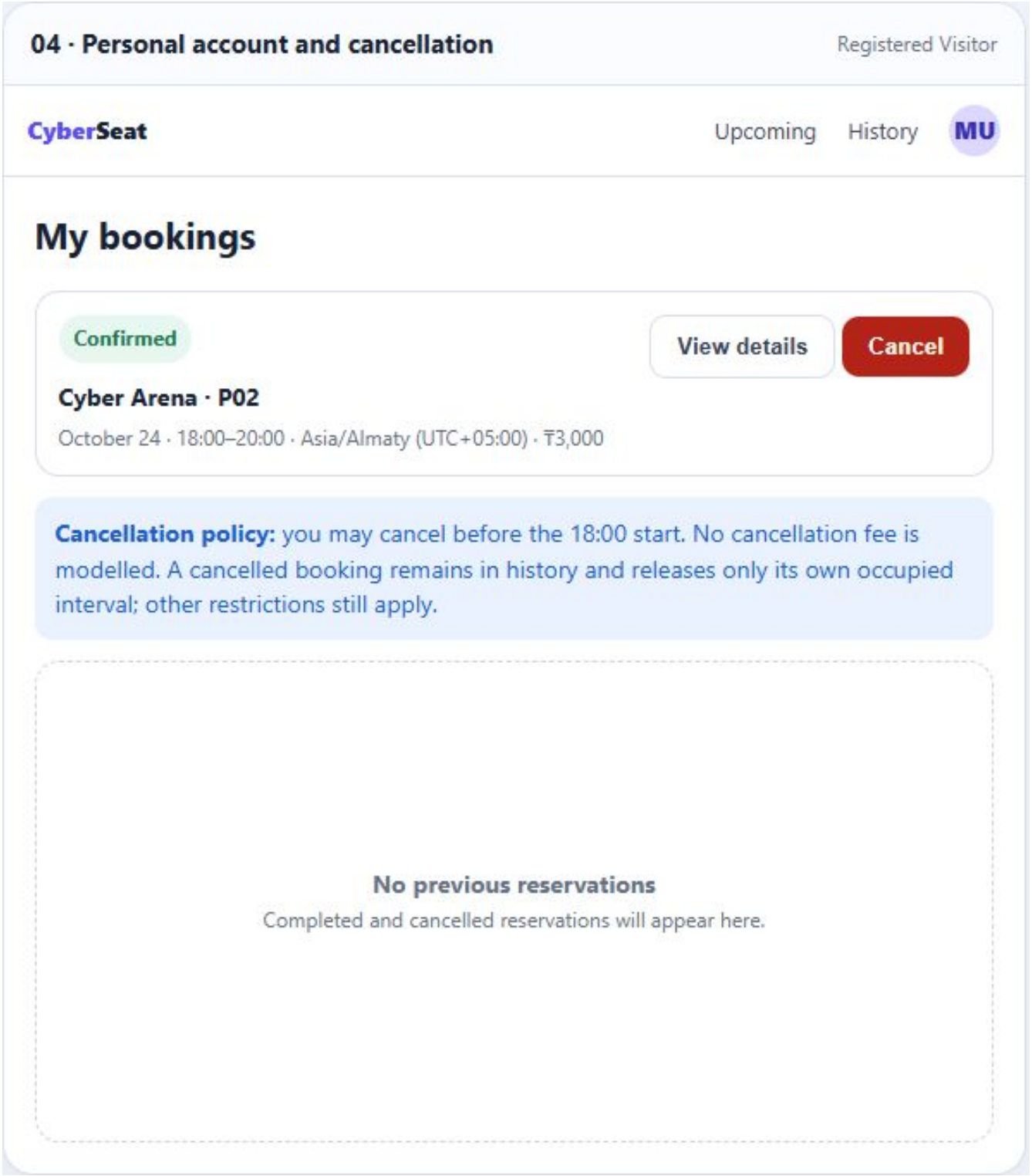
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 04: historical screen design

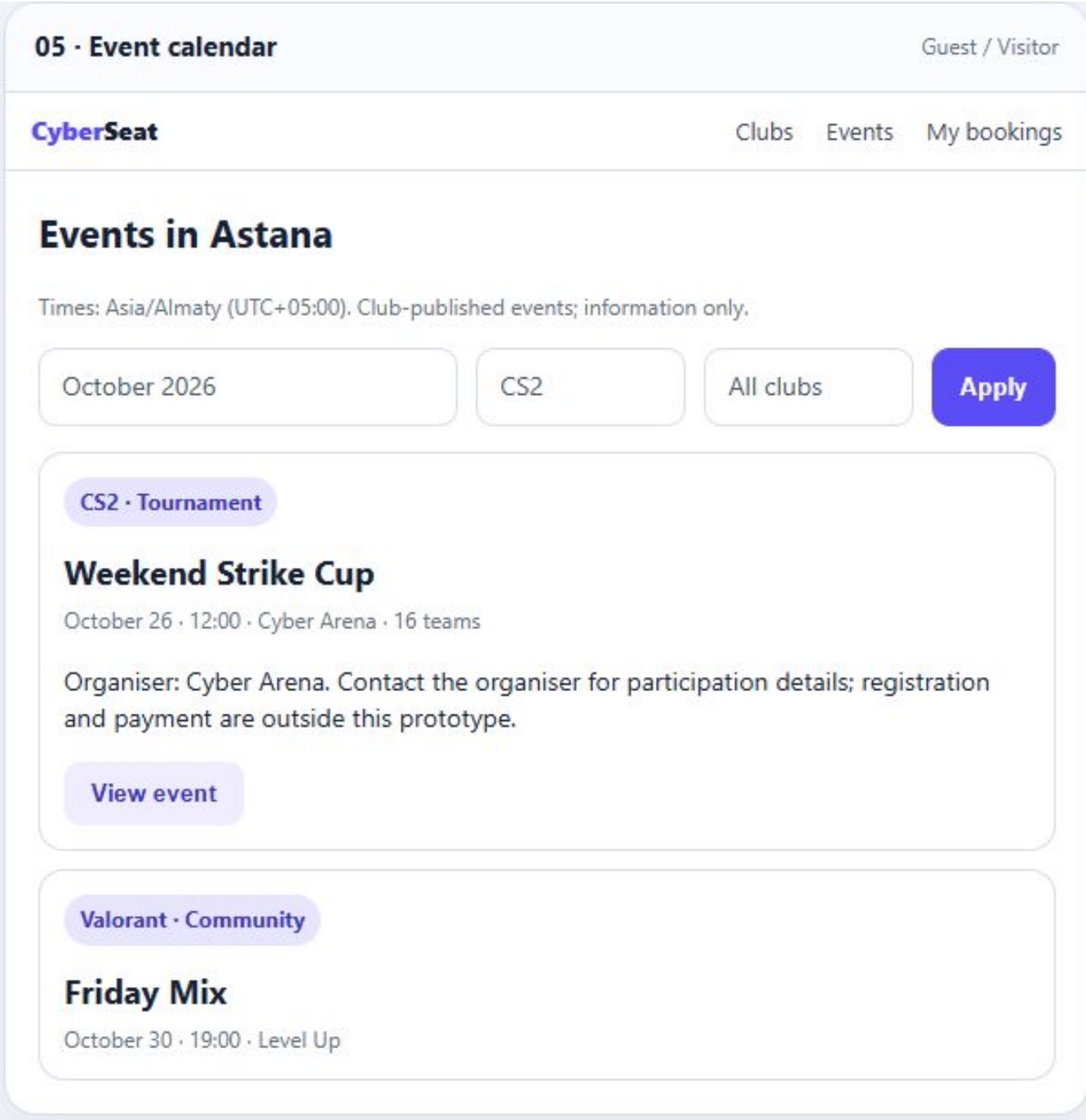
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 05: historical screen design

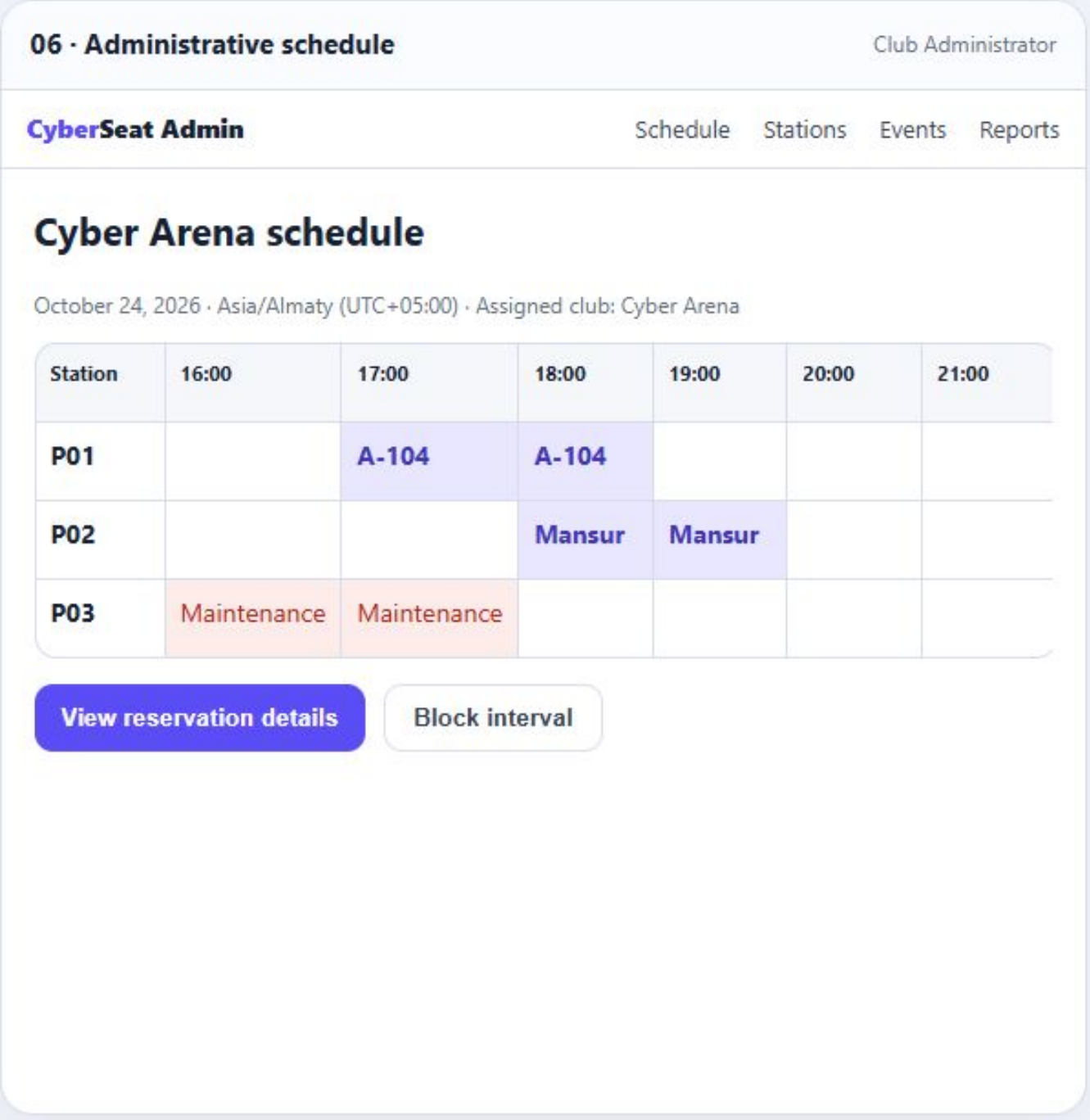
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 06: historical screen design

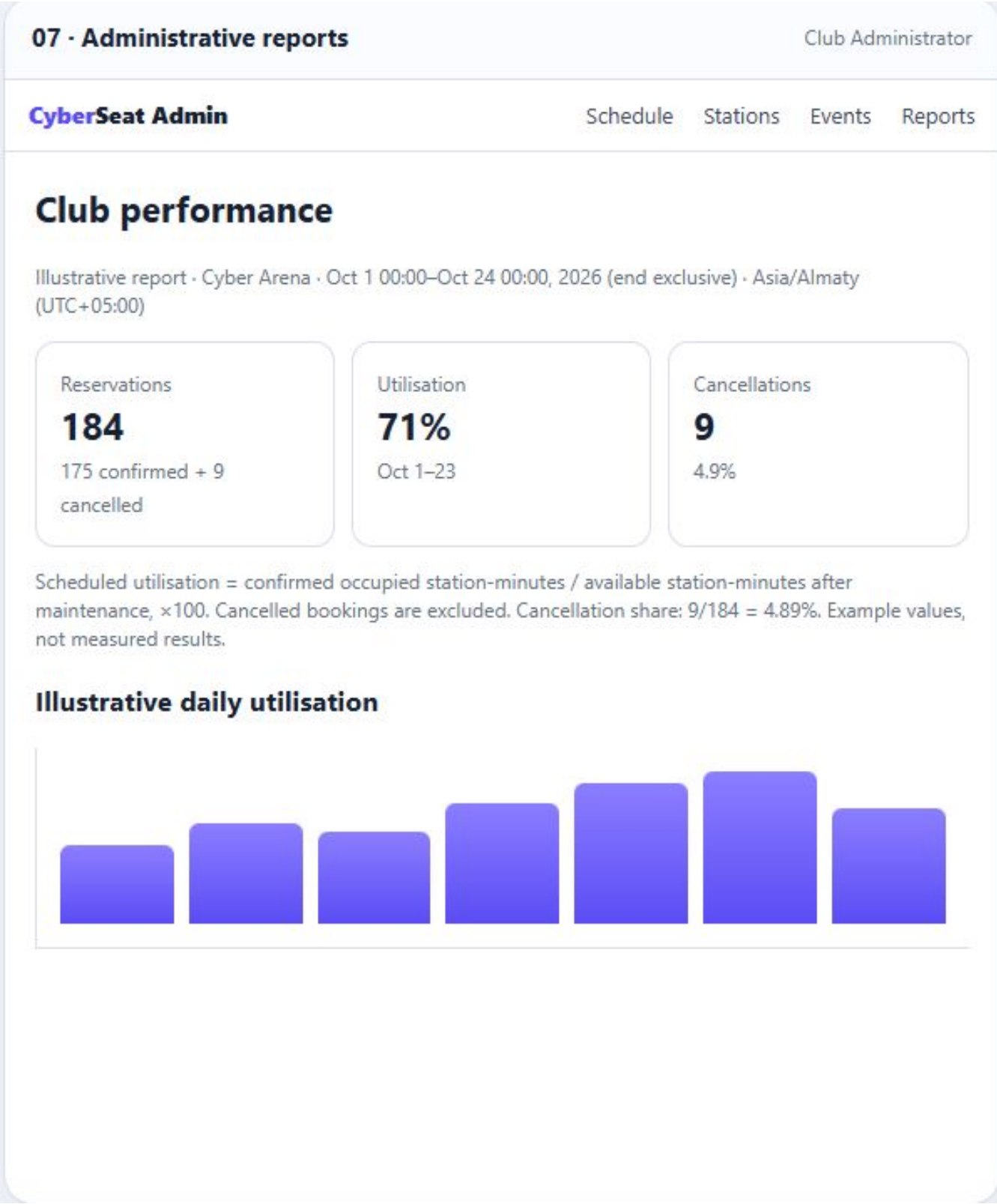
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 07: historical screen design

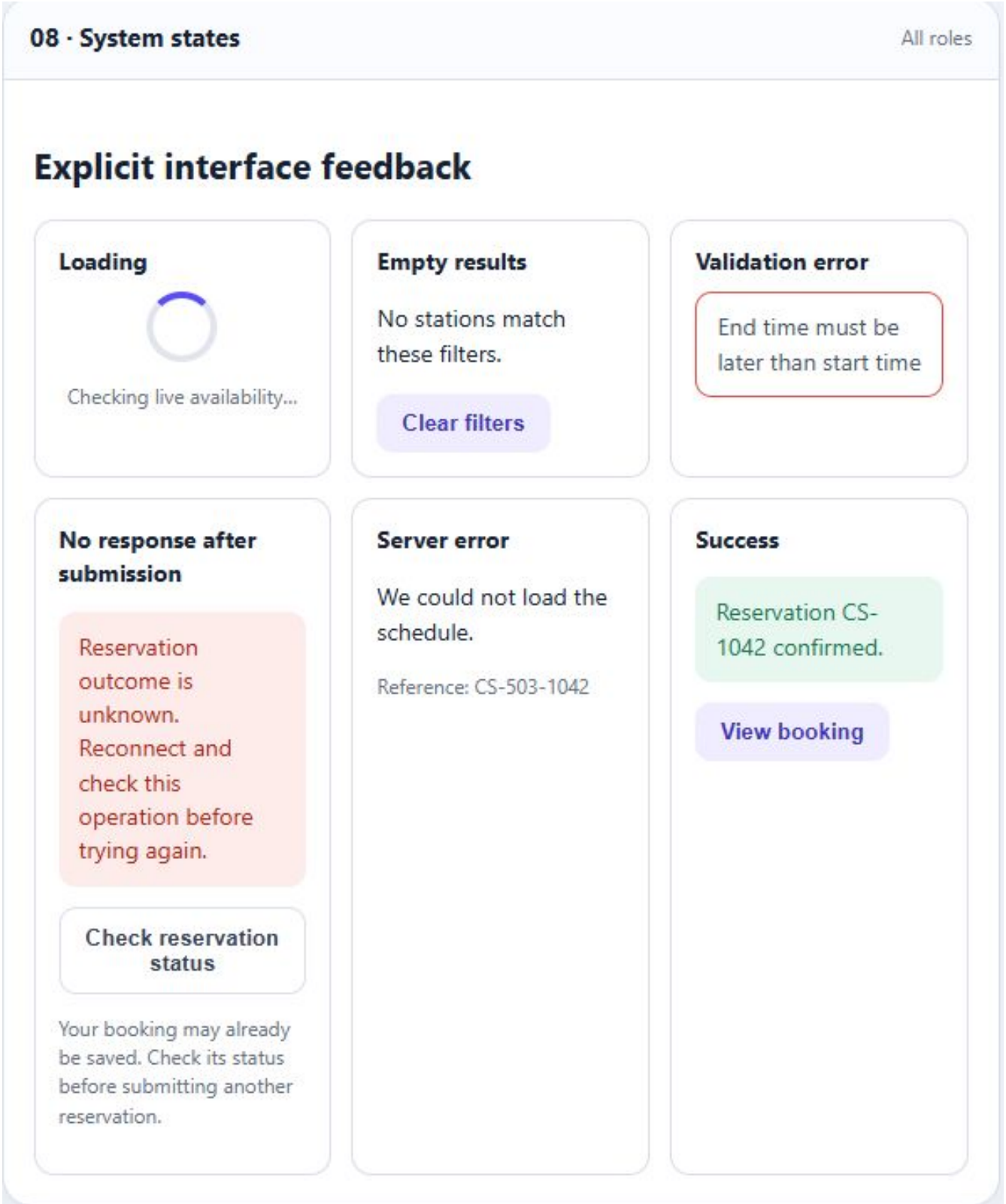
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 08: historical screen design

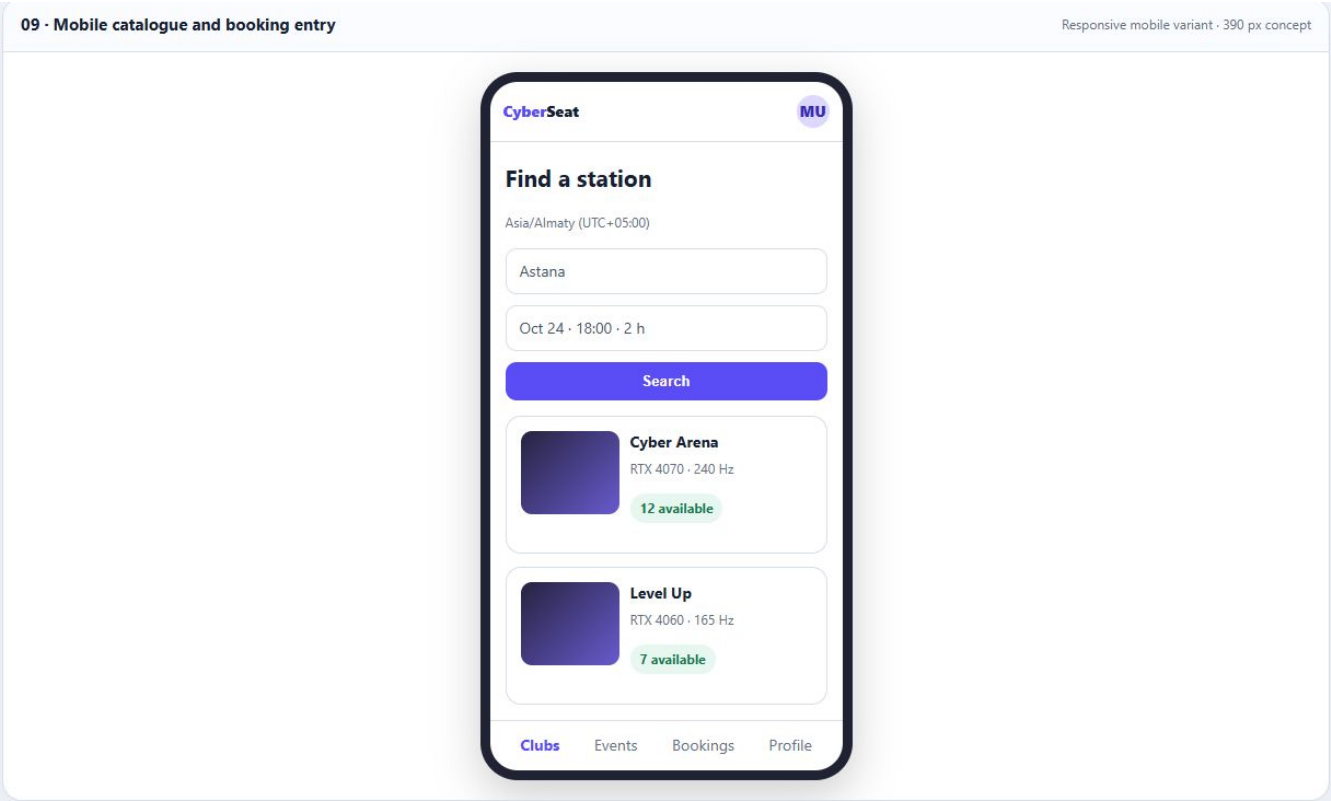
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 09: historical screen design

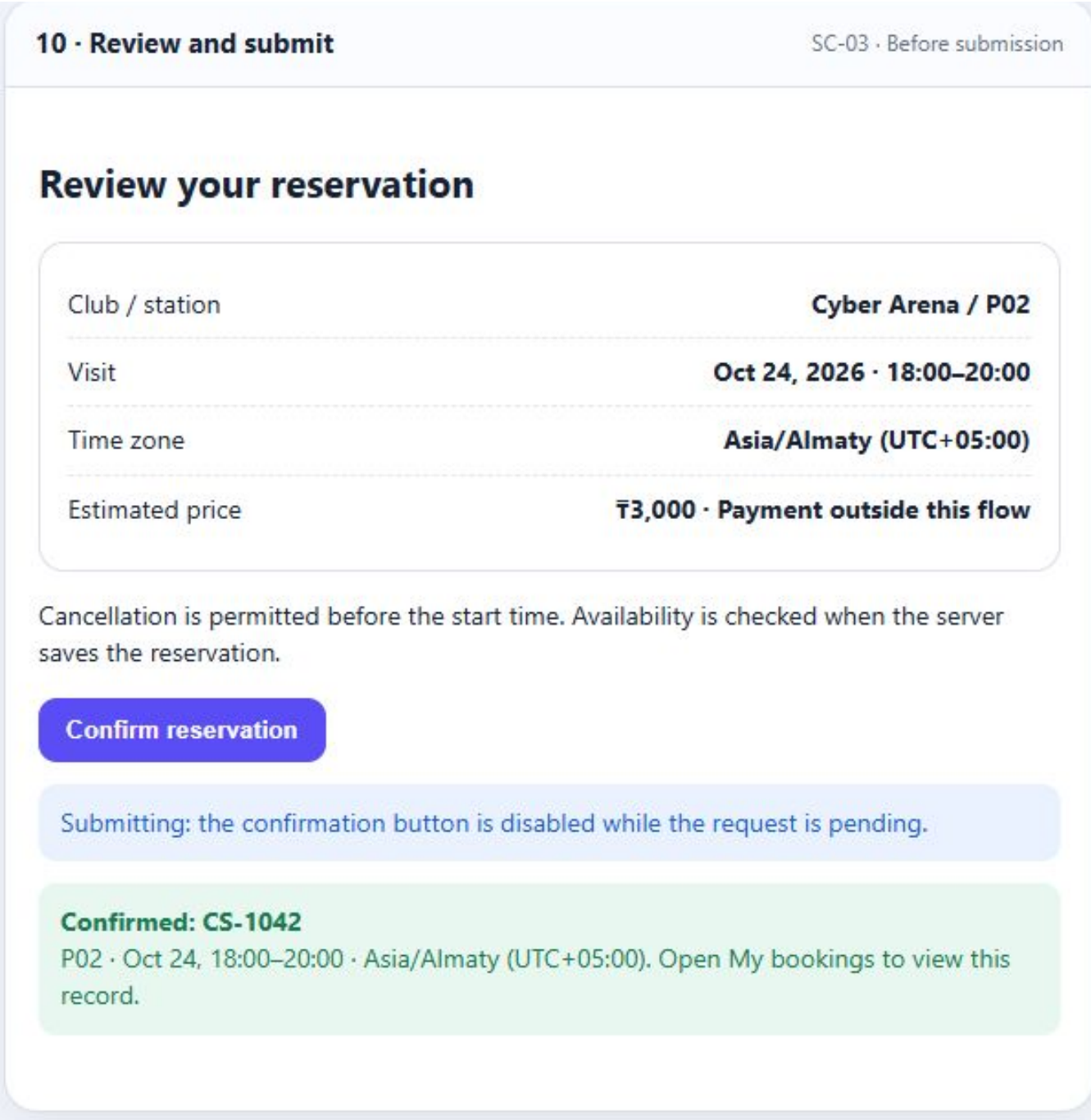
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 10: historical screen design

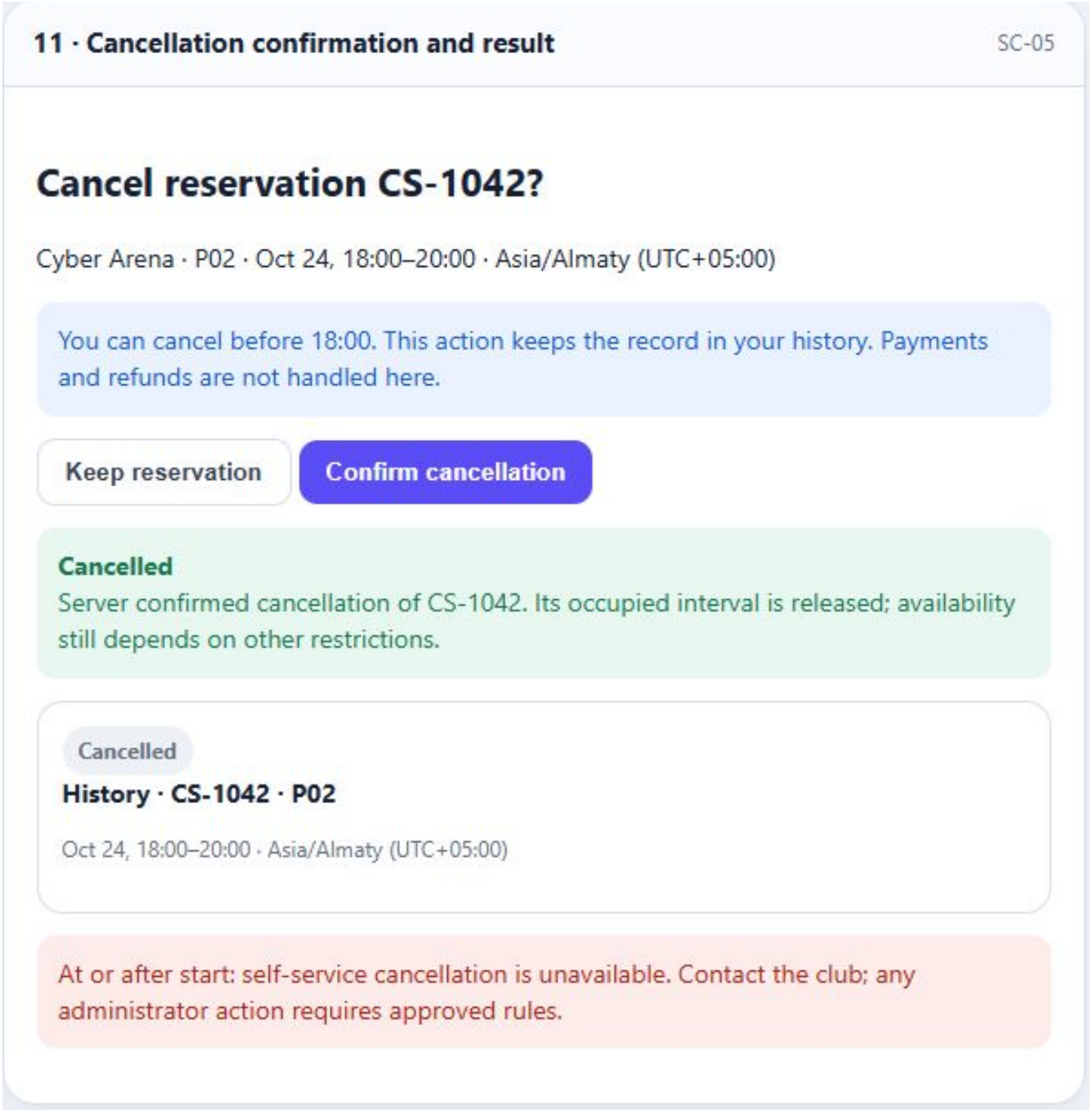
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 11: historical screen design

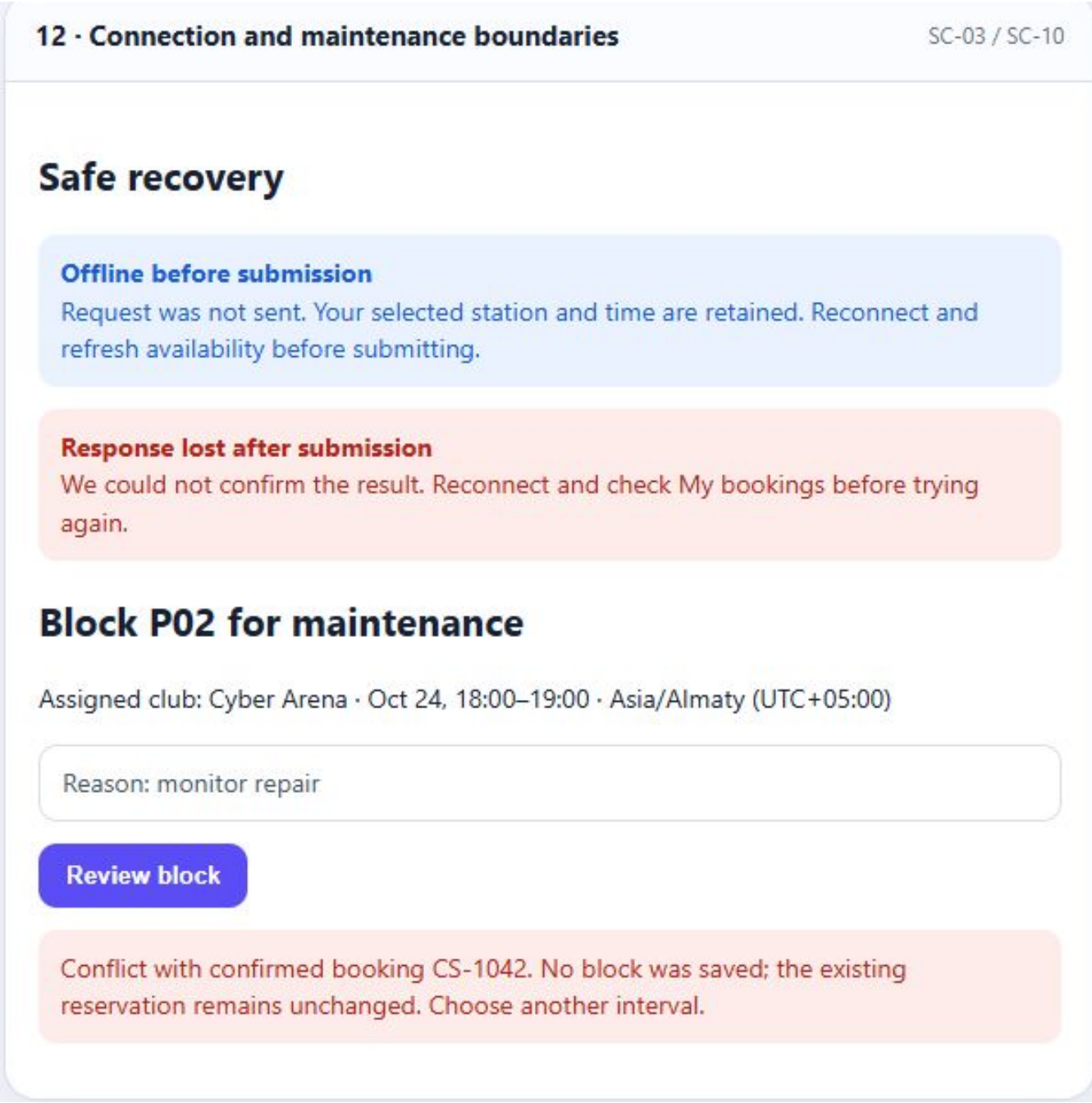
Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

UX plate 12: historical screen design

Preserved static export v1.1, October 23, 2026. This is not a screenshot of v2, a live service, or an executed usability study.



Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/ux/cyberseat-ux-prototype.png

9. Verification and evaluation plan

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/testing/verification-and-evaluation-plan.md

Version 1.0 · October 23, 2026 · ECR-13 · Document coordinator: Akhmetov Nurali

Status: planned; no application tests or user studies have been executed under this plan. Numerical examples are expected values, not observed results. This plan has NeryBR's approval of PR #6 (commit `eee6c80b0134a252eb4a1f862df9701e30f378d7`); see the October 24 review and agreement record (new package record; see chapter 11). Approval concerns the plan, not execution of its tests.

1. Submission scope and objectives

This document addresses university item 9, Verification and evaluation plan, for the first analysis/design/planning submission. It defines future checks and measurement procedures, not final results from a finished product. Required submission: **October 24, 2026 before 18:00 in the university-specified time zone**. October 22, 2026 21:00 is the team's internal planning target, not a separate university deadline.

Verification asks whether the implementation will meet its specified behaviour; evaluation asks whether representative users can complete the intended tasks and how the proposed workflow compares with a controlled manual workflow. A working frontend/backend and executed final tests are not prerequisites for this October document.

The planned scope covers booking integration/concurrency, permissions, administrative safeguards, reports, UX/responsiveness, persistence, reproducibility and performance. Payments/revenue, loyalty, hardware control, event registration and deferred post-start cancellation rules are excluded. Informational events must not automatically occupy stations.

2. Inputs, baseline and responsibilities

The plan is based on the repository after requirements merge 2c1c419. At every future execution record the actual implementation SHA, requirement revision and fixture version; changed inputs require an impact review before comparing results.

User scenarios

Baseline: ECR-1, v1.0

Use: SC-01-SC-14 and shared interface states

Booking and NFR specification

Baseline: ECR-7, v1.1

Use: BR-01-16; NFR-01-10; invariants I1-I6

Role matrix

Baseline: ECR-8, v1.0

Use: RBAC-01-09; four roles and assigned-club boundaries

Administrative specification

Baseline: ECR-8, v1.0

Use: ADM-01-15; protected changes and report definitions

Architecture

Baseline: ECR-10, v1.1

Use: React/TypeScript, FastAPI REST API, PostgreSQL, shared transaction coordinator

UX coverage, editable prototype, screen export

Baseline: ECR-9, v1.0

Use: Screen walkthrough and later user-task basis; static screens are not a working reservation service

Aibekuly Alibi

Future work: INT booking checks, CON races, persistence, PostgreSQL fixtures and k6 load execution

Cross-review: Nurali reviews reservation-safe administrative races; Mansur checks API/UI failure handling

Akhmetov Nurali

Future work: SEC permission checks, ADM integration checks, REP hand-calculated oracles, audit; coordinates this plan

Cross-review: Alibi checks data/snapshot assumptions; Mansur checks understandable report/error presentation

Mansur Urynbassar

Future work: UX task protocols, responsive/keyboard checks, browser integration and formative user sessions

Cross-review: Nurali checks role-specific navigation; Alibi checks booking-state consistency

Owners coordinate boundary cases together; assigning an owner does not record completed work or approval. Tools are planned: Pytest with real PostgreSQL for integration/races, Vitest/Testing Library for client components, Playwright for browser flows, and k6 for load. Pin actual versions and executable commands when a runnable build exists; no unverified setup commands are presented as working instructions.

3. Environments and synthetic data

Dataset/environment	Exact planned conditions
F-CORE	Resettable two-club fixture: C-A and C-B, PC/console stations, active/inactive resources, weekly hours plus dated closure, overnight hours, CONFIRMED/CANCELLED bookings, maintenance and DRAFT/PUBLISHED/CANCELLED events. Use a documented positive club minimum, e.g. 30 minutes for this fixture only.
F-AUTH	Guest, visitors V-A/V-B, administrator of C-A only, administrator assigned to both clubs, revoked administrator, and superadministrator without a club assignment. Give each protected resource an owning club. Also test an explicit club assignment added to/revoked from the superadministrator.
F-TIME	Freeze the server clock at a recorded instant; place create/cancel boundaries relative to it. Default club zone Asia/Almaty; add UTC round trips and an IANA zone with an offset transition, with pinned time-zone data. Test unresolved ambiguous/nonexistent local times. Never hard-code future bookings that become past as the calendar advances.
F-REPORT	REP-01-10 below. Hand-calculated expected records are versioned independently of production report code. Historical capacity revisions are mandatory inputs. Corrupt/missing-history examples use isolated test fixtures, not production writes.

Dataset/environment	Exact planned conditions
F-LOAD	10 clubs with 50 stations each (40 PC/10 console); all active and open 24 hours, minimum duration 30 minutes, no maintenance in the load seed (maintenance is exercised separately). Seed 20261023; a future 30-day UTC horizon begins at the next UTC midnight after the frozen test clock. Each station has 200 one-hour reservations spaced three hours apart from horizon start: first 160 CONFIRMED, remaining 40 CANCELLED, for exactly 100,000 records (80,000/20,000). Distribute ownership deterministically across 50 synthetic visitors. Use free gaps for valid creates and owned future confirmed rows for cancels; never count fixture exhaustion as normal business conflict. Restore this seed for each run; record counts at measurement start/end, since warmup and load mutate records. No real personal data.
E-INTEGRATION	Real PostgreSQL with applied migrations; independent connections and at least two independent API workers/processes for races. The load environment remains one API service, as NFR-04 specifies. In-process mocks/SQLite cannot establish the PostgreSQL concurrency guarantees.
E-LOAD	Dedicated 4-vCPU/8-GB host; local Docker network; one API service plus PostgreSQL, no unrelated workload. Record OS, CPU model, DB/API worker counts, resource limits, connection pools, dependency versions and k6 host placement. Keep these constant in baseline and target runs.
E-UI	Record browser/OS versions. Planned Chromium desktop viewport 1440x900; mobile 360x800 and 390x844; tablet 768x1024. Also repeat core flows in current stable Firefox and WebKit at 360px and desktop width, pinning actual versions at execution. Real-device checks are supplementary and explicitly labelled.

All intervals use [start,end); adjacent endpoints are allowed. Reservation statuses remain CONFIRMED/CANCELLED; past end is not an attendance status. Reset synthetic databases between independent runs, but preserve durable state deliberately during restart/replay checks. Use deterministic seeds and independent request/operation IDs; capture the seed with any failure.

4. Integration and administrative checks

All cases are **NOT RUN**. Every table row expands into explicit test parameters from the linked requirements. Must cases must pass all listed variants; Should features may be deferred with a recorded reason.

INT-01

Owner: Alibi

Procedure and expected outcome: Search F-CORE: free PC, booked PC and blocked console give correct statuses. Submit missing fields, reversed/empty/past intervals, below-minimum duration, out-of-hours, inactive resource and mismatched club/station IDs: reject without writes.

Requirement trace: BR-01/02; SC-01/02

INT-02

Owner: Alibi

Procedure and expected outcome: Create on a valid free interval; assert committed owner comes from session, station/instants match, one stable ID is CONFIRMED and success follows commit.

Requirement trace: BR-03; SC-03

INT-03

Owner: Alibi

Procedure and expected outcome: Book adjacent intervals on one station and identical intervals on different stations: both succeed. Partial overlap, containment and equality on one station fail.

Requirement trace: BR-04/05; SC-02/03

INT-04

Owner: Alibi

Procedure and expected outcome: Retry identical operation key/body: same ID and one record; changed body conflicts. After cancellation, replay old key returns original cancelled record; a new logical creation uses a new key.

Requirement trace: BR-06; I4

INT-05

Owner: Alibi/Nurali

Procedure and expected outcome: List/view own history; change visitor ID, detail ID and filters to another owner: no disclosure. Cancelled history remains linked to original station.

Requirement trace: BR-07; RBAC-04; SC-04

INT-06

Owner: Alibi

Procedure and expected outcome: Cancel before start, repeat cancellation and test exact start/after start. Only permitted cancellation changes status, preserves history and releases its own occupancy; maintenance remains effective.

Requirement trace: BR-08; I6; SC-05

INT-07

Owner: Alibi/Mansur

Procedure and expected outcome: Drop response after commit; refresh own list/reconcile and retry same key: original ID, no duplicate. Drop before submission: no success. Missing operation result alone does not prove an in-flight request failed.

Requirement trace: BR-11; SC-03; shared states

INT-08

Owner: Alibi/Mansur

Procedure and expected outcome: Another client takes a displayed slot; first gets conflict and refreshed choices. Expire session before confirmation: no commit; sign-in requires revalidation. Alternatives are not represented as reserved.

Requirement trace: BR-12/16; SC-02/03

INT-09

Owner: Alibi

Procedure and expected outcome: F-TIME: offset/UTC/club-zone round trips, overnight hours, exact schedule boundaries and server start-time boundary. Reject unresolved invalid local times; show consistent zone labels.

Requirement trace: BR-02/14; SC-02-06

INT-10

Owner: Nurali

Procedure and expected outcome: Validate club creation, active state, minimum duration, assignments and duplicate pairs; create/edit station labels/types. Duplicate label within one club fails, same label across clubs succeeds; resources cannot be reparented.

Requirement trace: ADM-01/02/03; SC-08/14

INT-11

Owner: Nurali/Alibi

Procedure and expected outcome: Change hours, dated exceptions, maintenance, club/station activation and zone around future/in-progress bookings. Conflicting whole mutation fails with no silent move/cancel/delete. Safe block removal preserves other restrictions/history. Increased minimum affects new requests only.

Requirement trace: BR-09/10; ADM-04/05/06/08; SC-08-10

INT-12

Owner: Nurali/Mansur

Procedure and expected outcome: Admin pre-start cancellation needs assignment/reason and is replay-safe; at/after start reject. Validate draft/event publication/cancellation: guests see published/cancelled records, never drafts; events never change station occupancy.

Requirement trace: ADM-09/10/11; SC-05/06/11/13

5. Concurrency protocol and cases

Owner: Alibi, with Nurali reviewing administrative/assignment races. Use real PostgreSQL and independent API requests, connections and workers. A synchronisation barrier releases contenders together; record actual arrival/start/commit times. Instrument lock acquisition or a test-only barrier when necessary to prove the intended overlap. Exercise both orderings in two-writer races through controlled scheduling, as well as free races. A sequentially processed run is not concurrency evidence.

Run **100 fresh repetitions of each named race variant**, as required by NFR-01; record winner/order and query committed DB state after every repetition. CON-01 has three separate variants, each with 100 runs. Restore fixtures/keys between repetitions. Infrastructure failures are recorded separately and replaced with valid repetitions; they do not count as passes and a persistently unstable environment makes the suite inconclusive.

CON-01

Concurrent stimulus and expected committed outcome: 20 different visitors, distinct operation keys, one free station: identical [14:00,15:00); partial-overlap starts 14:00 through 14:19 with 60-minute duration; containment uses ten [13:00,16:00) and ten [14:00,15:00) requests. All contenders in each variant mutually overlap. Exactly one creation succeeds per valid repetition; all others receive domain conflicts; no overlapping confirmed rows.

Trace: BR-04/05; NFR-01; I1

CON-02

Concurrent stimulus and expected committed outcome: Two same-owner requests with same key/body: same booking ID, exactly one row. Same key with different bodies: at most one logical payload accepted, other conflicts. Repeat for replay after a dropped response.

Trace: BR-06/11; I4

CON-03

Concurrent stimulus and expected committed outcome: Booking vs overlapping maintenance creation/update: one wins, other observes conflict; never committed booking plus conflicting block.

Trace: BR-09; ADM-06/08; I2

CON-04

Concurrent stimulus and expected committed outcome: Booking vs club closure, schedule shrink and station deactivation: each is a separate 100-run variant. Booking first means unsafe change fails; change first means invalid booking fails. No partial multi-station update.

Trace: BR-10; ADM-04/05/08; I3

CON-05

Concurrent stimulus and expected committed outcome: Cancel existing booking vs create in its interval: create may fail if ordered before cancellation, or succeed after it. Cancellation preserves old history. After cancellation returns, a fresh valid operation can book unless another restriction applies. Race new booking and block removal separately: no restriction is bypassed.

Trace: BR-08/09; NFR-01; I6

CON-06

Concurrent stimulus and expected committed outcome: Revoke assignment vs protected admin write: write either commits before revocation or fails after revocation. Every request begun after revocation commit is denied. Test stale sessions and direct resource requests.

Trace: RBAC-05; ADM-02; architecture authorisation guard

CON-07

Concurrent stimulus and expected committed outcome: Two admin changes based on one resource version: one succeeds, the stale one gets 409 and cannot overwrite it; cancellation retries remain harmless.

Trace: ADM-14; transaction/version contract

After each run independently inspect: no overlapping CONFIRMED intervals per station; no confirmed/maintenance overlap; valid schedule/state at the transaction's effective time; one mapping/booking per logical operation; unchanged rejected resources and preserved histories; authorised actor/scope; durable audit of success. Compare historical validity with the effective revision at commit, not a later configuration.

Baseline: run the same fixtures and logical requests sequentially under the same production protocol. It must already satisfy invariants; concurrency must not introduce violations. Do not benchmark against an intentionally unsafe implementation or claim that a finite suite proves absence of all races. Failure count threshold is zero; one integrity/access violation fails the affected Must suite.

6. Permissions and data handling

Owner: Nurali. For every protected operation in the role matrix run an allowed control, then guest, visitor, assigned/unassigned/revoked admin and unassigned superadministrator variants where applicable. Test direct HTTP requests and browser navigation; hidden buttons alone are not enforcement. Record the complete actor x operation x resource-scope matrix, not just one representative endpoint.

SEC-01

Procedure and expected outcome: Guest protected request 401; visitor admin route 403; forged role/owner ignored. Own-record ID/filter substitution gives 404 without foreign details.

Trace: RBAC-01/04; BR-07/08; NFR-02

SEC-02

Procedure and expected outcome: Admin C-A changes path/body/nested station, booking, event, block, list/report scope to C-B: denied without disclosure. Mixed-club aggregate scope is rejected; counts, audit reads and any implemented export obey scope.

Trace: RBAC-02/03/06; ADM-07; NFR-02

SEC-03

Procedure and expected outcome: Admin cannot grant privileges or edit system club identity/zone. Superadministrator without assignment manages clubs/assignments but cannot read operational reports; explicit assignment enables only its club. Revocation removes it.

Trace: RBAC-05/06/07; ADM-01/02

SEC-04

Procedure and expected outcome: Inspect successful/error response bodies, report output and scoped audit entries: only specified service fields; no unrelated visitor profile, credentials or tokens. Audit includes actor/action/resource/club/time/result/reason/correlation ID.

Trace: RBAC-08/09; BR-15; ADM-13

SEC-05

Procedure and expected outcome: In controlled deployment confirm HTTPS, secret exclusion from repository/config examples and credential-free logs/errors. Authentication-specific checks are added after mechanism is agreed; unresolved auth blocks full security acceptance.

Trace: NFR-09

Baseline/oracle: the allowed/denied cells in ECR-8, plus successful same-club/own-record controls. Threshold: every expected denial and allow outcome matches, zero unauthorised reads/writes and zero credential disclosure. A universally broken API that denies everything fails its positive controls.

7. Report verification with independent expected values

Owner: Nurali; Alibi reviews fixture/history design. Use one committed snapshot T, display club/zone/[from,to), T, generation time and definition version. Hand-calculate the fixture first and store expected values without importing production report functions. Baseline is this independent arithmetic, not the implementation's previous output.

Capacity A is open active station-minutes minus the union of effective maintenance; bookings are not removed from capacity. Occupied O is CONFIRMED scheduled time clipped to period/capacity. Free=A-O; utilisation=100*O/A, otherwise N/A when A=0. Count distinct CONFIRMED/CANCELLED IDs whose original visit intervals overlap the period; cancellation share is cancelled/total, otherwise N/A. Aggregate exact durations before two-decimal half-up formatting; club utilisation uses total O/A. Missing history or invalid occupancy must not be disguised as a valid zero/percentage.

REP-01

Controlled fixture and expected result: P 10:00-12:00; two active/open stations; S1 maintenance 10:00-10:30 and confirmed 10:30-11:30; S2 confirmed 11:00-12:00 and cancelled 10:00-11:00. A=210, O=120, free=90 min; utilisation=57.14%; confirmed=2, cancelled=1, total=3, cancellation share=33.33%.

Trace: ADM-12; NFR-10; SC-12

REP-02

Controlled fixture and expected result: One station open 10:00-14:00; booking 10:30-11:30; P 11:00-12:00. A=60, O=30, free=30; utilisation=50.00%, confirmed=1.

Trace: Period clipping/counts

REP-03

Controlled fixture and expected result: P 10:00-12:00; one open station, blocks 10:00-10:45 and 10:30-11:00; no bookings. Maintenance union=60, A=60, O=0, free=60; utilisation=0.00%; counts=0; cancellation share=N/A.

Trace: Union denominator; empty data

REP-04

Controlled fixture and expected result: All stations closed/inactive/blocked during P. A=O=free=0; utilisation=N/A with no-capacity explanation. Calculation failure is a separate error.

Trace: Zero denominator

REP-05

Controlled fixture and expected result: P 10:00-11:00: cancelled visits ending exactly at 10:00 or starting exactly at 11:00 do not count. A booking with positive overlap counts once regardless of creation/cancellation timestamp.

Trace: Half-open boundaries; visit-period counts

REP-06

Controlled fixture and expected result: Historical P had 120 active/open minutes; later deactivate station, change future hours or remove a future block. Historical A stays 120 under retained effective history. Missing required revision returns insufficient historical data.

Trace: Historical capacity

REP-07

Controlled fixture and expected result: Isolated corrupt fixture: overlapping confirmed bookings or booking intersecting maintenance/out-of-hours. Raise data-integrity error and withhold valid utilisation/free result.

Trace: Invalid-data detection

REP-08

Controlled fixture and expected result: Station S1 A=30/O=30; S2 A=90/O=0. Club A=120/O=30, utilisation=25.00%, not the unweighted 50%. Repeat disjoint periods: durations add under the same snapshot; spanning distinct counts are not blindly summed.

Trace: Aggregation/rounding

REP-09

Controlled fixture and expected result: Report spans a controlled reservation cancellation. Snapshot before cancellation gives REP-01; new snapshot after cancelling S1 booking gives A=210/O=60/free=150, utilisation=28.57%, confirmed=1/cancelled=2/share=66.67%. Each report must match one whole snapshot, never mixed-query values.

Trace: Consistent snapshot; restatement

REP-10

Controlled fixture and expected result: Convert a one-hour UTC interval to each fixture's club zone; elapsed duration remains 60 min. Inclusive date range ends at next local midnight; test an offset-transition day using pinned zone data and explicit offsets. End <= start or unresolved local ambiguity is rejected.

Trace: BR-14; ADM-12; club-local boundaries

Threshold: exact ID counts and exact expected interval/duration values before formatting; exact displayed two-decimal strings/N/A/error outcomes. Do not loosen tolerance to hide a wrong denominator. No financial revenue or actual-attendance inference is evaluated because those inputs are outside scope.

8. UX, responsiveness and formative comparison

Owner: Mansur, with Nurali on admin/report tasks. In October a walkthrough can inspect prototype coverage; record its findings separately as design-review observations. A static mockup cannot establish server enforcement, real persistence, end-to-end performance or successful production booking. Application UX acceptance is executed later on a runnable integrated build.

UX-01

Procedure and expected outcome: Walk SC-01-06: catalogue/filter, interval/station, confirmation/conflict, own history/cancellation and public calendar. Verify club/zone and selected context persist. Optional adjacent-seat mockup does not imply an implemented multi-seat feature.

Trace: BR-01/03/07/08/14/16; SC-01-06

UX-02

Procedure and expected outcome: Inject loading, empty, invalid, expired session, forbidden, conflict, stale version, server failure, offline-before-send and unknown-after-send states. Distinct textual message, safe next action, retained valid input; never false success/empty-data substitution.

Trace: BR-11/12/13; ADM-14; common states

UX-03

Procedure and expected outcome: Execute core visitor/admin flows at all E-UI viewports and browser combinations. No page-level horizontal overflow, clipped critical control, overlapping unreadable text or inaccessible modal action. Tables may scroll inside a labelled contained area.

Trace: NFR-06; SC-01-14

UX-04

Procedure and expected outcome: Keyboard-only catalogue/book/cancel and admin block/report flow: meaningful labels, visible focus, logical order, reachable actions, no keyboard trap. Statuses convey text as well as colour. Automated accessibility checks supplement manual checks; no WCAG certification claim.

Trace: NFR-06; BR-13

UX-05

Procedure and expected outcome: Formative visitor study using the paired baseline protocol below: find a specified free station, create one reservation and cancel it. Score final state independently. Goal at least 80% unassisted completion per task.

Trace: NFR-07; SC-01-05

UX-06

Procedure and expected outcome: Assigned admin selects correct club, reviews bookings, attempts conflicting block, then requests report. Out-of-scope controls/data are absent, conflict preserves booking, period/denominator/N/A are understandable. If ADM-15 preview exists, verify its summary and later conflict handling.

Trace: ADM-07/12/15; SC-07/10/11/12

Controlled manual-workflow baseline

Recruit at least five relevant visitors outside the development team; record sample size and experience, obtain consent for anonymised timing/error notes, and allow withdrawal. Add two representative club staff/admin participants if available for a separate formative UX-06 round; if unavailable, label team admin walkthrough as expert inspection, not a user study.

Compare the runnable CyberSeat workflow with a **simulated manual booking workflow**: participant requests station/time from a moderator acting as club operator; the operator checks an equivalent reservation spreadsheet and records/cancels the booking. Use synthetic data and an in-session conversation; do not send real messages or make real club reservations. This is a reproducible comparison condition, not a measured description of a particular real club.

Use matched but different task data for two conditions, same devices, identical club hours/availability/conflicts and a frozen moderator script. Counterbalance order: for five visitors, three manual-first and two CyberSeat-first; alternate for further participants. Give equal short neutral introductions and one practice task outside scoring. Reset data between conditions; keep moderator workload and response script fixed, record actual response delays. Do not remove manual operator waiting time from the end-to-end measure; log it separately so its effect is visible.

For each task start timer when instruction is shown; stop at independently verified intended final state. Task cap is 180 seconds, proposed for this study. Timeout, incorrect state or facilitator intervention means not unassisted success; record failure/censoring, not an invented completion time. Offer assistance only after scoring the task. For create, verify one correct reservation; for cancellation, verify retained CANCELLED history and released occupancy in the system, and equivalent retained cancelled row in the manual baseline.

Record per participant/condition/task: success, assistance, completion time, errors, wrong station/time choices and duplicate/conflicting entries. Report per-task counts, paired outcomes and individual times, not only a pooled percentage. Compute time reduction only for paired successful attempts: 100*(manual_time-system_time)/manual_time; report median individual reduction and how many pairs qualify. Report failures separately to avoid survivorship bias. With five people the 80% target means at least 4/5 unassisted successes on each task. Samples are formative and cannot establish population-wide productivity gains.

9. Metrics, thresholds and comparison conditions

Existing NFR-04/07 targets are retained. Additional usability-time targets below are explicit proposed evaluation goals, not user-derived or already approved numerical findings. Changes require a versioned rationale before execution, never after seeing results.

Functional correctness

Conditions, data and measurement: All Must INT/SEC/REP/OPS cases and mandatory browser flows on reset fixtures; assert responses and durable state.

Planned threshold: 100% defined Must cases pass; zero critical/high defects; blocked/not-run does not pass.

Baseline/comparison: Requirement outcomes and positive/negative fixture controls.

Concurrency integrity

Conditions, data and measurement: CON suite, independent workers and synchronised starts, 100 valid runs per named variant. Count DB invariant violations per run.

Planned threshold: Zero violations; CON-01 exactly one successful creation in every valid run.

Baseline/comparison: Same requests sequentially under the same safe implementation; zero violations there too.

Access control

Conditions, data and measurement: Full F-AUTH operation/scope matrix through direct API and UI; verify no unauthorised data or mutation.

Planned threshold: All expected allow/deny outcomes match; zero unauthorised operations.

Baseline/comparison: ECR-8 matrix, with permitted own/assigned-club controls.

Report correctness

Conditions, data and measurement: F-REPORT at fixed snapshot; compare independent arithmetic, counts and formatted values.

Planned threshold: Exact expected counts/durations and displayed values/errors/N/A.

Baseline/comparison: Hand-calculated REP fixtures, not production-code reuse.

API latency

Conditions, data and measurement: E-LOAD/F-LOAD; k6 50 VUs, 80% availability/10% create/10% cancel, 2-min warmup excluded + 10-min measured run; three independent restored runs. Measure client HTTP elapsed time by operation, including lock/retry waits. Use nearest-rank p95 on successful responses; classify conflict/error latency separately.

Planned threshold: In every run p95 availability <=500 ms; create/cancel <=1000 ms. At least 100 successful samples per operation per measured run, otherwise inconclusive.

Baseline/comparison: Three additional 1-VU baseline runs with same data/mix/durations and fixed environment. Report absolute and relative latency; baseline does not replace absolute thresholds.

Reliability and contention

Conditions, data and measurement: Same measured workload: unexpected 5xx / all completed requests; report transport failures separately and include them in overall failure rate. Count bounded retries and timeouts; record achieved request/s and operation mix.

Planned threshold: Unexpected 5xx <1% in each run; no integrity loss. Zero successful response for a rolled-back operation; configured lock timeout 2 s, at most 3 full-transaction retries after initial attempt.

Baseline/comparison: 1-VU run and explicit injected-fault checks; expected business 409 conflicts are reported separately, not unexpected failures. No invented throughput threshold.

Viewport/keyboard usability

Conditions, data and measurement: E-UI browser/viewport matrix, core flows and defined UI states.

Planned threshold: Zero blocking layout/keyboard defects; all required textual status/error states visible and actionable.

Baseline/comparison: Same task outcomes in desktop 1440x900; mobile must retain required functions.

Visitor task success

Conditions, data and measurement: Paired study, >=5 visitors, three tasks, 180 s cap, independent final-state scoring.

Planned threshold: NFR-07 Should goal >=80% unassisted for each task; show numerator/denominator and failures.

Baseline/comparison: Matched simulated manual workflow under counterbalanced order.

Visitor task time and errors

Conditions, data and measurement: Same paired study; task/participant times and wrong/duplicate/conflicting entries.

Planned threshold: Proposed Should improvement goal: median paired time reduction >=20% per task with at least 4 valid paired successes; otherwise descriptive/inconclusive. Proposed guardrail: no more duplicate/conflicting reservations than manual condition.

Baseline/comparison: Manual workflow times/errors. No benefit claim if baseline unavailable, failed or incomparable.

Persistence/reproducibility

Conditions, data and measurement: Commit/restart/rollback fixtures and independent clean setup.

Planned threshold: All expected IDs/statuses/mappings survive restart; no pre-commit residue; teammate reproduces migrations/seed without hidden manual changes.

Baseline/comparison: Pre-restart snapshots and documented fresh-install inputs.

Keep all three load-run summaries and raw samples; do not choose the best run. Use the same fixed bounded pacing/think-time configuration in target and single-VU baseline and record it with the load script. Prepare enough nonoverlapping slots and owned cancellable records to exercise valid create/cancel paths; report achieved operation mix rather than assuming configured weights were achieved. A timeout/failure is not silently omitted from reliability reporting, even though success-only latency is a separate distribution.

10. Operational/failure checks

OPS-01

Owner: Nurali

Procedure and expected outcome: Verify creation/cancellation/admin success audit is durable with mutation. Force rollback: neither success mapping nor partial business/audit success remains. Rejected conflicts/access failures retain safe correlated operational evidence.

Trace: BR-15; ADM-13; RBAC-09

OPS-02

Owner: Alibi

Procedure and expected outcome: Commit booking, restart API, reconcile same key/read history: same ID/status. Inject failure before commit: no booking or success mapping; lost committed response resolves original result.

Trace: NFR-03; BR-06/11

OPS-03

Owner: Alibi

Procedure and expected outcome: Another teammate installs pinned dependencies, applies migrations and loads versioned synthetic seed in fresh environment using recorded commands; no undocumented schema edits.

Trace: NFR-08

OPS-04

Owner: Alibi/Nurali

Procedure and expected outcome: Isolated fault harness induces lock wait/deadlock/serialization failure and exhaustion. Check 2 s lock-timeout configuration, whole-transaction retries <=3, safe rollback and explicit retryable failure; known 409 business conflict is not retried as infrastructure failure. This bound is not a promise of <=2 s total request time across retries.

Trace: NFR-05; architecture protocol

11. Requirement-to-check coverage

This is planned traceability, not a pass report. Case IDs refer to the procedures above; every source Must requirement has a planned check. PERF-01 is the three-run load protocol in section 9. Optional/deferred cases must be labelled rather than silently removed.

Source requirement	Planned checks
BR-01, BR-02	INT-01, INT-09, UX-01
BR-03	INT-02, UX-01
BR-04, BR-05	INT-03, CON-01
BR-06	INT-04, CON-02, OPS-02
BR-07	INT-05, SEC-01/02
BR-08	INT-06, CON-05, SEC-01
BR-09, BR-10	INT-11, CON-03/04/05
BR-11, BR-12	INT-07/08, CON-02, UX-02, OPS-02
BR-13	UX-02/03/04

Source requirement	Planned checks
BR-14	INT-09, REP-10, UX-01
BR-15	OPS-01, SEC-04
BR-16 (Should)	INT-08, UX-01/02
RBAC-01, RBAC-04	SEC-01, INT-05/06
RBAC-02, RBAC-03	SEC-02, INT-10/11/12, REP-01-10 with foreign-club variants
RBAC-05	SEC-03, CON-06
RBAC-06, RBAC-07	SEC-02/03
RBAC-08, RBAC-09	SEC-04, OPS-01
ADM-01, ADM-02, ADM-03	INT-10, SEC-02/03, CON-06
ADM-04, ADM-05, ADM-06, ADM-08	INT-11, CON-03/04/05
ADM-07	INT-05, SEC-02, UX-06
ADM-09, ADM-10, ADM-11	INT-12, SEC-02, UX-01/06
ADM-12	REP-01-10, SEC-02, UX-06
ADM-13	OPS-01, SEC-04
ADM-14	CON-07, INT-07, UX-02
ADM-15 (Should)	UX-06 if preview is implemented; otherwise record deferral
NFR-01	CON-01-07; DB invariant oracles
NFR-02	SEC-01-04, INT-05, CON-06
NFR-03	OPS-01/02
NFR-04	PERF-01, section 9 conditions and thresholds
NFR-05	OPS-04, PERF-01 contention/failure counters
NFR-06	UX-01-04/06, E-UI matrix
NFR-07 (Should)	UX-05; counterbalanced manual baseline
NFR-08	OPS-03
NFR-09	SEC-05, SEC-01/04
NFR-10	REP-01-10
SC-01-06	INT-01-09, INT-12, UX-01/02/05
SC-07-14	INT-10-12, SEC-01-04, REP-01-10, UX-06

12. Execution gates, evidence and evaluation limits

October plan acceptance: inputs/owners exist, cases have expected outcomes, metrics have conditions/data/threshold/baseline, traceability is complete and no executed-results claim is made. A named teammate

must review the new plan against its commit; record actual agreement source (PR review or reported team agreement), without impersonating another reviewer.

Future execution entry: integrated build available; schema/auth/API contracts agreed; shared booking/admin protocol implemented; migrations and deterministic seed verified; dependency/browser/time-zone versions and test commands pinned; measurement environment and manual baseline prepared. Until these exist, corresponding tests remain NOT RUN or BLOCKED, never PASS. Prototype-only inspection is labelled separately.

Execution order: clean setup and smoke checks -> integration/report/access checks -> controlled races/faults -> load -> browser/responsive checks -> formative paired user evaluation. Stop affected testing on data loss/access leakage, retain evidence, fix and rerun affected cases plus dependent regression checks.

Future exit: every defined Must case passes under recorded conditions; zero unresolved critical/high defects; mandatory latency/reliability/integrity thresholds met in all valid runs. Record all Should failures/deferments separately. Critical = unauthorised access, data loss or double booking; High = core booking/cancellation/report flow unusable or materially wrong; lower-severity cosmetic issues require an owner and disposition. Failed or missing evidence cannot be converted to PASS by merging this document.

For each future run retain: run ID/date, commit/requirements SHA, dataset seed/revision, environment manifest, commands, case IDs, expected and actual outcomes, status (NOT RUN/BLOCKED/PASS/FAIL), defect links, request/operation IDs, sanitised traces and DB assertions. Load evidence includes raw per-request samples, percentiles by operation, rates, errors/conflicts, retries/timeouts and all three run summaries. UX evidence uses anonymised participant IDs, condition/order, task scripts, assistance, timing, failures and qualitative notes. No credentials or unnecessary personal data enter artifacts.

Evidence locations will be docs/testing/results/<run-id>/ with a run manifest and machine-readable results plus a readable summary; these paths are planned, not files claimed to exist. Summarise planned vs executed vs passed vs failed vs blocked counts with explicit denominators. Keep design-review observations apart from application test results.

Limitations to report: convenience sample and learning effects; simulated rather than observed real-club baseline; hardware/network dependence of latency; finite race repetitions; mockup limitations; absent attendance/financial data. Compare only runs with matching conditions or explain the change. Quantified efficiency claims require the actual baseline and corresponding raw results.

13. Task completion evidence

Recorded in source: Plan defines integration/concurrency work for Alibi, permission/report checks for Nurali and UX/responsiveness work for Mansur.

Recorded in source: Metrics specify conditions, synthetic data, thresholds and explicit baselines.

Recorded in source: Requirement IDs map to planned checks, using requirements, architecture and UX inputs.

Recorded in source: Plan clearly distinguishes expected values and future execution from actual results.

Recorded in source: Record the actual reviewing participant, plan commit and approval outcome: NeryBR's approval of PR #6 (commit `eee6c80b0134a252eb4a1f862df9701e30f378d7`). Any newly identified blocking issue still requires resolution.

Recorded in source: The plan permalink, reviewed commit, PR #6 and actual NeryBR review link were published in ECR-13 on October 24 with Mansur’s explicit permission; the saved comment was verified.

Recorded in source: Matched the live ECR-13 criteria to owners/cases, metric conditions/data/thresholds/baselines and requirement traceability on October 24; the card was already Done. No planned test is marked executed.

PR #6 supplies approval of ECR-13 itself. No final tests or measured improvement are asserted by this plan.

10.1 Project stages and dependencies

Source: New package record prepared October 24, 2026

Version 1.1 · October 24, 2026 · ECR-15 · Coordinator: Akhmetov Nurali · University item 10: Project management and risks

Status: proposed plan approved by Mansur in PR #10, with subsequent team agreement reported by Mansur. See the October 24 review and agreement record (new package record; see chapter 11). This document records responsibilities, delivery gates and dependencies; it does not establish that a milestone, review, submission or application test has been completed. Read with the risk register.

1. Sources and deadline boundary

Sources inspected: current README, contribution rules, project proposal, documentation map and the task description supplied by Nurali. The repository inspected on October 24 includes analysis/design documents, an interactive local UX simulation and an implementation scaffold; it does not claim a runnable integrated backend/frontend or final evaluation results.

The authenticated October 24 follow-up verified ECR-15, its assignee Akhmetov Nurali, acceptance criteria and existing Done status. ECR-8 and ECR-13 were also inspected and already Done. This does not assert the status of other referenced cards. See the closure evidence (new package record; see chapter 11).

Date/control	Meaning
October 21, 2026 at 20:00	Historical internal target for this management task, supplied in its description. It is not a university deadline or evidence of completion.
October 24, 2026 BEFORE 18:00	Mandatory submission of the October package in the university-specified time zone. Sending at 18:00 does not satisfy a before-18:00 deadline.
Post-submission development	Separate gated work; dates require later team/supervisor scheduling. No later university deadlines are invented here.

Mansur confirmed the submission form and that he will send the package himself. He also confirmed Aibekuly Alibi as backup and the October 24 before-18:00 deadline in Astana time, UTC+5. These are user-confirmed release controls, recorded in ECR-15. Do not infer the zone from GitHub timestamps or a local computer clock. Retain the authoritative notice/channel reference with submission evidence; see the release record.

2. Responsibilities and handoffs

These assignments follow the README/proposal ownership. Coordinating this plan does not transfer the others' implementation work to Nurali or record their agreement. Reviewers shown are proposed; record who actually reviews each result.

Shared data and booking

Primary owner: Aibekuly Alibi

October contribution: Proposal; literature; technology comparison; booking/NFR specification; system context and component architecture. Review common data/transaction decisions with Nurali.

Later implementation / handoff: ECR-12 detailed model, migrations, booking create/cancel, overlap prevention, integration and load checks; agree API with Mansur.

Administration and reporting

Primary owner: Akhmetov Nurali

October contribution: Role/access matrix; administrative specification; verification plan coordination; project stages/risks; supervisor-feedback record coordination.

Later implementation / handoff: ECR-23 roles, assigned-club administration, schedules, blocks, events, reports and audit; coordinate all booking-affecting writes with Alibi.

Visitor/admin interfaces and evidence

Primary owner: Mansur Urynbassar

October contribution: User evidence; existing-system comparison; scenarios; use cases; UX prototype and responsive states.

Later implementation / handoff: ECR-21 screens, shared API integration, event/calendar UI, browser/responsive checks and user-task evaluation.

Cross-component contract

Primary owner: Alibi and Nurali, with Mansur accepting the client boundary

October contribution: Agree identifiers, ownership, interval/status semantics, safe errors and scope; log open decisions.

Later implementation / handoff: Schema/API/configuration changes are announced and versioned before dependants use them.

Package coordination

Primary owner: Nurali (proposed coordinator), all owners supply their sections

October contribution: Assemble an index/checklist; identify missing evidence/review; confirm university channel/time zone; coordinate ECR-17 access/final-tag checks.

Later implementation / handoff: Each owner remains accountable for correctness of their material. Aibekuly Alibi is the submission backup, as confirmed by Mansur on October 24.

Submission

Primary owner: Mansur Urynbassar (confirmed by him on October 24)

October contribution: Personally fill and submit the university form after package checks.

Later implementation / handoff: Backup: Aibekuly Alibi, as reported by Mansur; retain receipt and exact submitted version.

Each person commits from their own account. Another participant reviews the affected area. Agreement conveyed by Nurali must be labelled as reported agreement; no fabricated commits, reviews, respondent records or supervisor approval.

3. October materials versus subsequent development

Problem, objectives and bounded proposal

Reference / location: ECR-2; proposal

Dependency and October exit condition: Align team scope and evidence limitations with README. Do not claim validated local demand without real evidence.

User needs and alternatives

Reference / location: ECR-3/4/5/6; research

Dependency and October exit condition: Trace needs to real evidence or explicitly mark assumptions; cited comparisons support requirements/technology decisions. Respondent access is R-01.

Scenarios and requirements

Reference / location: ECR-1/7/8; requirements

Dependency and October exit condition: Four roles, own-record/assigned-club boundaries, intervals, booking protection and report formulas are consistent and checkable.

Architecture and interaction design

Reference / location: ECR-9/10/11; diagrams and UX

Dependency and October exit condition: Editable sources/readable exports and prototype flows match requirements. Browser-memory simulation is clearly distinguished from a live service.

Verification/evaluation plan

Reference / location: ECR-13; testing

Dependency and October exit condition: Assign future integration/concurrency, access/report and UX checks; specify conditions, fixtures, thresholds and baselines; no final-result claims.

Management and risks

Reference / location: ECR-15; this plan and risk register

Dependency and October exit condition: Three owners, gates, dependencies, risk treatment and review/evidence procedure are explicit.

Repository and configuration

Reference / location: ECR-14; repository guides

Dependency and October exit condition: README/module ownership/configuration boundaries are documented. Scaffold presence is not runtime implementation.

Feedback and final package

Reference / location: ECR-16 feedback; ECR-17 access/final-tag checks

Dependency and October exit condition: Supervisor feedback received October 24: no comments; see the feedback record. Verify complete-package access, exact version and submission receipt separately; do not infer them from a merge.

This is a coordination map derived from the task and repository, not a replacement for the university's complete official submission checklist. The coordinator checks all official items, including any presentation/report requirements, before release. No implementation requirement is added merely to make the October package appear more complete.

Outside the mandatory October implementation scope: runnable API/frontend, detailed ER/migrations where assigned to later work, production hosting, completed integration/concurrency/load studies and final UX results. Existing scaffold or prototype work may accompany the design package but cannot replace mandatory analysis/design documents. Online payment, loyalty, hardware control, automatic tournament registration and multi-station group booking remain excluded/deferred as in the proposal.

4. Milestones and acceptance gates

The following October 24 checkpoints remain proposed internal controls in the Astana time zone (UTC+5) confirmed by Mansur, not additional official deadlines or statements that work was completed. If a proposed checkpoint has already passed when this plan is accepted, execute its gate immediately and replan the remaining buffer without moving the mandatory deadline.

G0 - Authority and ownership

Proposed checkpoint / ordering: First action, before scheduling release

Owner and prerequisite: Nurali; university notice and available team

Evidence needed to pass: Time zone/channel and submitter/backup are recorded in the release record; verify the official checklist and package coordination before passing G0.

G1 - Content/dependency freeze

Proposed checkpoint / ordering: October 24, 15:00

Owner and prerequisite: All section owners; G0 and completed mandatory drafts

Evidence needed to pass: One versioned package index; consistent roles/statuses/API assumptions; each missing item has an owner and disposition. Scope additions require explicit impact review.

G2 - Cross-review

Proposed checkpoint / ordering: October 24, 16:00

Owner and prerequisite: Relevant teammate per section; G1

Evidence needed to pass: Actual reviewer, reviewed commit and outcome recorded. Resolve blocking inconsistency, unsupported evidence or unreadable artifacts. Actual supervisor feedback is now recorded as no comments; any later unreceived response stays pending.

G3 - Package and access check

Proposed checkpoint / ordering: October 24, 16:30

Owner and prerequisite: Nurali coordinates; ECR-17 owner verified at G0; Alibi is submission backup; G2

Evidence needed to pass: Verify links and viewer/download access from the intended recipient perspective, readable exports, current index and exact commit. Confirm all official items, not only Alibi's dossier.

G4 - Send with buffer

Proposed checkpoint / ordering: Target October 24, 17:00; receipt check by 17:30

Owner and prerequisite: Mansur Urynbassar; G3

Evidence needed to pass: Upload/send through the university channel; retain receipt/time and exact artifact/version identifiers. If upload fails, trigger R-05 immediately.

G5 - Mandatory cutoff

Proposed checkpoint / ordering: October 24 BEFORE 18:00

Owner and prerequisite: Mansur Urynbassar and backup Aibekuly Alibi

Evidence needed to pass: Submission accepted/received before cutoff according to the university's channel. A PR merge or local upload attempt alone is insufficient evidence.

G6 - Post-submission baseline

Proposed checkpoint / ordering: After confirmed submission

Owner and prerequisite: All three; G5

Evidence needed to pass: Record the submitted version, feedback and remaining backlog. ECR-17 manages the final tag; do not move an existing submission tag.

No gate is marked passed by creation of this document. A critical missing mandatory item is escalated to the team immediately; submit the best complete approved package within the rules and disclose unresolved evidence rather than fabricating it. A deadline extension is not assumed without explicit university confirmation.

Subsequent development gates (proposed sequence, dates to be agreed)

D1 - Foundation

Dependency: Submitted design baseline; resolve model/auth/config questions

Lead / contributors: Alibi for data; Nurali for access; Mansur for client contract

Exit condition: Agreed ER/API contracts, migrations/seed, authentication/catalogue and verified setup commands on synthetic data.

D2 - Core MVP

Dependency: D1; shared booking/admin transaction protocol

Lead / contributors: Alibi booking; Nurali administration; Mansur client

Exit condition: Find/create/view/cancel flow integrates; overlap, maintenance, schedule and assigned-club checks pass defined integration/race cases.

D3 - Diploma features

Dependency: Stable D2 and versioned report/event contracts

Lead / contributors: Nurali reports/events/audit; Mansur UI; Alibi data support

Exit condition: Event calendar, station layout and specified report/audit features integrated; historical report capacity retained.

D4 - Final evaluation and correction

Dependency: Implemented required flows plus executable ECR-13 plan

Lead / contributors: Owners execute their suites; all review limitations

Exit condition: Reproducible actual correctness/load/UX evidence, resolved critical defects, documented limitations and demonstrable system.

5. Dependency register

DEP-01

Upstream -> dependent work: User/client evidence -> proposal, scope, requirements and UX

Handoff contract / blocker response: Mansur supplies dated real evidence or a clearly labelled assumption; lack of respondents triggers R-01, not invented interviews.

Owner: Mansur

DEP-02

Upstream -> dependent work: Scenarios + booking/role rules -> architecture and UX

Handoff contract / blocker response: Agree four roles, one-station intervals, cancellation, informational events, errors and uncertain outcomes; inconsistent documents block G2.

Owner: Alibi + Nurali; Mansur confirms UX

DEP-03

Upstream -> dependent work: Shared data/transaction model -> booking and admin writes

Handoff contract / blocker response: Stable IDs, UTC/club-zone rules, shared locking/revalidation, operation identity and history. Schema/API changes require joint review.

Owner: Alibi; Nurali reviews

DEP-04

Upstream -> dependent work: Versioned API contract -> frontend/backend integration

Handoff contract / blocker response: Agree request/response/error examples and role scope before client integration; keep mocks tied to contract revision. Contract drift triggers R-03.

Owner: Alibi + Nurali; Mansur accepts

DEP-05

Upstream -> dependent work: Historical schedules/activity + booking statuses -> reports

Handoff contract / blocker response: Agree snapshot and station-minute definitions, preserve effective history and use independent control fixtures. Missing history blocks a valid utilisation claim.

Owner: Nurali; Alibi supports data

DEP-06

Upstream -> dependent work: Requirements + architecture + UX -> evaluation plan -> future results

Handoff contract / blocker response: ECR-13 maps requirements to checks. Execution waits for runnable build, migrations/seed and documented environment; planned checks are not results.

Owner: Nurali coordinates; all execute owned checks

DEP-07

Upstream -> dependent work: All mandatory materials + teammate review -> package -> access/receipt

Handoff contract / blocker response: Index points to the exact release version; ECR-17 access/tag gate follows complete-package verification. No status is inferred merely from file presence.

Owner: Nurali coordinates; Mansur submits; Alibi is backup

DEP-08

Upstream -> dependent work: Actual supervisor feedback -> revised scope/plan

Handoff contract / blocker response: ECR-16 now records the October 24 WhatsApp reply and no-comments decision; see feedback record (new package record; see chapter 11). Revisit dependencies after material changes.

Owner: Nurali coordinates; all owners respond

The October 24 closure check (new package record; see chapter 11) reconciles the post-start cancellation rule and references to existing visitor evidence/product comparison. It records the confirmed submitter, backup and time zone without certifying any gate as passed. The October critical chain is mandatory drafts -> consistency/review -> package/access verification -> accepted submission. The implementation chain is agreed data/API/auth contracts -> integrated booking/admin/UI -> reports/features -> executed evaluation. These chains must not be conflated.

6. Change, risk and status control

Use one reviewable branch/PR per outcome, following CONTRIBUTING. Record the verified task ID in branch/commit/PR and YouTrack when known. A proposed scope change records reason, requirement impact, owner, effort, dependency impact and effect on G1-G5; all affected owners agree before implementation. Freeze nonessential additions after G1. Preserve the original baseline and decision history.

At each handoff, owners report artifact/commit, next step, blocker and needed decision. On October 24 review the risk register at G0-G4 and immediately when a trigger fires. After submission, review it weekly and when requirements/contracts change; this is a proposed team cadence, not a scheduled automation. A material booking-integrity, access or submission risk is escalated immediately to all three participants and, where needed, through the university's established supervisor/channel process.

Done means the stated acceptance evidence exists, blocking review comments are resolved and the task contains result/commit/PR plus the actual reviewer. PR merge is version-control evidence; it is not proof of respondent participation, supervisor approval, successful tests or university receipt.

7. Acceptance and closure record

Task criterion	Evidence in this plan
Responsibilities of all three	Section 2; README/proposal ownership and handoffs

Task criterion	Evidence in this plan
Milestones and dependencies	G0-G6, D1-D4 and DEP-01-08
Required five risks with owners and mitigation	R-01-05 in risk-register.md; additional R-06-08 protect access/report/evidence consistency
Development after first submission separated	Section 3 and separate D-stage gates
Result, commit/PR and reviewer	Attach both files and the final reviewed PR/commit to the actual management task; PR #10 reviewer approval recorded

Recorded in source: Verified the exact task: ECR-15, “Оформить этапы, риски и зависимости проекта”, assigned to Akhmetov Nurali; authenticated inspection on October 24.

Recorded in source: A named teammate approved the plan/risk register against the recorded commit: NeryBR's approval of PR #10 (commit `84602e21fe42c5f31bf91b70fdd96aa992271288`). Outstanding release decisions and any new blocking comments remain separate.

Recorded in source: Mansur confirmed the submission form, himself as submitter, Aibekuly Alibi as backup and the October 24 deadline BEFORE 18:00 Astana time (UTC+5). See the release record and ECR-15 comment; no form submission is claimed.

Recorded in source: Both file permalinks, reviewed commit, PR #10 and actual NeryBR review link were published in ECR-15 on October 24 with Mansur’s permission; the saved comment was verified.

Recorded in source: Matched the live ECR-15 planning criteria to sections 2-5 and R-01-R-05; the card was already Done, with Nurali’s result/commit/PR/reviewer record. No status change or full-project submission is claimed.

10.2 Risk register

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/report/risk-register.md

Version 1.0 · October 24, 2026 · Register coordinator: Akhmetov Nurali · University item 10

Status: initial proposed assessment approved by Mansur in PR #10; subsequent team agreement reported by Mansur. See the October 24 review and agreement record (new package record; see chapter 11). Risk scores and residual targets are planning judgments, not measured probabilities or evidence that treatment has been implemented. Read with the project management plan (new package record; see chapter 11), its G0-G6 release gates, D1-D4 development stages and DEP-01-08 dependencies.

1. Assessment method and ownership

Likelihood L: 1 = unlikely under current assumptions; 2 = plausible; 3 = likely without action. Impact I: 1 = local rework without a gate at risk; 2 = several deliverables/components affected or milestone threatened; 3 = mandatory submission, booking integrity, access protection or credibility of evidence compromised. Score S=L*I: 1-2 Low, 3-4 Medium, 6-9 High. These ordinal scores prioritise discussion; they are not statistical forecasts.

An owner monitors the trigger and organises treatment; collaborators keep their README responsibilities. Open means the risk is being tracked, not that an incident already occurred. A residual target is valid only after the owner supplies treatment evidence and the team reassesses it. Acceptance of residual risk is explicit; it is not implied by a lower number in this table.

2. Prioritised register

R-01

Risk / consequence: Target respondents unavailable; user-need evidence and UX validation remain incomplete.

L / I / score: 3 / 2 / 6 High

Owner: Mansur Urynbassar

Exposure and initial state: October visitor evidence is documented in ECR-3 (three owner-supplied discussions); administrator research and the later user study remain open. No independent authentication of interviews is claimed.

R-02

Risk / consequence: Concurrent booking/admin operations create overlapping reservations or invalidate existing bookings.

L / I / score: 2 / 3 / 6 High

Owner: Aibekuly Alibi

Exposure and initial state: October design consistency; D1-D4 implementation; Open, no implementation guarantee claimed.

R-03

Risk / consequence: Frontend and backend API contracts diverge, delaying integration or misrepresenting booking outcomes.

L / I / score: 3 / 2 / 6 High

Owner: Mansur Urynbassar

Exposure and initial state: October contract review; D1-D3 integration; Open. Alibi/Nurali own server contract decisions.

R-04

Risk / consequence: Scope expands into optional features, displacing mandatory materials or core implementation.

L / I / score: 2 / 2 / 4 Medium

Owner: Akhmetov Nurali

Exposure and initial state: G1 freeze through D4; Open.

R-05

Risk / consequence: Missing materials, review/access or submission failure causes late/incomplete delivery.

L / I / score: 3 / 3 / 9 High

Owner: Akhmetov Nurali

Exposure and initial state: Immediate October release; Open. High rating reflects deadline proximity and unverified release evidence, not a claim that submission is already late.

R-06

Risk / consequence: Incorrect ownership/club checks expose another visitor's or club's data or permit unauthorised changes.

L / I / score: 2 / 3 / 6 High

Owner: Akhmetov Nurali

Exposure and initial state: Requirements review and D1-D4; Open.

R-07

Risk / consequence: Report denominators/history/snapshots are inconsistent, producing misleading utilisation.

L / I / score: 2 / 2 / 4 Medium

Owner: Akhmetov Nurali

Exposure and initial state: Data design and D3/D4; Open. Alibi supports retained history.

R-08

Risk / consequence: Prototype/scaffold, planned tests or reported agreement are presented as implemented, measured or independently approved evidence.

L / I / score: 2 / 3 / 6 High

Owner: Akhmetov Nurali

Exposure and initial state: October package and later reports; Open.

R-01-R-05 are the five risks explicitly required by this task. Additional risks follow the current role/report requirements and the repository's evidence boundaries.

3. Triggers and treatments

R-01

Observable trigger: No confirmed relevant respondent before the evidence checkpoint; missed session or unavailable replacement; required user evidence missing at G1.

Prevention / mitigation: Mansur proposes alternative relevant visitors/club staff, remote sessions and a short interview protocol; request participation early and record actual dates/sources/limitations. Separate assumptions from respondent evidence.

Contingency when triggered: Use another genuine participant or remote slot. Record actual contact attempts and evidence gaps; seek guidance through the established supervisor/channel process. Do not invent interviews, use

teammates as target-user evidence or mark unmet mandatory evidence complete. Replan the later formative study explicitly.

Evidence and residual target: Dated anonymised evidence record and confirmed session/replacement; target L=2/I=2/S=4 after real availability is established. DEP-01, ECR-3 and UX evaluation.

R-02

Observable trigger: Any two confirmed overlaps, booking/maintenance overlap, silently changed booking, or admin writer bypassing common guards; contradictory design rules at G2.

Prevention / mitigation: Alibi and Nurali agree half-open intervals, UTC/club zones, shared transactional guards/revalidation, operation identity and database overlap protection; review all booking-affecting writes together. Plan real-PostgreSQL integration and synchronised races in ECR-13.

Contingency when triggered: Block the unsafe mutation/release, retain synthetic reproduction and DB evidence, fix the shared protocol, then rerun affected integration/concurrency checks. Do not hide conflicts in the UI or silently cancel valid bookings. In October resolve the design contradiction; do not claim unexecuted tests passed.

Evidence and residual target: Reviewed contract plus later zero-violation test evidence under recorded conditions; target L=1/I=3/S=3 only after implementation verification. DEP-02/03, BR-04/06/09/10.

R-03

Observable trigger: Request fields/statuses/errors differ from client assumptions; changed schema without notice; UI claims success after missing response; contract example fails client/server check.

Prevention / mitigation: Mansur coordinates a versioned API contract accepted by Alibi/Nurali: IDs, offset-aware intervals, CONFIRMED/CANCELLED semantics, operation keys, errors, permissions and report fields. Tie mocks to the same revision and review contract changes before implementation.

Contingency when triggered: Pause the affected integration, agree one documented version, update mocks/client/server together and rerun contract plus end-to-end cases. Keep a clearly labelled simulation if backend is unavailable; do not describe it as live integration.

Evidence and residual target: Contract revision and examples accepted by affected owners; later client/server checks on same revision; target L=1/I=2/S=2. DEP-04 and D1/D2.

R-04

Observable trigger: New payments/loyalty/hardware/group booking or other optional work enters the October critical chain; tasks grow without owner/impact estimate.

Prevention / mitigation: Nurali maintains the proposal exclusions and Must/Should boundary. Require reason, effort, dependency and deadline impact for changes; affected owners agree. Freeze nonessential additions at G1.

Contingency when triggered: Defer optional work to a separate backlog and restore the required package/core workflow. If a mandatory requirement changes, version the baseline and replan with all owners; do not silently delete an official requirement.

Evidence and residual target: Versioned scope/change record and a package checklist free of optional blockers; target L=1/I=2/S=2. DEP-02/07, G1.

R-05

Observable trigger: Mandatory artifact/review missing at a checkpoint; university time zone/channel unconfirmed; broken recipient access/link; no receipt by the 17:30 proposed control; submitter unavailable or upload fails.

Prevention / mitigation: Nurali confirms official time/channel/checklist at G0, names submitter and backup, assigns every remaining item, checks exact package/readability/recipient access at G3 and targets 17:00 submission with buffer. Track proof of receipt separately from PR merge.

Contingency when triggered: Activate the named backup, Aibekuly Alibi (confirmed by Mansur); retry through the authorised university channel or an explicitly university-approved alternative. Immediately notify the team and seek university instructions for channel failure; preserve timestamps/errors. Do not assume an extension or treat an upload

attempt as receipt. If a checklist item is missing, escalate and transparently record the gap.

Evidence and residual target: Exact artifact/commit, recipient access check and receipt BEFORE October 24 18:00 in confirmed zone. Target L=1/I=3/S=3 after readiness checks; close this release-specific risk only after actual receipt evidence. DEP-07, G0-G5, ECR-17.

R-06

Observable trigger: Foreign record/club data returned; admin expands own assignments; stale revoked session still mutates; sensitive data appears in logs.

Prevention / mitigation: Nurali enforces server-side owner/club checks per operation, default deny, current assignments, revocation ordering and minimal data. Use allowed/denied matrix fixtures and direct API checks, not just hidden UI controls. Keep secrets out of source.

Contingency when triggered: Disable the affected route in the controlled release, preserve safe evidence, correct scope checks and rerun negative and positive controls. Treat any real exposure through the team's actual incident process; do not paste sensitive details into public issues.

Evidence and residual target: RBAC-01-09 matrix and later SEC/CON-06 results showing zero unauthorised operations; target L=1/I=3/S=3. DEP-03/04, D1/D2.

R-07

Observable trigger: Hand calculation differs, maintenance double-counted, A=0 yields 0% instead of N/A, history missing or mixed snapshots change one report inconsistently.

Prevention / mitigation: Nurali agrees available/occupied station-minute and cancelled-count formulas with Alibi, retains effective activity/schedule/maintenance history and uses one committed snapshot with independent expected fixtures.

Contingency when triggered: Withhold misleading metric, show insufficient-history/data-integrity error where defined, correct model/query/formatting and rerun all affected controlled calculations. Never replace an error with a valid-looking zero or call scheduled usage attendance/revenue.

Evidence and residual target: Reviewed data/formula contract and later REP fixtures matching expected values; target L=1/I=2/S=2. DEP-05/06, ADM-12, NFR-10.

R-08

Observable trigger: Wording says tests passed without run artifacts, static/local UX is called the complete app, reviewer name lacks actual review source, supervisor approval/Done inferred from file existence.

Prevention / mitigation: Nurali asks each owner to label artifact stage, actual vs expected values, limitations and review source. Cross-review claims against source files/commits. Keep pending feedback/review and execution gates explicit.

Contingency when triggered: Correct the affected report/PR/task record before G3, identify what is actually evidenced and retain correction history. If new evidence is needed, assign an owner and deadline; do not manufacture it to satisfy a checklist.

Evidence and residual target: Traceable review/commit records, actual-feedback source and executed-run evidence only when available; target L=1/I=3/S=3 after cross-check. DEP-06/08, ECR-13/16, G2/G3.

4. Monitoring, escalation and closure

Review R-01-08 at G0-G4 on the submission day and immediately after a trigger. After confirmed submission, review weekly and at material scope/data/API changes as a proposed team practice; no automated reminders are created by this document.

For a triggered risk, record: risk ID, observed date/time and time zone, evidence link, affected gate/dependency, action owner, action due point, contingency decision and outcome. Escalate a High risk immediately to all three participants. The owner can change a score only with an explanation and evidence. Do not reduce impact merely because mitigation is proposed.

Use states Open -> Treatment in progress -> Monitoring -> Closed, or Escalated when a gate is threatened. Closure needs evidence and teammate acknowledgement; a PR merge alone is not closure. For later technical risks, agreed design can reduce uncertainty but cannot substitute for implementation tests. R-05 must be reassessed at the next real release even after this October submission succeeds.

The authenticated October 24 follow-up verified ECR-15, assigned to Akhmetov Nurali and already Done, with the plan/register, PR #10, commit and reviewer identified in its result record. See the closure evidence (new package record; see chapter 11). This is planning-task evidence, not package receipt or execution of risk treatments.

5. Review record

Prepared by Akhmetov Nurali with assistance in drafting, based on the supplied task and repository material. Proposed reviewers: Alibi for booking/data risks and Mansur for respondent/API/UX feasibility; all three agree submission ownership and timing. Mansur's actual GitHub approval is recorded in PR #10; additional agreement was reported by him on behalf of the team. See the October 24 review and agreement record (new package record; see chapter 11).

Recorded in source: Record the actual reviewer, reviewed commit, date and decision source: NeryBR's approval of PR #10 (commit `84602e21fe42c5f31bf91b70fdd96aa992271288`), submitted October 24, 2026.

Recorded in source: Treatment owners and proposed G0-G4 checkpoints are covered by the recorded team agreement reported by Mansur. He subsequently confirmed Aibekuly Alibi as submission backup and Astana time (UTC+5); ECR-15 and the release record retain the confirmation. Actual gate execution and risk treatment remain separate.

Recorded in source: Check published PR comments and correct identified documentation inconsistencies. The October 24 check found no issue/review comments on PRs #5, #6, #10 and #11; scenario cancellation wording and research references were reconciled. See the closure evidence record (new package record; see chapter 11). Submission receipt and actual execution of the planned gates remain separate.

Recorded in source: Existing ECR-15 result/commit/PR/reviewer evidence was verified; direct file and review permalinks were added with permission on October 24. The task was already Done before this supplemental comment; no earlier closure chronology is rewritten.

11. Supervisor feedback and response

Source: New package record prepared October 24, 2026

Record updated: October 24, 2026. Coordinator: Akhmetov Nurali.

Supervisor and receipt

Supervisor: Myrzakerimova Alua, as identified in the project README and by Nurali when supplying the original conversation image.

Channel: WhatsApp. Date received: October 24, 2026, confirmed by Nurali. The screenshot shows the request at 17:54 and the supervisor's reply at 17:55; it does not display a calendar date or time zone. The date and channel are therefore attributed to Nurali's explicit confirmation rather than inferred from the image.

Request and reviewed materials

The visible request links Pull Request 13 and asks the supervisor to inspect the consolidated PDF and requirements checklist for the October analysis-and-design submission. It explicitly separates later implementation and final tests. The supervisor states that she checked YouTrack, the PDF, and GitHub.

Verbatim reply in the original language

The following Russian text is reproduced verbatim as primary evidence; the surrounding record and its translation are in English.

Здравствуй Нурали! Да я проверила ваши работы в ютреке и пдф и гитхаб, мне все понравилось, не вижу никаких недочетов, так что я считаю что замечаний нет.

English translation

Hello Nurali! Yes, I checked your work in YouTrack, the PDF, and GitHub. I liked everything, I do not see any shortcomings, so I consider that there are no comments.

Decision and response to comments

The supervisor reported no shortcomings and no comments on the materials she reviewed. No substantive correction was requested. The disposition is therefore no corrective action required for that feedback. This is supervisor feedback on the October materials, not a GitHub review or a claim that the final application was implemented or tested.

Nurali subsequently relayed a joint message from Mansur Urynbassar and Aibekuly Alibi requesting that the reply, supervisor name, date, and channel be recorded without embedding the screenshot in the report; that stale missing-feedback statements be updated; and that the PDF be rebuilt. This record implements that documentation request. It does not claim that either participant has already reviewed the subsequently rebuilt PDF.

Evidence preservation

The original supplied image is retained separately as `supervisor-whatsapp-original.jpg` in the local evidence archive. Its SHA-256 is `989c31282072dc4711cd3f82a23d8dc47e6f552effe1376488ac7d50af9ef50c`. The original conversation should also remain available in Nurali's WhatsApp history. No screenshot of the private conversation is embedded in the report or committed to the public repository.

The screenshot was supplied by Nurali, who identified the respondent as the supervisor and confirmed the date and channel. The image shows the message exchange but no contact header. This provenance is retained rather than represented as independent access to the supervisor's account.

Change log

Supervisor feedback

Action: Replaced the earlier pending-feedback entry with the actual reply and receipt metadata

Evidence and limit: Original image retained; quotation transcribed and English translation supplied

Comments

Action: Recorded no comments and no required substantive corrections

Evidence and limit: Supervisor's actual reply; no invented correction or approval

Report and checklist

Action: Updated current feedback status and rebuilt the consolidated report

Evidence and limit: Technical checks and revised-PDF review are recorded separately

Historical evidence

Action: Retained the chronology of earlier missing evidence and earlier PDF reviews

Evidence and limit: Earlier records do not certify a later PDF revision

Task disposition

The ECR-16 feedback criterion is now supported by the supplied conversation: materials were presented, a real dated response was received, and no comments required correction. Record the delivered record and revision in YouTrack before changing the issue to Done. Final submission, access/tag controls, and review of the rebuilt PDF are separate tasks.

Appendix A. Eleven-component package audit

Source: New package record prepared October 24, 2026

Audit date: October 24, 2026. Coordinator: Akhmetov Nurali. Audited source baseline: `0ac6fb0f1b662896910127040ba6a3e0876ddb45` (PR 13 merged). Updated with the supervisor response supplied on October 24.

Method and boundary

All eleven components listed in ECR-18 were inspected against the current repository. Related live October YouTrack descriptions, assignees, priorities, and stages were inspected. This is a document and package audit assisted by Codex. The supervisor response is now recorded separately as primary evidence. Earlier PDF reviews and the submitted GitHub approval of PR 13 apply to their identified revision; they do not certify the rebuilt PDF. No application test results or submission receipt are claimed. Nurali coordinates the audit; he relayed the participation of Alibi and Mansur through their earlier full-document reviews and their joint request to incorporate the feedback.

The original university requirements document was not provided to this work. The eleven-component structure follows the live ECR-18 acceptance criteria. The existing closure record reports an earlier inspection of the university document and distinguishes it from the separately reported deadline and submission arrangements.

Eleven-component checklist

1

Component: Project proposal and scope

Source: project-proposal.md

Finding: Present: identity, problem, objectives, bounded scope, stages, and evidence limitations.

2

Component: Evidence of relevance and user needs

Source: user-needs-evidence.md

Finding: Present with limitations: three owner-reported anonymised visitor discussions; exploratory sample. Original exchanges were not independently authenticated; administrator research remains outstanding.

3

Component: Critical literature review

Source: literature-review.md

Finding: Present: source selection, 13-source critical matrix, limitations, and engineering implications. This audit checks the supplied synthesis, not independent replication of every cited study.

4

Component: Existing systems comparison

Source: existing-systems-comparison.md

Finding: Present: SENET, ggLeap, and LANGAME against nine criteria with source traceability. Unknown features remain unconfirmed rather than absent.

5

Component: Technology comparison

Source: technology-comparison.md

Finding: Present: backend, database, communication, client, and deployment trade-offs; decisions are proposed, not benchmarks.

6

Component: Requirements specification

Source: booking-and-nfr.md

Finding: Present: booking/NFR requirements, scenarios, administrative requirements, and four-role access matrix; priorities and acceptance criteria are documented.

7

Component: System Context, Use Case, and Component Diagrams

Source: architecture.md

Finding: Present: editable Mermaid/PlantUML sources and SVG exports. All three exports are embedded in the consolidated report; detailed ER design remains ECR-12 future work.

8

Component: UX prototype

Source: README.md

Finding: Present: interactive v2 local simulation and preserved v1.1 static screens. The report labels the historical screens correctly; real API integration, user studies, and v2 real-browser validation are not certified.

9

Component: Verification and evaluation plan

Source: verification-and-evaluation-plan.md

Finding: Present: owners, cases, fixtures, metrics, thresholds, comparison baselines, traceability, execution gates, and limitations. Future tests remain unexecuted.

10

Component: Project management and risks

Source: project-management-plan.md (new package record; see chapter 11)

Finding: Present: three participants, October gates, later development stages, dependencies, and an owned risk register. Planned checkpoints are not treated as executed.

11

Component: Supervisor feedback

Source: supervisor-feedback-record.md (new package record; see chapter 11)

Finding: Present: Myrzakerimova Alua replied through WhatsApp on October 24, 2026 at 17:55, reporting no shortcomings or comments. The original wording, English translation, provenance, and evidence checksum are recorded. No substantive correction was requested.

Repository and release controls

Repository

Observation: Public repository cloned successfully at 23047b9566a376f6c463b5111831946cd12bf429

Disposition: Verified technical availability; the supervisor reports checking GitHub; committee account access remains a separate ECR-17 verification

Individual contributions

Observation: Git history contains substantive source changes under Akhmetov Nurali, Alibi-stack, and NeryBR

Disposition: Authorship metadata inspected; identity and account ownership are not independently authenticated

Pull Request evidence

Observation: Source approvals are recorded in the team-review record. PR 13 has NeryBR approval #5638090721 and merge 0ac6fb0f1b662896910127040ba6a3e0876ddb45; the subsequent feedback update has its own revision

Disposition: Distinguish actual submitted reviews from comments and reported agreement

Review timing

Observation: The source review record notes some approvals after merge

Disposition: Do not claim universal compliance with review-before-merge requirements

Final submission version

Observation: No pre-defense-1 tag was present in the cloned baseline

Disposition: ECR-17 remains a separate unresolved release gate; no tag created or moved

Priorities and deadlines

Observation: Live October descriptions retain the before-18:00 cutoff; internal targets are explicitly separate

Disposition: Stage, owner, and priority observations recorded below; not all issue field dates are exposed in the inspected UI

Deadline/channel

Observation: Existing closure record reports Mansur's confirmation of October 24 before 18:00 Astana time, UTC+5, and the Microsoft form

Disposition: Reported source, not newly confirmed by the university in this work

Consolidated PDF

Observation: Team_3_Approbation_Report_24_October.pdf compiled from this baseline, with new audit and feedback-status records

Disposition: The report is rebuilt with the actual feedback. The previous PDF was reviewed by Alibi and Mansur; review of the changed sections in the rebuilt file is recorded separately and must not be inferred

YouTrack screenshot PDF

Observation: ECR-19 is Queue in the inspected live card; no final screenshot PDF was found in this repository

Disposition: Separate required artifact remains unverified; this consolidated report does not substitute for it

Submission receipt

Observation: ECR-20 is Queue; no receipt supplied

Disposition: No submission performed or claimed

Supervisor feedback

Observation: Actual response and no-comments decision are transcribed in the feedback record; date and channel confirmed by Nurali

Disposition: Feedback evidence present; no escalation is needed for an answered request

Team participation

Observation: Nurali coordinates; he reported Alibi and Mansur's full review of the preceding PDF and relayed their joint instruction to incorporate the supervisor reply

Disposition: Historical reviews remain revision-specific; rebuilt-PDF review is separate

Live task observations

The following observations are a point-in-time snapshot from October 24. Titles are reproduced in English for this audit. The underlying ECR-19 and ECR-20 cards are outside this editing scope.

ECR-1

Assignee: Mansur Urynbassar

Priority: Major

Stage: Done

ECR-2

Assignee: Aibekyly Alibi

Priority: Major

Stage: Done

ECR-3

Assignee: Mansur Urynbassar

Priority: Major

Stage: Done

ECR-4

Assignee: Aibekyly Alibi

Priority: Major

Stage: Done

ECR-5

Assignee: Mansur Urynbassar

Priority: Major

Stage: Done

ECR-6

Assignee: Aibekyly Alibi

Priority: Major

Stage: Done

ECR-7

Assignee: Aibekyly Alibi

Priority: Major

Stage: Done

ECR-8

Assignee: Akhmetov Nurali

Priority: Major

Stage: Done

ECR-9

Assignee: Mansur Urynbassar

Priority: Major

Stage: Done

ECR-10

Assignee: Aibekyly Alibi

Priority: Major

Stage: Done

ECR-11

Assignee: Mansur Urynbassar

Priority: Major

Stage: Done

ECR-13

Assignee: Akhmetov Nurali

Priority: Major

Stage: Done

ECR-14

Assignee: Aibekyly Alibi

Priority: Major

Stage: Ревью

ECR-15

Assignee: Akhmetov Nurali

Priority: Major

Stage: Done

ECR-16

Assignee: Akhmetov Nurali

Priority: Major

Stage: Queue

ECR-17
Assignee: Aibekyly Alibi

Priority: Major

Stage: Queue

ECR-18
Assignee: Akhmetov Nurali

Priority: Major

Stage: Queue

ECR-19
Assignee: Mansur Urynbassar

Priority: Major

Stage: Queue

ECR-20
Assignee: Mansur Urynbassar

Priority: Critical

Stage: Queue

ECR-24
Assignee: Akhmetov Nurali

Priority: Critical

Stage: Queue

Audit conclusion and separate release controls

All eleven required components have been checked and linked. The previously missing supervisor response is now documented. The checklist explicitly identifies the remaining limitations; completion of this audit is not a declaration that every release task is complete.

ECR-18 can be completed as the eleven-component verification task once this updated checklist is published and its evidence is attached. Open controls remain visible rather than being described as complete:

1. ECR-24: verify the rebuilt PDF and obtain revision-specific confirmation of its changed feedback and checklist sections; retain the earlier full-PDF review history.

2. ECR-17: complete committee access and exact-version/tag checks; preserve any established submission tag.

3. ECR-19: complete the separate YouTrack screenshot PDF and confirm form limits.

4. ECR-20: submit through the authorised channel and retain the actual receipt.

The historical Alibi-only PDF remains a dated five-deliverable dossier. Future implementation in ECR-12 and ECR-21-ECR-23 is not an October completion prerequisite. The live-task table above is the earlier point-in-time snapshot, not a claim that those statuses are unchanged after this update.

Appendix B. Existing reviews and reported agreement

Source: New package record prepared October 24, 2026

Baseline and source

This record concerns the analysis/design materials at baseline commit `bba6485e70b03d456c4be994232e99baf373086d`. It reconciles stale review checklists with existing GitHub reviews and a new team-agreement report; it does not claim that the edits in this record received a new independent GitHub approval.

On October 24, Mansur reported in the working chat that the team had gathered, read the repository-audit notes and agreed with them. He requested that the agreement be recorded across the documents. English translation of his confirmation: "We have all read your notes and comments, and we agree with everything."

This is agreement reported by Mansur on behalf of Aibekuly Alibi, Akhmetov Nurali and Mansur Urynbassar. It is not a separately authenticated statement from each participant, a claim that each performed every named technical check, or an AI approval issued in place of a teammate. Named-action checkboxes without that evidence have been clarified as reported design agreement rather than left as unsupported claims.

Existing submitted GitHub reviews

The audit verified these public review records and compared applicable file content with the reviewed revisions (ignoring line-ending-only changes). Several approvals were submitted after merge; they are not represented as pre-merge gates.

#1

Scope: Original scenarios (historical)

Approval: Alibi-stack's approval of PR #1 (commit `ba1f58e8dcff85885dec1aea45a607d35a325d40`)

#3

Scope: Alibi design dossier

Approval: NeryBR's approval of PR #3 (commit `0fc2c4555821922e0dae7b233e091281e36a89f3`)

#4

Scope: Initial research/UX/UML (historical)

Approval: Alibi-stack's approval of PR #4 (commit `8830fc956bd17590ad7870c163452a51cc20d8f7`)

#5

Scope: Administrative requirements and permissions

Approval: NeryBR's approval of PR #5 (commit `834f87e1a78a5ebc1b5c29097b856427dd43f993`)

#6

Scope: Verification/evaluation plan

Approval: NeryBR's approval of PR #6 (commit `eee6c80b0134a252eb4a1f862df9701e30f378d7`)

#7

Scope: Corrected scenarios, research, UML and UX v1.1

Approval: Alibi-stack's approval of PR #7 (commit `d7bae8ea9931bbdb8e4caec01c243f33c8cea493`)

#8

Scope: Repository scaffold and guides

Approval: NeryBR's approval of PR #8 (commit `d087177856d5577db669ba207a2f507fbcc629e8`)

#9

Scope: Interactive UX v2.0

Approval: NeryBR's approval of PR #9 (commit `723579d61ac025b50f59ff64cb0086cb03698a0d`)

#10

Scope: Management plan and risk register

Approval: NeryBR's approval of PR #10 (commit `84602e21fe42c5f31bf91b70fdd96aa992271288`)

Remaining actions

Agreement with the documents does not select a missing submitter/backup, establish the university channel/time zone, prove that task links were added, close a YouTrack card, upload a PDF, confirm submission receipt or execute tests. Those action checkboxes remain open until verified. The later closure check (new package record; see chapter 11) records live task verification and subsequent release decisions without changing the provenance of the agreement above.

The following audit observations remain follow-up work where relevant: reconcile the post-start cancellation wording across scenarios and administrative rules; refresh cross-links from the older proposal/literature text; update or clearly date the historical Alibi-only PDF; verify actual-browser responsive and keyboard behaviour; and confirm the final package, access and submission evidence. Acceptance of these observations does not assert their resolution.

Future backend/frontend implementation, real database/concurrency/load tests, administrator research, and participant usability sessions retain their existing planned/unverified status. Supervisor feedback has subsequently been received; see the dated update below.

Subsequent package and supervisor evidence

PR 13 was approved by NeryBR in a submitted GitHub review against commit `f473efa9b683970bfbe81a868b62ce3ea36c1bb6` and merged as `0ac6fb0f1b662896910127040ba6a3e0876ddb45`. The later NeryBR comment reports a Codex-assisted inspection performed at Mansur's request. It is a comment, separate from the submitted approval.

Nurali supplied the supervisor's actual WhatsApp reply and confirmed its date as October 24, 2026. Myrzakerimova Alua reported that she checked YouTrack, the PDF, and GitHub and had no comments. See the verbatim feedback, translation, and provenance (new package record; see chapter 11). The image is retained separately and is not embedded in the report.

Nurali identified the accompanying instruction to record the response and rebuild the report as a joint message from Mansur and Alibi. Earlier full-document reviews concern the preceding PDF checksum. They do not automatically certify the rebuilt file; its verification record states the revision-specific review position.

Appendix C. Prior closure evidence and release controls

Source: New package record prepared October 24, 2026

Baseline: `a590e465c3ea5fd2fb52516e77ab70cda23fd78d` (PR #11 merged).

Repository checks completed

The public GitHub issue-comment and inline-review-comment endpoints for PRs #5, #6, #10 and #11 returned empty lists on October 24. This is a point-in-time observation about those PRs; it does not establish the absence of YouTrack comments or later feedback. Existing submitted approvals are listed in the review record (new package record; see chapter 11).

- SC-05 and the cancellation decision now explicitly prohibit post-start status changes in the current baseline, matching ADM-09 and the permission matrix. Future exceptions require separately approved club rules.

- The proposal and literature review now link the existing ECR-3 visitor discussions and ECR-5 comparison rather than describe them as missing deliverables.

- R-01 distinguishes the documented visitor discussions from outstanding administrator research and later usability recruitment. No risk score was reduced and no interview authenticity was certified.

- The management plan and risk register share this record; release decisions remain separate.

Official requirements source

The locally supplied university document “До 24 октября.docx” was read for its requirements. It lists the eleven evaluation components and repository requirements, including review before merge and exact submission tags. Its text does not specify 18:00, a time zone, a submission form URL or a named submitter/backup. The live YouTrack descriptions specify the before-18:00 cutoff. Mansur subsequently confirmed the applicable Astana time zone (UTC+5) and form in this conversation. The release controls below rely on that explicit confirmation; the DOCX itself is not represented as the source of those additional details.

The historical Alibi PDF remains a dated five-deliverable dossier. The current Markdown records and subsequent review/closure records supplement it; it is not a newly assembled full-team submission.

Authenticated YouTrack verification

After Mansur signed in, the live cards and their result comments were inspected on October 24. All three were already Done and assigned to Akhmetov Nurali; no status change was needed.

ECR-8

Acceptance cross-check: Four roles and assigned-club limits in role matrix; ADM-01-15, booking protection, priorities/criteria and report formulas in administrative specification.

Existing task evidence: Nurali’s October 23 comment links PR #5 and merge `2c1c419`, and names Alibi as the agreeing teammate. The independently visible GitHub approval is by NeryBR; the task’s attribution to Alibi is reported evidence, not a second GitHub approval.

ECR-13

Acceptance cross-check: Owners/cases in sections 2-8, metric conditions/data/thresholds/baselines in section 9, traceability in section 11, planned-versus-executed boundary in section 12.

Existing task evidence: Nurali’s result comment links PR #6, `eee6c80`, and names Mansur; the latest state is Done after an intervening reopen.

ECR-15

Acceptance cross-check: Responsibilities, checkpoints and dependencies in plan sections 2-5; required five risks R-01-05 each have an owner and mitigation; later development has separate gates.

Existing task evidence: Nurali's October 24 comment names both files, PR #10, `84602e2`, and reviewer Mansur.

Submission ownership

Mansur Urynbassar confirmed in this conversation that he will submit the final package himself through the Microsoft submission form. The form has not been filled or submitted by Codex. He subsequently confirmed Aibekuly Alibi as backup and the deadline of October 24, 2026 BEFORE 18:00 Astana time, UTC+5. Existing internal checkpoints (15:00 content freeze, 16:00 review, 16:30 package check, target 17:00 send and 17:30 receipt check) remain planning controls in that same zone; this record does not assert they were executed. Confirmation is reported by Mansur, not a separately obtained statement from Alibi. No submission receipt is claimed.

Disposition of the eleven remaining checklist items

Item	Result of this check
ECR-8: attach both document links and commit/PR	Completed: direct file permalinks, reviewed commit, PR #5 and actual GitHub review link published and verified in ECR-8 on October 24.
ECR-8: verify criteria and Done	Verified against live criteria; already Done.
ECR-13: attach file and commit/PR	Completed: plan permalink, reviewed commit, PR #6 and actual GitHub review link published and verified in ECR-13 on October 24.
ECR-13: verify criteria and Done	Verified against live criteria; already Done.
Management: verify task ID	Confirmed ECR-15 and assignee Akhmetov Nurali.
Management: channel, time zone, submitter and backup	Completed: Mansur submits through the supplied form; Alibi is backup; before 18:00 Astana time (UTC+5), confirmed by Mansur.
Management: attach file/commit/PR/review links	Completed: both file permalinks, reviewed commit, PR #10 and actual GitHub review link published and verified in ECR-15 on October 24.
Management: verify criteria and Done	Planning criteria verified; already Done. Submission itself remains separate.
Risks: owners/checkpoints/backup	Completed as a planning record: existing owners/proposed checkpoints retain the reported team agreement; Mansur confirmed UTC+5 and backup Alibi. Actual execution remains separate.
Risks: resolve blocking comments and inconsistent documents	No blocking comment was visible on the inspected PRs/cards; cancellation/research inconsistencies corrected in this change.
Risks: evidence in task before Done	Completed: existing ECR-15 evidence verified and supplemental direct links published with permission. The task was already Done; no earlier chronology is rewritten.

All eleven checklist items are resolved in this change. With Mansur's explicit permission, supplemental comments were published from his account in ECR-8, ECR-13 and ECR-15 and verified in the rendered task activity. These comments document the checks and existing review evidence; they are not newly submitted GitHub approvals of PR #12. Application tests, the final tag, the complete-team PDF, supervisor feedback and submission receipt are not certified by this checklist update.

Subsequent supervisor response - October 24

The pending-feedback statements above describe the evidence available during the earlier closure check. They are superseded for current status by the supervisor-feedback record (new package record; see chapter 11): Myrzakerimova Alua replied through WhatsApp, with a displayed time of 17:55 and an October 24, 2026 date confirmed by Nurali, reporting no shortcomings or comments. PR 13 has also been approved and merged. Earlier reviews remain tied to the original report revision; the subsequent rebuilt PDF has its own verification record. Submission receipt, committee access, and the final tag are not established by the supervisor response.

Appendix D. Repository verification

Source: https://github.com/NeryBR/cyberseat-booking-system/blob/0ac6fb0f1b662896910127040ba6a3e0876ddb45/docs/project/repository-verification.md

Verification date: October 24, 2026. Responsible participant: Aibekuly Alibi (`Alibi-stack`). Reviewer designated by ECR-14: Mansur Urynbassar (`NeryBR`). Review of this scaffold is recorded in NeryBR's approval of PR #8 (commit `d087177856d5577db669ba207a2f507fbcc629e8`). See the October 24 review and agreement record (new package record; see chapter 11).

Authorship and continuity

The existing English README was prepared by `NeryBR` in commit a6a0a29 and merged through PR 2. Alibi's five design deliverables were merged through PR 3, including the improved PDF. This task verifies and extends those materials; it does not attribute the original README to Alibi.

Acceptance coverage

ECR-14 criterion	Result
Preserve and identify existing README authorship	Original commit and PR linked above
Working YouTrack and documentation links	Existing board/scenario links retained; component, contribution, configuration and documentation maps added
Responsibilities, scope, status and proposed stack	Existing team/scope/stack sections preserved; scaffold responsibilities and pending implementation made explicit
Backend, frontend and test structure	Purpose/owner READMEs committed in the planned directories and architectural boundaries
Configuration examples	Root `.env.example` contains placeholder values and distinguishes server secrets from public browser configuration
`.gitignore` without committed secrets	Local environment files, keys, dependencies, caches and raw test output excluded; examples, source and lockfiles remain eligible for version control
Startup instructions only after a runnable prototype is verified	No runtime or application-test commands are claimed; implementation entry points and manifests remain later work
Mansur's review	Approved in PR #8; reviewer and commit linked above

Repository checks

Executed on October 24, 2026:

Check	Observed result
Relative links across the 47 Markdown files	122 relative targets resolved; zero broken paths

Check	Observed result
Ignore-rule cases for root/nested `.env`, example configuration, keys, environments, dependencies, caches, test artifacts, databases, lockfiles, source and PDF	All 25 cases matched the expected ignore/keep behaviour
Environment example	12 unique keys; secret/password fields use explicit replacement placeholders; public browser configuration contains no secret key name
Git whitespace validation for new scaffold content	Passed; inherited Markdown hard line breaks are preserved

The ignore checks used Git's actual rules; they did not create credentials or install dependencies. These results verify the committed scaffold and do not certify a running system or substitute for human review.

The final merge preserves the team's UX guide and the materials added through PRs 6 and 7. Markdown, Mermaid and SVG line endings are explicitly standardised to LF, matching `.editorconfig`; inherited Markdown CRLF line endings are normalised without changing document text or intentional hard line breaks.

No backend, frontend or database implementation is included, so application unit/integration/build/load tests are not claimed. This verification concerns repository structure and configuration boundaries.

Remaining work outside ECR-14

ECR-17 still requires actual participant contribution/access evidence and a verified final submission version before `pre-defense-1` is created. This scaffold does not claim that supervisor/committee access, the final tag, outstanding team deliverables or final submission have been completed.

Appendix E. Exact-source manifest

All existing source chapters were read from repository baseline 0ac6fb0f1b662896910127040ba6a3e0876ddb45. The feedback-status record and eleven-component audit were prepared for this package. The PDF hash is recorded outside the PDF to avoid a self-referential checksum.

docs/report/project-proposal.md
f9d73e543d24b29775ee30f4d2f843b9fbbb31ffdbacd0040a2504d4457aa95d

docs/research/user-needs-evidence.md
a6246880ec35943efb716f9e70b89d960fae3fe2205b1c38fbae6d83e6157654

docs/research/literature-review.md
30f4346e848c4ad20f12bd69c3ccf11e4117e4fd61ec9b32ff3050eb02d339eb

docs/research/existing-systems-comparison.md
ff1c472d15cdd004e1bebe5321a2b8281feb215613560fc6a7f4bdba88673d84

docs/research/technology-comparison.md
b55a16f7ea6c2ba9cc727bb2d3af5150aa1b24d27842b5e4714c6b852e07cabc

docs/requirements/user-scenarios.md
35ad24f1bce334afb4d1e526d8e99f9bba454ed886f75bc04dc128a6b5d5cf85

docs/requirements/booking-and-nfr.md
6915e71ed08cd6524bb8ee1f6b2c7cc922568b84da3ce867a51b52d966b4b28b

docs/requirements/roles-and-permissions.md
fb6a5efabdabd8dc789dc4297503d2f5e86c6977793366c55c2de95a21ef234b

docs/requirements/administrative-requirements.md
af340331664a4c743a76ffc1d9341d48d836a5c0a2c9435a4817aa2cfce26a31

docs/diagrams/architecture.md
f4e64fd9a85fb6019c3d8bed153e3e0069bcc20b3f27f72aaaf6db25bb2f71a1

docs/diagrams/README.md
da1d671530a43cf8bc29e1a829d7d8c6ae3dbba180458772bef3602c0da1c4d2

docs/ux/README.md
047513014865e1861c0f17baa3db8562b2afa291fe8ca59a9026ed85fdc6c977

docs/testing/verification-and-evaluation-plan.md
5aea254a61e7ab4d53e66f5d897943d4ffe88c3017eca344800cce41c5dd365b

docs/report/project-management-plan.md
0b91002fcee543d9536eca7be1f167a2df361160e9cf793d8485c4df6b66bfa4

docs/report/risk-register.md
32b9b7825805d25b2188755d473c1b29f65c34c2a50f20d17df523cbfea74ebc

docs/report/supervisor-feedback-record.md
4e19703c874d6a6f8b23aba5a0bdcaa1e5a8fa0ca57ee8a8f847b3e9a5885d9c

docs/report/october-submission-checklist.md
bab4deb7d20bd1258f198bdc1617d40fa883bfbb5d8f96f6bf352e996b47ded6

docs/project/team-review-2026-10-24.md
ab0070380a8a788a5ef889d804585383685121dc4cf74c382e78a6eb069cb8a8

docs/project/closure-evidence-2026-10-24.md
0b14d67460e41101004e8fbf0f4bc38c8e07ee43a8c785fe764f04898f4d102a

docs/project/repository-verification.md
d5281e3f4a29801bc4fea45db48bf9d2ccb56f969a30215dd9186affd473869d

Final human-review gate

Earlier full-document reviews by Alibi and Mansur are recorded against the preceding 121-page PDF (SHA-256 8498ccedb377999dc19c53ba91641f045663edc976c37b593bbe2c3f794fdcac). This rebuild adds the supervisor reply and updates the affected status records. Confirmation that the changed sections were checked in this new file must be recorded against its new checksum before ECR-24 closure. The supervisor response and PR 13 approval do not automatically approve later edits.